

A Brief History of Teleportation

Introduction

At 10:35 this morning, I, along with fourteen other guests, witnessed the first successful light field decoherence, and spatially directed recoherence of a living organism. This process goes by the more popular name: teleportation. Long the realm of science fiction, watching the ebullient scientists and engineers who had brought teleportation into the realm of science fact react as the enterprising white laboratory rat reappeared at station B, I was struck with a tremendous sense of awe and wonder.

It is probably little wonder why Aurelia Veracruz invited a chronicler like me to the test, but no one had explicitly asked me to tell any story. I couldn't help but reflect on how we had gotten to this point. For any student of scientific history, this milestone may have seemed inevitable, but I think it's important to recognize that it was not. I felt strongly then, that a simple article would not capture the moment. A breakthrough like teleportation warranted a greater story, one that captured to some extent the myriad breakthroughs that allowed it to take place.

It is a story of science, no doubt, but there are so many pieces to the puzzle that I imagine by the end we'll have touched on a rather significant portion of the human endeavor. Even within the scientific endeavor, we will cross many fields. Simply the fact that I am writing this from a hotel in Lima, Peru, is something that would have been nigh impossible just a century ago, and warrants explanation.

Where to start is a valid question. We could certainly go all the way back to the Greeks, as many have done before, or perhaps start with Newton's laws of motion. Maybe we should go 400 years later to the development of the 20th Century's crowning theories: General Relativity

and Quantum Mechanics. How about 100 years later with the discovery of the Higgs Boson, rounding out the Standard Model of Physics? Or maybe we should start at the end of the 21st century with the discovery of the Dark Field, one of the most critical discoveries for teleportation.

All of these seem like worthy starting points, but I will forgo them for two reasons. For one, there have been great books in the literary lineage of this humble work that have told those stories. Second, this is a story that spans the whole of the human endeavor, and it makes sense to start at a point that allows us to dip into all those waters.

There is one more consideration, and it's one that will play a not insignificant role in our story, and that is the audience of this book. It's hubristic in the extreme to try and write a book that's for *everyone*. Certainly not everyone will be as interested in the history of teleportation as others, but one of the things I hope to do is make this work relevant to everyone alive today. 200 years ago, the human life span was just under a century, now the oldest amongst us are approaching 200. Writing a history for them is a different calculus.

So for the above I've chosen our starting point just under 200 years ago with the year 2062, far enough back to hold interesting history for even the most well-read bicentigenarians. Four events that year contributed largely to the eventual development of teleportation. The first sets us on our Physics course and takes us to the fifty year anniversary of the discovery of the Higgs boson. The second kicks off Biochemistry with the launch of a revolutionary product. The third brings us into space with the first manned mission to Mars. And the fourth begins our socio-economic shifts: Post-peak fossil fuels. This chronicle is laid out in fifty year chunks, following

the developments of each of these four tracks. Though these tracks do overlap at times, they can be read as laid out, or independently. I leave it to the reader to choose their own path.

This history will encompass so much more than those four tracks, but my hope is that that starting point is compelling enough to get us going. Beyond Physics, Biochemistry, Space, and Socioeconomics, we'll touch on large and small scale energy production, the global anti-war movement, Planet Nine, Dark Matter and Dark Energy, the end of aging, Mahd Arrays and the rise of the southern hemisphere, large scale carbon capture, Climate Change, MAGIC, disease nullification, and more.

Over the course of this history there will be a lot of names, it's impossible to tell the story of the human endeavor without them. Some you'll have heard of, but many may be new. My hope is that by sharing these names and a bit of their backstory I will also begin to convey some of the sociopolitical changes of the last 200 years. The world looks very different now than it would have looked to someone in the mid 21st century, and had not such shifts happened it's doubtful we would be as far along as we are.

Well, enough prattling on about what's to come. Let's get started.

Dark Field 1

Theoretical Physics changed direction in October of 2062 when a 28 year old theorist named Solomiya Diduch presented one of the most important papers of the 21st century, and perhaps the most important paper in our history of teleportation.

Geneva was a balmy 64° on Thursday October 26, 2062. A group of 120 or so scientists were gathered in Switzerland for the celebration of the 50th anniversary of the discovery of the Higgs boson. Named for Scottish physicist Peter Higgs who was the first to suggest the mechanism which necessitated the particle, its discovery was, even fifty years later, one of the most important discoveries that the CERN supercollider had made.

The Higgs was an essential component of the theory known as the Standard Model,—the theory which explained why the particles we observe exist. In the decades leading up to the Higgs discovery in 2012, the Standard Model had successfully predicted the existence of a medley of particles, all of which had been found by accelerators and/or cosmic ray detectors. So in 2012, the discovery of the Higgs capped off what had been one of the most successful predictive models of all time. Not ones to rest on their laurels, the CERN supercollider continued its research into the next decade, but it would actually be the astrophysicists who would take the next step forward.

Launched on Christmas Day 2021, and made operational in 2022, the James Webb Telescope had an almost immediate impact on our understanding of the universe. Two observations made in its first five years of operation would fundamentally affect the generation of physicists coming of age in the first quarter of the 21st century. The first plays a direct part in

what would occur at the fiftieth anniversary of the Higgs discovery conference. The second would play a much more dramatic role on the human endeavor, but less on Physics and more on Philosophy. I'll start with the second one first.

The first exoplanet, the term given to any planet found outside our own solar system, was found in 1992 by Aleksander Wolszczan and Dale Frail who were studying a pulsar about 2300 light years from Earth. In the thirty years between that discovery and the first experiments run on the James Webb Space Telescope (JWST), more than 5,000 planets had been found. Planets are interesting and all, but the burning question that was at the forefront of every new discovered planet was, "is there life?"

One of the first experiments run on the JWST was an attempt to start to find an answer to that question. Exoplanets are all much too far away for us to resolve a picture of what's happening on their surfaces so we have to look for signs of life in a more indirect way. One of those ways was to try and determine the composition of the planet's atmosphere. The JWST did this using spectroscopy. Stars give off light at certain wavelengths called a spectra. If something comes between you and the star that lets light through, like a planet's atmosphere, some of those lines may be absorbed. It turns out that that absorption is predictable based on which elements are present in the atmosphere. So by looking at how the spectra of a star changes when a planet passes in front of it, we can discover the chemical makeup of that atmosphere. If we were to find certain gases, like ammonia, nitrous oxide, or the jackpot oxygen, it would be indicative of life.

In 2022, the JWST delivered back the first of its spectroscopic investigations into the atmosphere of an exoplanet. The planet in question was much larger, and much closer to its star (a so-called "hot Jupiter") than one suspected to harbor life, but it gave scientists a lot of

information about how to perform such a study. By 2026, three exoplanets were selected for spectroscopic study that had a better chance of having the life supporting gases. When the data was collected, Gliese 581g astonished astronomers. Not only did it have a spectral signature indicating oxygen, but the data suggested its atmosphere was almost identical to that of the Earth's.

This first direct evidence of life elsewhere in the universe would have a profound impact on the philosophies of those of us here on Earth. The discovery also supercharged interest in experimental astrophysics worldwide. We'll get into more of that later on. For now I'd like to turn to the other crucial discovery of the JWST, the one that has more bearing for the fiftieth anniversary conference.

In 2022 it was well established that in nearly all galaxies, the normal everyday matter that we see was not massive enough to hold the galaxies together. There were two prevailing theories. Dark Matter was the leading theory. It held that there was some kind of matter that did not interact with the electromagnetic field—the field that gives us light—that was sufficient to explain galactic formation. Alternatively there was the theory of modified Newtonian Dynamics (MOND), which explained galactic formation by bending the laws of dynamics and gravity.

The most well known evidence for Dark Matter was found in a pair of colliding galactic clusters collectively known as the Bullet Cluster. In the Dark Matter scenario, the collision of two or more galaxies involve three things, stars and their stellar systems, interstellar gas, and dark matter. Surveys of the Bullet Cluster's gravitational lensing are consistent with what we would expect from the Dark Matter interpretation.

Predicted by Einstein's Theory of General Relativity, gravitational lensing provides a way for us to see the mass of a region of space. General Relativity tells us that gravity works by curving the four dimensions of Space and Time (three spatial dimensions and one time dimension). This curvature can act as a magnifying glass for light found behind something of sufficient mass. Galaxies and clusters of galaxies provide sufficient mass for gravitational lensing to produce visible results, and by analyzing these results we can understand the structure of the mass in those galaxies and clusters. Such an analysis of the Bullet Cluster found that the lensing we see is consistent with a model of dark matter where the dark matter interacts only through gravity.

In 2025, the JWST began the deep field dark matter survey, a yearlong sweep into the dark regions of our night sky to try and find more cluster collisions that might support the findings from the Bullet Cluster. By looking into the deep field, we are able to look back in time to when the universe was a smaller, and more chaotic place. If more colliding galaxies were found that supported Dark Matter it would help focus theorists and experimenters efforts towards finding the elusive motes.

When the data was collected and processed in the first half of 2026, scientists found twelve new colliding clusters. The data found fit a dark matter theory in all cases but one, with two colliding clusters showing lensing that pointed to dark matter so well it put MOND on the retreat. While the MOND theorists scrambled to find a way to explain this data in their theory, dark matter enthusiasts were renewed in their search for a theory that would extend the standard model to accommodate particles that avoided three of the four known forces, electromagnetism,

and the weak and strong nuclear forces (the fourth being gravity, the one force that dark matter did interact with).

In the decades from the JWST's deep field dark matter survey to the fiftieth anniversary of the discovery of the Higgs boson, a number of theories popped up to explain the existence of dark matter. Perhaps the most well known is that of SDS, an abbreviation that requires a story to explain.

At the turn of the twenty-first century, physicists were trying mightily to fill in gaps in the standard model. The standard model had proved itself wildly successful during the second half of the twentieth century predicting all sorts of particles that would be found experimentally while also predicting values for characteristics of those particles which were right in line with what was found through experiment. Still, it had some gaps. One of the genres of theories trying to fill in those gaps was known as Super Symmetry or SUSY for short.

Particles in the known universe were split into two types. There were the matter particles like the electron and quarks which were called fermions, and there were the force carrying particles like the photon and gluon which were called bosons. The SUSY theories held that for every fermion there was a symmetric boson (these were denoted by a prefixed s as in selectron and squark), and vice versa. SUSY theories had a number of problems, but the greatest of them was simply that no experimental evidence for them could be found. Still the theories, often combined with the other great theory (which also lacked experimental evidence) of the time: String Theory, lived on into the first half of the twenty-first century.

The JWST results from its deep field dark matter survey renewed interest in SUSY theories. In particular, theorists sought out ways for the mathematics to allow for fermions and/or

bosons to interact only gravitationally. In early 2032, rumor started to spread that an enterprising post-doc named Terrence Jackson had found such a mechanism.

Jackson was born in 2004 in an affluent suburb of Denver, Colorado. His last two years of high school were spent bouncing between in-class and zoom classes due to the COVID-19 pandemic of the early 2020s. Unhappy with the at-home learning, he started looking around for other classes to take online. He found MIT's Open Courseware project online and started diving in to the college content, mostly the sciences with a special dose of Physics. When it came time to apply to colleges, studying Physics at MIT had become his number one goal.

After four years at MIT, he decided to head back West for his graduate work, eventually earning his PhD from CalTech in 2030. His doctoral thesis was an attempt to apply SUSY to dark matter. While fundamentally flawed, it laid the groundwork for his work two years later. The gist was to take SUSY and raise it up one more level to show that each fermion had a boson, *and* a dark matter partner (denoted by a prefixed d as in a delectron). His thesis defense granted him his PhD, and the almost tongue-in-cheek name for his theory when his thesis advisor William Macon remarked, “You’ve taken Super Symmetry and turned it into some sort of Super Duper Symmetry!”

Super Duper Symmetry (SDS) introduced a new type of symmetry onto the universe. Traditionally physicists thought of symmetries as either binary or infinite. Binary symmetries are like those displayed in humans where our right side more or less looks like our left side—so-called mirror symmetry. Infinite symmetries are like that of a circle where turning the circle any number of degrees results in the same picture of the circle, what we call radial symmetry. Until

SDS, all the symmetries of SUSY were either binary or infinite, but SDS introduced a third kind of symmetry which was trinary.

If we look at an equilateral triangle, we can see that it is radially symmetric, but only when we turn the triangle 60° . It is with this kind of three-part symmetry that SDS uses to describe what dark matter is. Fermions run alongside one side of the triangle, bosons along the second side, and dark matter along the third side. Through a mechanism called spontaneous symmetry breaking, we can learn about the properties of and make predictions about dark matter.

Jackson stuck to SDS after getting his PhD and moving on to post-doc work back at MIT. In late 2031 he had a breakthrough and emailed a friend saying, “I have hit upon a theory, which may describe all matter in the known universe. When at last I sat back and looked at what I had conjured, I found the dark matter had appeared clearer than it ever could in the telescopes of the world. More finishing touches are needed, but I believe this theory almost ready.” Jackson’s friend decided to skip the almost part, and started spreading the word.

Afraid of getting scooped on his discovery, Jackson decided to present his theory at the annual Dark Matter Symposium held at Cal Tech (so chosen as it was the location where another Cal Tech professor Fritz Zwicky coined the term dark matter in 1933) after he had submitted it to *Science*, but before it had been accepted. The atmosphere at the symposium was one of energetic optimism, as is often the case at conferences focused on the hotter parts of the latest science. Hooking into that optimism, the SDS talk was an immediate success. Jackson was inundated with requests for his paper, which he filled after requesting expedited publication from the editors of *Science*.

As we now know, SDS was not the ultimate answer to the dark matter quandary, but it was a powerful step in the right direction, and as is often the case in science, even incorrect theories can bring important advancements to the table. Even at the time though, two problems quickly arose with Jackson's formulation of SDS.

The first problem was a natural extension of one of the problems with SUSY theories. In SUSY theories, the predicted super symmetric particles had masses many many times more massive than their known counterparts. In SDS the masses of the dark matter particles were orders of magnitude more massive than those. This caused not only a theoretical problem as there was no good explanation as to why one group of particles would be so much more massive than the particles we know, but also an experimental problem as no reasonably predicted accelerator could ever hope to hit the energy levels required to find them.

Where the problem of too much mass could be hand-waved away to a degree, the second problem was more troublesome. In SDS the dark matter particles interacted with the known bosons, photons, Ws, and gluons. Astrophysicists had long been quite certain that dark matter didn't interact with the three non-gravity forces. Far from being a death knell to SDS however, this quirk of the theory inspired others to search for solutions that would allow for dark matter to not interact electromagnetically.

The search for answers beyond SDS to address these and other smaller failings turned out to be fertile ground for theorists for the next thirty years. There were a few other theories, but none captured the imagination the way that SDS did. Extensions to and modifications of SDS came fast and furious from Jackson's talk and paper in 2032. Likely the interest would have

continued if it were not for Solomiya Diduch, and a marginally successful science fantasy writer named Owen Rogers.

Rogers was born in 1991 in an unassuming corner of the American midwest. He was a bright student with a knack for creative ideas found in various writing assignments during his school years. When he was twelve or thirteen he was given a copy of Stephen Hawking's *A Brief History of Time*, which sparked for him a deep rooted love for science and science history. In 2010 he enrolled at Reed College in Portland, Oregon, intent on studying Physics.

In his junior year, one of his classes was Quantum Mechanics, and here we need to dive back into the science. Developed in the early parts of the twentieth century, Quantum Mechanics was used to explain the plight of electrons within an atom. At the time it was believed that electrons orbited the nucleus of an atom, but if that were the case the electrons should lose energy over time and eventually collapse into the nucleus. Niels Bohr saved the atom from this ignominious end by proposing that electrons could only gain or lose energy in discrete amounts. These discrete amounts, called quanta, turned out to be a fundamental component of the realm of the very small.

A few years later, Erwin Schrödinger provided the mathematical framework for thinking about the plight of quantized particles, and so by 2012 when Rogers sat down in Quantum Mechanics, the average physics student started with one of the simplest implementations of the Schrödinger Equation: the infinite square-well potential. The square-well potential is a problem that takes place on a one dimensional line segment. On either side of the segment is an infinite potential, effectively an impenetrable wall (although as those students would find, walls in

quantum mechanics aren't entirely impenetrable). The Schrödinger equation describes the likelihood that a particle will be found somewhere along the line.

Later in the semester, as the class progressed from one dimension to the four dimensions of Space-Time, Rogers had an insight. Special Relativity tells us that nothing can move faster than the speed of light in a vacuum. This meant that if we treat particles as points, then we can draw a sphere around them of radius ct (c being the speed of light, and t being some specific time), and that that sphere would act as an infinite potential for that particle.

The infinite spherical potential scenario was as well-known a result as the infinite square-well potential so Rogers plugged his assumption into that derivation, and what popped out was the De Broglie wavelength. Every particle has a wavelength given by the equation $\lambda = h/p$ where λ is the wavelength, h is Planck's Constant, and p is the relativistic momentum of the particle. This result was surprising to Rogers's undergraduate self, and it stuck with him.

Normally the story of Physics would take us into graduate and post-graduate schooling with conferences and papers, but Rogers, as it turned out, wasn't a great physics student. Instead at Reed he found his work passion in computer science, and that led him to later pick up his first passion writing as a hobby. When the COVID-19 pandemic hit in 2020, Rogers took the time he used to spend commuting to and from work, and started in on a sweeping science fantasy series called *The Advancement*. In it he revitalized his college recovery of the De Broglie wavelength, and expanded it into a theory held by an advanced alien culture called the Azurdans. He called the theory Discrete Space-Time.

Obviously a "theory" in a novel is very different than a scientific theory with proofs and predictions, but it's worth exploring Discrete Space-Time for a moment. Remember the infinite

spherical potential. Rogers called these spheres Exclusion Spheres, a nod to Pauli's Exclusion Principle, which states that no two particles can have the same quantum state. Rather than having these spheres exist along the space-time continuum, Rogers thought what if the spheres were space-time itself. For the Azurdans, space and time existed as discrete pockets in a matrix foam. This was radical enough, but Rogers added an appendix to one of his books that explained that in this discrete space-time matrix, time actually progressed in discrete jumps based on interactions between particles. Rogers's computer science background was probably showing as he explained time as a series of discrete clock ticks, but it's where we get to get back to Solomiya Diduch.

Diduch was born in 2036 in the Ukrainian city of Donetsk. In 2022, Russia invaded Ukraine seeking to win back some of the land it had lost after the end of soviet occupation. The aggression led to incredible damage across much of the country. Donetsk was particularly hard hit in the southeastern corner of Ukraine where Russia had focused much of its invading effort. Starting shortly after the war, a rebuilding campaign, fueled by aid from Europe and the US, led to a modernization of Ukraine. In 2027, Ukraine joined the European Union, and the economic foundations the rebuilding had laid slingshotted Ukraine towards the front of the European scene. One of the biggest beneficiaries of this advancement was the sciences, in particular physics, a particular need of a former soviet bloc country equipped with multiple nuclear reactors.

By the time Diduch was looking at colleges, Ukraine was becoming a leader in theoretical physics. She chose the National University in Kharkiv, a university known for its work on theoretical dark matter, and in particular for SDS. During her sophomore year, she took a creative writing class as an elective where she met an energetic Lithuanian named Camilla Vidas. The two were fast friends, offering each other some intellectual relief from their studies.

Vidas was an avid reader, reading everything from historical non-fiction, to speculative science fiction. When she was fourteen she taught herself English because she was unsatisfied with the selection of translations in her native tongue. Vidas was always trying to get Diduch to read different books that she had liked, but rarely was there time. Then one day, while the pair were in their last year, Diduch was trying to explain some aspect of SDS, Vidas was reminded of a series she had read in high school. Two days later Vidas showed up with book four of Rogers's *Advancement, The Ballad of Sigurd*. Reluctantly Diduch took the book, and promised, if nothing else, to read the appendix on discrete time.

Diduch read the appendix as promised, and quickly dismissed it as the musings of an enthusiastic, but uneducated author. It wasn't until two years later, as Diduch was starting to think of topics for her doctoral thesis, that she would come back to that appendix. She was a little fried one night trying to think of some novel angle for dark matter theory. In her stressed state she had resorted to clicking through wikipedia articles looking for any kind of inspiration. She had stumbled on the entry for Minkowski Space-Time and was skimming it when a line stuck out to her. "Although initially developed by mathematician Hermann Minkowski for Maxwell's equations of electromagnetism, the mathematical structure of Minkowski spacetime was shown to be implied by the postulates of special relativity."

Diduch couldn't help but be reminded of Rogers's appendix and the idea of discrete time emerging from interactions between exclusion spheres. Could it be that time was a necessary consequence of electromagnetism and special relativity? If so, what could that mean for dark matter that didn't interact with the electromagnetic field at all? Diduch dove into Minkowski's original papers on space and time. In a flurry of activity lasting three months she plumbed the

depths of twentieth century physics for clues as to the nature of time. By the end of those three months she had a working, but brittle framework from which to begin.

By late 2061, Diduch's work on emergent time was known to her university professors. No papers had been published yet, but when the invitations for talks for the fiftieth anniversary of the discovery of the Higgs boson conference were sent, one of her professors asked if she might be ready to present her work. Flustered by the request, Diduch was going to turn down the offer, but her friend Vidas encouraged her, and she accepted.

And so on that balmy afternoon, we got our first glimpse into Dark Time.

Biochemistry 1

If I was going to pick another year to start this book off on, it would probably have to be 2012. While the physicists were celebrating the finding of the Higgs boson, their compatriots in biochemistry were embarking on a brave new world of gene editing. Initially discovered in 1987, CRISPR (Clustered Regularly Interspaced Short Palindromic Repeat) would have to wait until 2012 when Jennifer Doudna and Emanuelle Charpentier delivered a paper on its biochemical process before it could be used as a mechanism for gene editing.

Back in 1987, scientists were enamored with sequencing DNA. In bacteria, scientists found repeated segments of DNA with short segments interspersed. These segments were the CRISPR segments, a name that would come to mean not only the phenomenon, but the process by which scientists would come to use it. In 2007 it was found that the repeated segments actually acted as a defense against viruses that invaded the bacteria.

If viruses have deadly potential for humans, they're a nightmare enemy for bacteria. The two microorganisms have been waging war for billions of years, so it should come as little surprise that the two have evolved interesting weapons against one another. Remember, a virus continues on by hijacking the machinery of a cell and telling it to replicate the virus. It does this through either DNA or RNA that confounds the cell's own operation. The CRISPR mechanism that some bacteria have is actually able to snip the offending viral DNA and replace it—effectively nullifying the virus.

It was this snip and replace that ignited the world of biochemistry. The Doudna-Charpentier paper ended with the portent, “[a certain mechanism of CRISPR] could offer

considerable potential for gene-targeting and genome-editing applications.” Those of us born in the era of embryonic editing can appreciate that considerable potential.

In the two decades after that pivotal paper, biochemists raced to devise uses for CRISPR. First was a focus on single gene diseases. Sickle-cell anemia, cystic fibrosis, hemochromatosis, and Tay-Sachs, while not completely cured, were at least arrested in trials of CRISPR-based treatments. In 2018, an enterprising young Chinese scientist named He Jiankui announced the birth of the first CRISPR-modified twin babies. For his trouble, He was fired from his university, and had his credentials suspended by the Chinese government, a move exemplary of the squeamishness of the early twenty-first century towards gene editing.

Two decades later, CRISPR was going strong with its single gene edits. Several publicly traded companies had sprung up to accommodate the new field, but science rarely moves at a pace that satisfies shareholders. Pressure began to mount on those at the forefront of CRISPR research to expand its capabilities. There were three avenues that looked promising.

The first was novel delivery. Genes are, of course, inherited. This means that for people that are already alive, if you edit some of their bone marrow genes, then those edits only get passed on to the daughter cells of those edited cells. Targeting *all* the cells that you needed to in order to get an edit to stick was a difficult problem, one that in 2032 hadn't been solved.

The second was error correction. As with any biological process, CRISPR sometimes caused errors in the genome it was editing. These errors could have potentially fatal consequences as errors in the genome often lead to cancer. And as with anything that has a certain percentage chance of occurring, the more you used CRISPR, the greater the chance of a

life altering mutation. Finding a way to lower the error rate and/or detect and fix errors would go a long way to building more trust in the CRISPR mechanism.

The third avenue was considered by most to be the next logical step in humanity's quest to control our own evolution, but at the same time was considered decades, maybe centuries off—multi-gene editing. For all of CRISPR's power, it could only focus on one piece of DNA at a time. Most of the things we care about in a genome are multi-gene processes with the genes on different chromosomes. To edit those, we could use CRISPR multiple times, so-called multiplexed CRISPR, but that increased the error rate, and was generally prohibitive. Some believed that if nature created a single gene editing mechanism then certainly there was a multi-gene one, we just hadn't found it yet. Others felt that multi-gene editing was simply beyond our current understanding.

As we'll see, all three of these avenues would come to be explored and crossed, but it is the last one that surprisingly got tackled first, and from an unlikely source.

That story starts in 2013 (missing 2012 by just a year). In August of that year, Mark Post, a professor at Maastricht University in the Netherlands brought to the world the first cultured meat. Grown from stem cells taken from your animal of choice, cultured meat was meat grown entirely on its own, without the need for the whole animal around it. Nowadays, of course, cultured meat is the world's preferred meat variety, but back then it was just an idea. An expensive idea at that. Professor Post's cultured hamburger patty cost around \$300,000.

Over the next 15 or so years, millions of dollars in capital were poured into bringing the cost of cultured meat down, and the quality up. Branching out from beef, there were startups

focused on chicken, fish, lamb, and pork. And it wasn't only meat too, several startups focused on cultured dairy, providing ice cream, butter, and milk that didn't require cows.

In 2027, the first cultured chicken was approved for sale in the USA and Europe. In classic USA fashion, the poultry lobby had worked mightily to prevent its sale in the US. The FDA (Food and Drug Administration) tried a compromise where the cultured meat could be sold, but couldn't be called chicken despite being biologically exactly the same as meat raised on farms (the EU held no such restriction). The other meats soon followed, and created an industry that doubled in size every year for the next seven years.

The influx of cash into the cultured food industry led many companies to branch out, and search for other places to apply their technology. And this is where we meet a convivial German named Werner Hartmann.

Hartmann was born in Dusseldorf Germany in 2008. He showed an early aptitude for the sciences, and was particularly interested in biology. In high school, his biology class did an experiment to isolate DNA, and Hartmann was hooked. He attended Heidelberg University, which was then a top school for biology in Germany. It was there that he discovered his second love, brewing beer.

While working on his doctorate in 2031 he took a lab position at Molktek, a German company focused on cultured dairy that was trying to branch out into the competitive world of German brewing. Energized by the opportunity to bring his two passions together, Hartmann brought that special combination of confidence and exuberance that young scientists seem to uniquely have.

Established in 1516, the German purity law for beer held that beer could only be made with barley, hops, yeast, and water. Thankfully for Hartmann and Molktek, 1516 was sufficiently before the advent of CRISPR and the age of gene editing. Nothing said you couldn't modify the barley, hops, or yeast. Most of Molktek's competitors were focused on modifying the yeast, so they took on the more difficult task of trying to modify the hops.

The task given to Hartmann's team was to modify Hallertau Hops' flavor profile. The task was open ended, any modification that resulted in a flavor change was sufficient for these early trials. The main group was exploring mechanisms for multiplexed CRISPR to change the genes sufficiently for the flavor change, but hadn't made much progress in the three years the project had been going on. Hartmann was part of a second group that was tasked with trying to find a better delivery method for the CRISPR system employed. This second endeavor was a long shot, but Molktek's deep pockets allowed it.

One of the main issues with multiplexed CRISPR was that not all of the gene edits were guaranteed to be taken up by the cell. Sometimes the partial edits were innocuous enough not to matter, but sometimes they had really nasty results. This problem of prolonged exposure was well known by the 2030s, and it's there that Hartmann put his attention.

The inside of cells are busy places. Organelles like mitochondria, ribosomes, and the eloquently named endoplasmic reticulum, sit suspended around the nucleus of the cell. The nucleus is well-guarded, but occasionally cellular things need to move into and out of the nucleus of the cell. To facilitate this, the cell has small structures called microtubules that act as highways from the interior to the exterior of the cell. Along these microtubules, motor proteins called kinesins and dyneins walk outward to and inward from the cell exterior respectively.

It had been known since the early twenty-first century that viruses were able to hijack these motor proteins to shuffle viral material around the cell. Hartmann felt that if he could find a way to use this viral mechanism to prolong the exposure of CRISPR's guided RNA (this gRNA is what performs the actual snip and replace on DNA), he could make the multiplexed delivery system more successful. To do so he turned to some parallel work that was going on in the cultured beef space.

Animals are remarkably capable of taking in all that their environments offer and depositing that in their cells. Some animal meats held on to this environmental signature more than others. Chickens for example only live for a matter of weeks before being slaughtered, hardly enough time to soak in the rare earth metals around them. Cows on the other hand lived long enough to take on these tastes, and reproducing that in the lab was no simple matter.

Esmerelda Rodriguez was working on putting that flavor back in cultured beef. Born in La Plata, Argentina in 1992, she rode the first wave of cultured beef to prominence in the South American biochemistry scene. Unlike its larger neighbor Brazil, which was hindered by a conservative government that refused to acknowledge the effect cultured beef would have on its own beef exports, Argentina worked to forge partnerships between the lab-cultural and agricultural sectors. Rodriguez's lab enjoyed a healthy endowment from the Argentinian government to research biochemistry that benefitted both real and fabricated beef.

In 2030, Rodriguez made a breakthrough. By utilizing the viral mechanism for shuffling material through cells, she was able to infuse cells with a cocktail of elements and molecules that mimicked what a cow would have collected eating grass on the Argentinian fields. In the grand

tradition of finding pithy sounding acronyms for biochemical processes, Rodriguez named the process Dynein/Kinesin-based Organelle Suffusive InfusiNG or DOSING.

Rodriguez's paper was published in 2031, and appeared jointly with patents filed around the world for the DOSING process's use in cultured beef. Faced with the task of prolonged CRISPR delivery, Hartmann seized upon DOSING, and wondered if instead of a cocktail of delicious molecules, the components of multiplexed CRISPR could be used instead. It took him only eight months to repeat Rodriguez's process with CRISPR components.

Hartmann called this new process Plentiful Incisive CRISPR-based Kinesin/Dynein-involved Recombinance or PICKR. Two months later, following Rodriguez's lead, Hartmann released a paper and patents for the new process in April of 2032. PICKR would, of course, go on to have a significant impact on our genetic futures, but that story would have to walk through several decades of bureaucratic red tape before coming to fruition. For 2062, however, we thankfully have a wonderful *in vivo* (meaning in a living organism as opposed to *in vitro* where embryonic CRISPR would take place) application to explore.

Melany Emerson was born in 2017 in London via surrogacy to two fathers. While not the death sentence it had once been, HIV was still a major concern in the first half of the twenty-first century. When Emerson was twelve, one of her fathers' HIV infection progressed to AIDS. Her father's decline, and eventual death imprinted on Emerson, and she vowed to devote the rest of her schooling to trying to find a cure for HIV or a vaccine to prevent it.

When she was in high school, she started searching for books on biology that could help her on her quest. Sophomore year she checked out Walter Isaacson's *The Code Breaker*, which told the story of Jennifer Doudna and the beginnings of the CRISPR revolution. Emerson

became convinced that *in vivo* CRISPR could be used to confer immunity to HIV, and she dedicated the next thirty years of her life to proving that.

After obtaining her PhD from Cambridge in 2042, Emerson moved to the states to join Intellia Therapeutics, the CRISPR company started by Doudna and others. She was described by others as driven and stubborn, often following experiments to obsessive ends trying to find the answers she sought. At the time Intellia was funding research on trying to develop an anti-viral CRISPR process, Emerson fit right in.

Developed a decade earlier, the PICKR process was well known, and had proved useful in conferring immunological responses *in vitro* that is in petri dishes. What Intellia was missing was some sort of delivery mechanism that could target the cells necessary to confer immunity. For example, HIV targets T cells, immune system cells that help the body recognize infections. How could we deliver PICKR to T cells directly?

Emerson worked on this problem for four years. Her hunch at the beginning, which proved to be spot on, was that the key lay in the viruses themselves. They already knew how to target the cells you were concerned with, so if we could use their finely evolved mechanisms to our advantage it might just do the trick.

Working with viruses was easier said than done though. Sure a whole class of them, adeno-associated viruses (AAV), were already in use to deliver CRISPR to cells, but the sheer variety of viruses made them difficult to work with outside of some simple cases. The task ahead of Emerson was to either modify one of the AAV viruses that we knew how to use to target T cells, or to modify the HIV virus itself to deliver PICKR instead of its regular deadly payload.

2045 was the breakthrough year, and the breakthrough came from some good old fashioned evolutionary biology. Emerson ran a months long experiment trying to modify a specific AAV virus to evolve the ability to target T cells *in vitro*. She started with a mixture of T cells, and cells that the virus could infect, and ran the virus through generation after generation where the T cells' percentage went up, and the other cells' percentage went down. It took 3,673 generations (luckily a virus's generation is much shorter than ours), but she finally got viruses targeting T cells. By continuing to ratchet up the T cell concentration, she was eventually left with a virus that could target T cells. Just like HIV, but one that could be used to deliver PICKR.

In 2046, Intellia was granted permission for a small human trial of Emerson's novel approach. The results were jaw-dropping. Of the 36 patients in the trial who received the PICKR process, 32 were able to reach zero viral load (a state where the virus cannot be detected in bodily fluids), and the remaining four had their viral loads drop precipitously. While not technically enough to call it a cure just yet, the trial exceeded most everyone's expectations.

The technique, cleverly named Virus-delivered Immunological Recombinant EXemplification (VIREX, it was a long established tradition to play fast and loose with one's acronyms in the biochemistry space), had been patented long before the results of the trial were delivered. The CEO of Intellia conferred with the board, and decided that VIREX would live better on its own. He asked Emerson for her recommendations on a team to start a new company. She chose Intellia's COO Julia Frense as the CEO of the new company, and together the two put together a founding team of six to launch the startup with.

The first task ahead of the new company Medemos, so named from a med for medical and demos, the ancient Greek word for the people, was to expand the VIREX process to new

diseases, and start the long process of seeking regulatory approval first in the US and then in the world. Keep in mind that this was the first time actual immunity was conferred on cells *in vivo*, Medemos had no idea how that would be received by the US's FDA. On the one hand, the prospect of curing HIV was so incredible it might move the needle with the regulatory body further, on the other hand, gene editing left many morally squeamish.

While Medemos's VIREX HIV treatment wound its way through the regulatory process, Emerson turned her attention to other viruses that could be treated using VIREX. She and her team identified three. Hepatitis C, a disease that affected the liver, and for which no vaccine had been developed. Measles, a disease for which there is a vaccine, but decades of anti-vaccine rhetoric had unfortunately led to outbreaks, often among children. And Rhinovirus B, one of the causes of the common cold, because who doesn't want a cure for the common cold? VIREX proved itself every bit as effective in these cases as it had in the case of HIV, and by 2050, Medemos had four treatments in regulatory review. The success in trials allowed the company to raise money for manufacturing, and it turned its attention to the construction of a machine that would dispense its four treatments.

In 2054, after eight years of regulatory review, Medemos unveiled its machine, the virex, taking on the mantle of the process that enabled it. The first virex machine was large and clinical. Its delivery method was intravenous so treatments had to be dispensed by trained medical personnel, but the demand was still huge. Priced at \$9,999, the virex was well within the budgets of hospitals around the country, and quickly became a necessity as cures for even just one of its four launch diseases would affect millions of Americans. By the time the virex launched in 34

more countries in 2056, sales of the virex were establishing Modemos on par with tech startups, investors around the world were taking notice.

Emerson and her team turned their attention to VIREX delivery, while others at the company were increasing the treatable viruses. She felt that the next step for the virex machines were to get them into homes, but to do so, the treatments had to be administered orally instead of intravenously. The human digestive system is designed to eliminate viruses that are ingested, and ingesting a virus is exactly what Emerson needed to happen with the modified AAV viruses that delivered the VIREX material. Fortunately, large amounts of cash can supercharge science.

Enteric viruses are a family of viruses that can survive in the hostile world of the human digestive track. They were a late comer to our understanding of viruses, they didn't really receive much attention from scientists until the late 2030s. Even by the mid 2050s when they became relevant for the teams at Medemos, their survival mechanism had garnered little attention. Emerson spent three years researching and then adapting that survival mechanism for use with the AAV viruses of the VIREX process.

The new oral delivery mechanism needed to go through the regulatory process as well, but with the unprecedented success of the virex machines, the FDA fast-tracked its approval. The manufacturing ramp up was considerable, but well-timed, and at a keynote presentation in 2062, Julia Frense unveiled the mVirex machine, a scaled-down version of the original virex machine made for the home. At \$5,999 it was priced like a high-end computer, and took up about the same amount of space in the home as an air fryer. The diseases cured had doubled in the eight years since the original machine had launched. The initial manufacturing run was slated to produce four million machines. They sold out in 24 seconds.

The new launch catapulted Medemos into the biotech company stratosphere. News of its success caused a run on Medemos's stock, which as the dust settled left the company at 17th on the list of most valuable companies in the world. Overnight, Emerson and Frense found themselves among the richest people in the world. Frense held a meeting with the founding members of the Medemos team to get their input on what to do with their newfound largesse. The minutes from that meeting would prove even more momentous than the launch of the cure for the common cold.

Frense: Current projections are putting our quarterly revenue north of \$30 billion with profit over \$7 billion. We think we can ramp up manufacturing, and double that in six quarters. We need a plan of what to do with all that cash. Any suggestions?

Emerson: There have been any number of vanity projects from men who've found themselves billionaires. I always thought those were a waste of time. If it was me, and I guess it sort of is, I would buy companies that are messing up our world, and de-evil them.

Frense: De-evil them?

Emerson: Yeah, you know? Take Nestle for example. What if new ownership had them give up their water claims and end child slavery picking cocoa beans? We could buy an evil company and undo all the shit that makes it bad. De-evil it into something good.

Frense: Interesting. Any thoughts on a company to de-evil?

Emerson: Well we're already in human health, maybe in the interest of diversification, we should think about branching out to plants. The PICKR process is perfect for plant innovation, and agricultural companies are almost universally bad. Monsanto for example has patented its

seed products and makes it their business to try and destroy any farmer who doesn't use their seed.

Frense: You want us to buy Monsanto and undo their bad practices?

Emerson: Yes. Exactly.

What would happen after that would change the game, for everyone.

Space 1

On January 1st 2062, humanity was an interplanetary species (Ok I'm fudging it a little here, the Mars landing happened in April of 2061 when Mars was at its closest to the Earth, but what's 8 months in the spirit of great narrative?). As Veronica Robinson, commander of Ares III stepped foot onto the red planet, she reported back to Houston, "By stepping on this red planet, we pull back the sling to the stars evermore. This destination is not the finish, but the first of many worlds to explore." The all woman crew had been selected out of practicality. Across the eight person crew, the choice saved around 120kg that could then be used for food, water, and fuel. At least that was the party line at NASA. It wasn't lost on the public, nor the media, that just a century ago, an all-woman mission to the moon would have been unthinkable.

Commander Robinson was American, as was the landing vehicle that brought the astronauts to Mars, but of the six crew that landed on Mars, four Americans brought a Russian cosmonaut and a Chinese astronaut with them. Thus, all three of the space-faring countries found themselves on the second planet to harbor human life. The six spent only moments taking in the Martian vista before trekking in the gravity, one third the strength of Earth's, to the subsurface base established by Ares II.

Mars had been an obvious next step for the space agencies of Earth ever since Neil Armstrong set foot on the moon, but getting to Mars, and more importantly getting back, presented a host of problems that took actual rocket scientists nearly a century to solve. First and foremost was the distance. At its closest approach, Mars is around 40 million miles from Earth, two orders of magnitude more distant than the moon. With the rockets of the early twenty-first

century, this distance represented a trip between 150 and 250 days, a far cry from the three days it took to reach the moon.

The time the trip took in space provided a host of challenges in itself. First was the simple logistics of bringing enough food, water, oxygen, fuel, and other essentials for such a long trip. Being in space for so long also presented a significant risk of radiation. Outside of Earth's atmosphere and magnetic field, charged particles rained down on space interlopers. Shuttles and space suits offered sufficient protection for a week-long trip to the moon, but a year-long round trip presented a different level of risk.

Once you got to the red planet, a new host of problems had to be dealt with. The martian atmosphere is too thin to provide the kind of braking that spacecraft enjoy when returning to Earth. Carrying enough fuel to brake a landing vehicle large enough to hold multiple people down to a speed slow enough to land human beings safely was a problem considering you wanted to be traveling as fast as possible to get to Mars in the first place.

If you were able to land safely, you then had to set up somewhere for astronauts to stay. The only reasonable power source on Mars is the sun, which is great on a planet with no clouds until you remember that the martian surface can see sustained dust storms that last months at a time. Hoping the weather holds isn't going to cut it when you're 40 million miles from any help.

Then there's getting back. Martian gravity is a little over a third of the Earth's, but that's still a significant amount of fuel that needs to be dragged 40 million miles with you. There's also the fact that Mars and Earth are in motion. They only get close to each other every 26 months. Do you send your astronauts for only a few days on Mars before turning around and heading

back, or do you have them stay there for 18 months before making the return trip. That latter plan carrying its own set of problems.

Probably worst of all has nothing to do with the logistics of going to Mars, but with the psychological well-being of the astronauts making the journey. A year in cramped spaceships is hardly the recipe for mental health. Coupled with the existential dread of being millions of miles from everyone and everything you've ever known, NASA had real concerns as to how people would fair on the trip. Studies could try to recreate the situation, but there's only so much that you can simulate being completely isolated by interplanetary space.

All these problems were known by the space agencies of the twenty-first century, unfortunately limited budgets and other priorities kept the space agencies from making much progress during the first thirty years of the century. By 2025, Mars looked like a very real possibility, but only for some future spacefarers, who were better funded by the governments behind them. As is often the case, however, hope came from a seemingly unrelated space project.

We've seen how in 2026, the James Webb Telescope found an atmosphere that indicated life on another planet. It had long been wondered if Mars had once had life. It was known that Mars had an atmosphere at one point in its history, and, though it was farther from the sun than the Earth, it might have just been in the goldilocks zone enough to harbor life. The Mars rovers that had been sent to explore the planet at the time had had no luck finding signs of life, and a consensus had formed that in order to find life, deeper digging was required. Deeper digging was the purview of people and not space robots so when the JWST discovered that atmosphere in 2026, Mars advocates seized upon it to drive up interest in the interplanetary mission. When the JWST team announced that two more planets had been found with atmospheres conducive to life

in 2028, the suggestion seemed to be that life wasn't rare, but in fact life could be *normal* in our universe. If that were the case, it greatly boosted the case for searching for life on Mars.

Funding a mission to Mars became a central theme of the 2028 presidential campaign. We'll get more into American politics in the next chapter. For now, suffice it to say that public sentiment was such that funding of the mission to Mars was going to happen, it was just a matter of how. With the matter of funding settled—the 2029 budget for NASA was a whopping \$60 billion, nearly three times what it had been a decade earlier—NASA began working on a Mars mission in earnest in 2029 with a mission target date of 2059. Three decades turned out to be a reasonable estimate, but at the time was more of an educated guess. It's hard to gauge the rate of scientific progress, and there was plenty to figure out in the intervening years.

Before launching into the advancements that enabled Robinson to step onto Mars, I briefly need to speak about the state of private space exploration at the beginning of the twenty-first century. Nowadays, commercial space travel is commonplace, but at the beginning of the twenty-first century it was strictly the purview of male billionaires. Three separate space ventures, none of which lasted past 2047, competed for space contracts, and did a reasonable job of taking low-hanging space fruit off of NASA's plate so that NASA could focus on the real science of space travel. But, as male billionaires are wont to do, they liked to boast about what they would accomplish, a mission to Mars foremost amongst those boasts, promises of outcomes that would never come.

By 2029, people's faith in the private space companies to accomplish the big things was waning, and the success of the JWST had bolstered support for the good old fashioned mission-driven science of NASA. With funding in hand, and a clear decades long mission, NASA got to

work solving the problems of travel to Mars. The first area of exploration was an obvious one: reducing the time spent traveling to and from the red planet.

It takes a lot of rocket fuel to get a multi-ton ship to leave the Earth's gravitational pull, and that's before you start loading the ship up with food, water, and fuel to get back. Carrying extra fuel to push you along between your starting point and your destination incurs a cost when you lift off in the first place. At the same time, moving a multi-ton spaceship around once it's in space isn't that difficult, and over such a long trip, even a small acceleration can add up. When you start adding up the numbers there's only one fuel that can provide a small constant acceleration that doesn't require you to add tons to your lift off weight—nuclear.

Those of us living in our post-fossil fuel world might find it hard to believe that the world once had a rather negative look on nuclear power, but that's precisely the case for the early twenty-first century. Meltdowns like Chernobyl and Three-Mile Island loomed large in the collective memory of the time, but even worse for the plight of nuclear power was decades of lobbying and propaganda from other sectors of the energy industry. Fortunately for NASA, the biggest proponent of nuclear power vessels was the one supplying a good portion of its astronauts, the US Navy. In the early twenty-first century, nearly a hundred submarines and aircraft carriers were nuclear powered, casually sailing around the world, visiting well-populated ports without incident.

For its part, NASA's work on nuclear thermal propulsion (NTP) began all the way back in the 1960s when the country had a rosier picture of nuclear power. The Nuclear Engine for Rocket Vehicle Application program actually succeeded in developing NTP rockets for use, but the program was ended in 1972. Still some work had occurred in the fifty years that followed so

when NASA picked up development of NTP rockets in 2029, a lot had already been done. In fact it looked like NTP rockets were ready to go except for one major problem.

Nuclear reactors create a lot of energy. Reactors are efficient at turning that energy into something useful, for a rocket that would be energy turned into thrust. Not all of the energy the reactor makes can be converted, that energy stays with the reactor as heat. Nuclear reactors create a lot of heat. On Earth, whether it's in a power plant or an aircraft carrier, the solution is pretty much the same, plop the hot part in a big pool of water and let the water cool it down. In space you don't have a giant pool of water, in fact you don't have anything at all. Without the convection provided by a fluid like water, heat doesn't dissipate faster than it's produced. Lots of heat on a spaceship with nowhere for it to go is a recipe for a bad time. NASA had to find a way to handle that heat.

In 2004, Andre Geim, and Konstantin Novoselov isolated and investigated a curious configuration of carbon atoms. The atoms formed a sheet just one atom thick, which presented all sorts of interesting and useful properties from an incredible strength-to-weight to the ability to conduct electricity and heat with great efficiency. The material would be called graphene, and since it was made entirely of carbon it had arguably the most important property for government use—it was cheap.

Richard Hamlin was born in 1998 to a single mother living in Des Moines Iowa. Hamlin's mother worked two jobs to ensure he would have every advantage possible, which included signing Hamlin up for all the extracurriculars she could find. One of the extracurriculars was a youth engineering outreach program run by the University of Iowa. Hamlin was hooked by the program, and with his mother encouraging him, pursued other engineering opportunities

through high school. The work got him in to the materials engineering program at the nearby University of Illinois. A humble student known for methodical work, he stayed at the university for his doctorate so he could stay near his mother. When it came time to look for jobs in the materials space after being granted his PhD in 2024, he joined the R&D team at John Deere in Moline, Il. about halfway between his school and his home in Des Moines.

When Hamlin turned 30, he had a bit of an existential crisis. As much as he loved living in the midwest, he felt that his skills were being wasted on developing better tractors and combines, so when the call came from NASA in 2029 for materials scientists, he had a long talk with his mother and together they formulated a plan to move to Pasadena, CA. and join the Jet Propulsion Lab.

There a group of 18 materials scientists and engineers were split into groups of three with the express mission of solving the heat problem for future nuclear spaceships. The division was to create a spirit of competitive collaboration, a very American approach to the problem. Hamlin was joined by Darnell Michaels and Emily Wu, and the three dove into the world of heat. NASA gave the teams 24 months to work on the problem before they would judge the results.

Wu suggested they look into graphene. She had worked on thermal applications of the material in the electronics world. At that point, most of graphene's applications had been found in the realm of small things, a commercial application for scaled up graphene had yet to be found, but the team recognized that the material's heat conducting properties had promise. The problem was actually that graphene was too good of a conductor of heat. There was no way to direct the heat where you wanted it, so while some of your heat was radiating out into space, the rest was convecting into your spaceship. The answer was to find some way to direct the heat

transfer within the graphene or some similar type of material. The only problem with that of course was that directed heat transfer of that kind hadn't been done on the macro scale of metal spaceships.

In 1975, Gordon Moore, co-founder and former CEO of the microchip manufacturer Intel, revised an earlier prediction and prognosticated that the number of transistors found on a silicon-based chip would double every two years. This prediction, dubbed Moore's Law, had roughly held for the next 50 years such that by the time Hamlin and team were searching for a solution to heat, chip manufacturers were making chips with transistors of only a few nanometers. By 2025, there were three big players in the American chip space: Intel, AMD, and Apple. All were racing for ever faster and smaller chips.

In 2026, Apple filed a patent for a modified version of the silicon manufacturing process used in making chips. One of the claims, added almost as an aside, was for the manufacture of silicon wafers only one atom thick. Apple said very little about this material other than how it could be made, and the patent garnered little attention at the time. Then in 2029, Jackson came across it while researching potential updates for graphene. It kicked off a period of innovation in materials that would catapult NASA forward.

Silicon is in the same atomic family as carbon, the element used in graphene (if you look at a periodic table, a column of elements is an atomic family. Families of elements have similar chemical properties). The one-atom thin configuration that made graphene possible, was also possible for silicon. The resulting material, dubbed silene by the NASA team, had similar properties to graphene, but not as extreme. In other words it too was a conductor of heat, but less efficient.

As it turns out, the next element in the Carbon family Germanium is also able to be configured in the same way. Gerene was harder to produce, but it conducted heat even less. The NASA team built a configuration of the three materials that acted as a gradient, taking heat in on one side and radiating out on the other side. When they ran tests on the heat transfer, they found the configuration radiated heat directionally in a three-to-one ratio. At this time it was June 2031, and the team was at the 22 month mark from the start of the materials project. They spent the next two months writing up the results of their findings, and testing new configurations.

When NASA reconvened the materials group, and the teams presented their findings, it was immediately clear that Hamlin and his team were further ahead than any other group. NASA asked the other teams to join Hamlin in developing the casige array, so-named for the three elements used: CARbon, SIlicon, and GERmanium. When Hamlin told his mother about the breakthrough and how it was received, she cried tears of joy for him.

Five years later, mid 2036, NASA had procured three nuclear powered vessels to start on missions to Mars. The next close approach of the red planet would be in 2037, and preparations were underway for the first nuclear powered missions to Mars. 22 years might seem like a long time, but it was a blink of an eye when considering missions that took years to accomplish. The unmanned missions, dubbed Haphaestus I-XXVII, had to establish a subsurface base on Mars, construct the energy infrastructure for the base, setup water mining capabilities, and build out a farming module for the eventual human inhabitants of the red planet.

While robotic missions were being flown, NASA scientists set about solving the next biggest problem: energy production and storage. As I mentioned earlier, the martian surface was occasionally beset by dust storms that could last months at a time. If one of these storms

happened while Mars was too far from Earth to allow for an emergency trip back home, something had to be done to provide the martian base with power for long enough to weather the storm. There was really only one energy storage possibility, and in the early 2030s it wasn't an attractive option—batteries.

Keeping all of the life support systems running on Mars during a dust storm for 100 days was projected to take 1 million kWh of energy (NASA was always generous with its estimates when it came to keeping its astronauts alive). The best commercial batteries of the time were the lithium-ion ones used in electric vehicles where a battery the size of a car could hold around 285 kWh of energy per kilogram. That's 3,500 batteries needed, at a weight of around 550 kilograms per battery. So, some time between 2029 and 2059, NASA had to figure out how to get 2,000 metric tons of batteries to Mars, or how to make more efficient batteries.

Of course, I said *commercial* batteries. This is NASA, no need to worry about just what is commercially viable. Chemists and material scientists had been working on various solid-state solutions throughout the 2020s. All batteries need a dielectric, a material through which positive ions can flow in order to free up electrons to make electricity. The lithium-ion batteries at the time used a liquid dielectric. Solid state batteries on the other hand used a solid dielectric. This had a few advantages that NASA wanted. For one, the solid state batteries didn't catch on fire when punctured, a very useful feature for anything traveling in space. Solid state batteries also worked better for long term energy storage, discharging at a much slower rate than the lithium-ion batteries at the time, an essential feature for any battery that would power a Mars base for 100 days. And finally, some studies suggested that certain solid state batteries operated at -20° C, far from the -60° C of the martian surface, but much closer than lithium-ion batteries could get.

By the early 2030s, solid state batteries had been found with energy storage more efficient than lithium-ion batteries, but which hadn't been scaled beyond a few grams of material. Other solutions had much longer storage capabilities, but without the energy efficiency. NASA estimated that in a reasonable scenario, it would take five missions to move the batteries needed if their efficiency and storage capacities were met. This meant that they had to get started moving the batteries in 2057, and because it's NASA that meant getting started in 2055. This gave the agency 15 years of battery development to find a viable solution. So in 2030 they wrote up their specifications for the batteries they needed, and put out a call for grant proposals. In December of that year they awarded three grants to three leading battery research organizations.

Enrique Rubero got his doctorate from Harvard University in 2028. He landed a post-doc fellowship at U.C. Berkeley where he focused on a particular solid state solution using glass anodes and cathodes using lithium instead of the liquid-based ones of typical lithium-ion batteries. By the time the call for grants came out from NASA, Rubero and his team felt they were close to solving one of the big problems NASA called out—the length of storage.

Sometimes scientific progress happens in inspired leaps, fully-formed breakthroughs appearing in a paper or a talk that we can point back to and say, “yes, this changed things.” More often, however, scientific progress happens in small incremental steps, supported by teams of dedicated scientists. Rubero and his team were those dedicated scientists. Over the next twelve years they refined their methods and approaches, and by the time the dust settled on their more than a decade's worth of progress, their lithium-based batteries delivered 425 kWh/kg worth of energy storage, that lasted six months of idle time, and could operate at -73° C. NASA eagerly

put in an order for 1,400 metric tons of batteries, and the power backup for the martian base was underway.

The new batteries created whole new applications for energy storage, for homes, businesses, and more efficient electric vehicles. While these applications were growing, the worldwide supply of lithium was stretched to a breaking point. In the next chapter we'll look at the solution to this problem, but for NASA, thankfully, their use case was small enough that they could afford the lithium the batteries used.

Numerous other problems had to be solved along the way to allow for the first (wo)manned mission to Mars to be solved. Many of them are covered much more eloquently than we have room for here in Jude Locto's excellent chronicle *Life on the Red Planet*. What matters is that NASA succeeded in putting a crew on Mars, and in particular we care about what Megan Pert, second officer on Ares III, found when she pulled out a particularly interesting rock from a drilled out hole in the martian surface.

She found the fossilized remains of a broad leafed plant—the first evidence of life found outside of Earth.

Socioeconomic 1

Any discussion about the twenty-first century would be incomplete without covering the issue that dominated the politics of the whole century—Climate Change. The Earth's atmosphere acts as an insulating blanket. Light and heat from the sun hits the surface of the Earth and bounces back into space, but some of it is trapped at the surface by the Earth's atmosphere. It's a good thing that it does too, without our atmosphere, our Earth would be as cold as the moon, which has an average temperature around -150°C .

This insulating effect is a result of the chemical composition of the atmosphere, which had, over four billion years of evolution, reached a sort of equilibrium that kept the Earth at an average temperature of about 15°C . This equilibrium lasted quite quaintly until the middle of the nineteenth century when human beings started burning energy stored in the carbon-based mediums of coal, oil, and natural gas. This upset the equilibrium such that, by our starting year of 2062, the Earth's average temperature had risen about 1.7°C , a rise that had cataclysmic effects across the globe.

Back in the twenty-first century, science deniers and just all-around bad actors were much more prevalent than they are today. There are plenty of examples of science denialism creating quite literally unlivable conditions for people around the world, but one of the most particularly egregious examples comes from the United States, and the story of the twenty-first century Republicans.

Beginning in the first decade of the twenty-first century, the American Southwest began a period of aridification that slowly but surely dried up the surface water of the American states of

Colorado, Utah, Arizona, Nevada, and California. The Colorado River, which supplied much of the water to these states (the river historically ran through two Mexican states as well, Baja California, and Sonora, but for most of the twenty-first century the river ran dry before reaching those states), with an annual discharge around 16 million acre-feet in pre-industrial times, was operating at half that capacity by 2025. Making this dry up even worse was the fact that the American Southwest was one of the fastest growing areas of the country. By 2025, the Colorado River basin was providing water for 40 million people, about 12% of the population of the country.

The aridification happened very quickly in geological time, but slow enough for the people of the affected states to be prepared. To compensate for the loss of water from the river, they began digging wells and tapping into groundwater underneath them. The groundwater cycle was poorly understood, however, and it wasn't at all clear to what extent the groundwater was a finite or renewed resource. Resources began to pour in, mostly from private companies speculating on water usage, to try and determine just how much groundwater there was and how long the region would be able to support its current population. By 2032 the general consensus was that groundwater could support the region for a conservative estimate of 20 years, but that beyond that the viability of the region was in question.

American politics of the time was incredibly polarized. The cities of the country were largely behind the more liberal democrats, while the rural areas backed the anti-science conservative republicans, with the suburbs and exurbs split between the two. Water conservation and management was an important point in the environmental platform of the democrats, while the republican stance was to deny any climate change was occurring and generally to hide their

head in the sand. So when the republicans took power in 2032 they looked at the environmental review of water usage in the American Southwest, and they took a look at how four of those five states voted for the democrats, and they pretty impolitely told them all to pound sand.

The water outlook for the region continued to worsen over the next years of the 2030s. Arizona, and Colorado, concerned with losing the precious agriculture that some 70% of their water usage went to, passed laws that took water from population centers and guaranteed it for agriculture. This began a mass exodus from those states. As water continued to dry up, people started leaving the other states affected. Some went to California, Oregon, and Washington, but the vast majority went East to where land was still cheap.

In the decade of 2030, the population of Texas grew by more than half a million people a year. This accelerated population growth was fueled largely in part out of the necessity of people fleeing the waterless homes they had in Colorado, Nevada, and Arizona. By 2040, the population change was sufficient to create the republicans' worse political nightmare: a blue (democratic) Texas. Long the largest state that would consistently vote republican, Texas held inordinate power in the republican party, but since the first quarter of the twenty-first century it had been voting more and more in favor of the democrats. With the influx of democratic voters left out to dry by the republican administration, Texas flipped.

Melissa Walters had won the democratic primary and eventually the presidency in 2040 with rehydration of the American Southwest a central pillar of her platform. By mid 2041, the democrats had passed the Rehydration and Invigoration of Valley Entrenched Revitalization (RIVER) Act, which carved out a trillion dollars over the next decade for the largest public works project the US had ever taken on. The idea was radical, but simple. Build a new river that

took salt water from the ocean and turned it into fresh water that would then flow back into the Colorado River.

Though nothing like the RIVER project had been attempted in its entirety, the individual components were generally well understood. First was the desalination process. In the first half of the twenty-first century, the leader in desalination was known as Sea Water Reverse Osmosis (SWRO). Two tanks are set up and separated by a membrane, one with a low concentration of chlorine, and the other with a high concentration. Normally, water would flow through the membrane from the low chlorine side to the high chlorine side. If you apply enough pressure however, water will flow from the high chlorine side to the low chlorine side. SWRO was well understood and employed in arid locations all over the world. The limiting factor for most utility-scale SWRO installations was the energy needed for the endeavor. Efficiencies by the early 2040s had reached about 3 kWh/m³. A cubic meter is a unit of volume just like an acre foot so we can convert one into the other at a rate of about 1,233 cubic meters to one acre foot. So to desalinate the 10 billion cubic meters (8 million acre feet) of sea water, we would need 30,000 GWh—the output of about four nuclear power plants.

The second was building the path for the man-made river. In general, water flows downhill, so the Americans had to find a starting point with sufficient elevation, and with favorable enough conditions in between the start and the connection point with the Colorado River. Then once the path had been identified, it had to be dug out sufficiently to accommodate the eight million acre feet of water they were trying to feed the river with. The mechanics of the path digging were not a problem, but the scale was.

Then there was the matter of getting the water from the ocean to the starting point. This was largely a problem of energy production. It takes 2,780 Wh to lift 1 cubic meter of water 1 km at 100% efficiency. To lift the eight million acre feet of water to the starting point height of 6.23 km would take about 173.2 GWh, about the output of three and a half nuclear power plants. Of course that was to lift the water once you got it to the Sierra Nevadas. To pump the water uphill for the 200 miles between the ocean and the Sierra Nevadas would take another 567.4 GWh, or another ten nuclear power plants.

Between the above concerns, adding in redundancies, and partly just because they could it seemed, the Americans pegged 20 GW as the energy output they needed for the project. Ten nuclear power plants were a tough sell for the American Southwest, but thankfully the region had two things in abundance, space and sunshine. 20 GW of energy was the output of about 60 million of the decade's hardest solar panels. The RIVER act commissioned ten new nuclear power plants, and 40 million solar panels. A number which provided a significant boost to the United States' green energy industry.

From the Sierra Nevadas, the river, named Potas from the ancient Greek for river, and a not so subtle nod to the office that advocated for it, would flow almost 700 miles to Lake Powell in Southern Utah. This would free up Colorado and Utah to use more of the Colorado River's output before it got to Lake Powell effectively doubling the amount of water available to the five states of the Colorado River region. It was even estimated that some of the water from the Potas would reach the Mexican states, a fact found as a gesture of good will for the many years America's water usage had run the Colorado River dry in its southern neighbor's regions.

At almost 900 miles, the Potas was almost 100 miles longer than the Alaskan oil pipeline, and more than twice as long as the Eerie Canal, North America's longest man-made waterway up until that point. The scale was daunting, but not insurmountable, and the time frame of 12 years for completion seemed doable. The Alaskan pipeline had been built in a quarter of the time, and was every bit as remote and daunting as the deserts of Nevada, Arizona, and Utah.

The project was a joint venture between the Departments of Energy and the Interior who picked Steven Ingulin as the project's manager. He was a practical leader, comfortable in both official settings, and amongst the workers who built his projects. Before working on the Potas river, he had been the head of building the high-speed rail upgrades which connected several East Coast cities called Atlantrax. This project was an order of magnitude or two larger and more complex, but it had favorable funding and time.

The building of the Potas river complex, with its 900 mile waterway, ten nuclear power plants, millions of solar panels, multiple river towns, tens of thousands of jobs created, hundreds of thousands of acres of new farmland created, and ultimate revitalization of the American Southwest has been covered extensively in media in the intervening centuries. Two of my favorite resources are the podcast *Let's Make a River*, and the book *From POTUS to Potas: the making of a RIVER* by Lashonda Watts. I'll leave it as an exercise for the reader to investigate that further. What's more interesting for us here is the impact the Potas had around the world upon its completion in 2050.

The American Southwest had hardly been the only region of the Earth affected by climate change. There were many other dry and arid places that had been feeling the effects of a warming planet. Many of them were already utilizing desalination to provide enough fresh water

for their populations and agriculture, and they were watching the audacious American project with interest to see how the Potas river would fare. For countries who had long lived around deserts, such as those in the middle east and northern Africa, being able to rehydrate a dry region with water from the nearby oceans would be an absolutely life-saving capability.

The Saudis were the first to announce their own RIVER project, followed closely by the United Arab Emirates. Egypt, long-time enjoyers of the longest river in the world, announced a project to revitalize the Nile by pumping water from the Red Sea up to Luxor. Next came Chile, whose coastal Atacama Desert provided easy access to sea water. These projects could not have come at a better time for a world, but before we get there we need to talk a little about the world's energy production in the first half of the twenty-first century.

In the mid-twenty-first century, the dominant source of energy on the planet was the same as it had been since the mid-nineteenth century: fossil fuels. Fossil fuels go all the way back to 360 million years ago with a period of geological time called the Carboniferous. During this period of time, plants existed, but nothing existed to break down plants so when they died, they just sat there on the ground. Sometimes that ground would get covered by geologic processes and those plants would end up underground. There they cooked and churned turning into coal, oil, and natural gas. The whole system was basically a planet-wide battery, taking energy from the sun and turning it into useful carbon-based molecules for enterprising animals like us to use. Yes, the fact that we used up millions of years worth of stored energy in 300 years is a shining example of our penchant for excess.

We've talked about climate change. The changing climate of the early twenty-first century led to calls to reduce our dependence on fossil fuels. These calls were largely ignored,

but they did cause some additional investigation into fossil fuels. In 2038, Muhammed Al-Sahir and Hamza Al-Asad, two geoscientists from King Fahd University of Petroleum and Minerals in Saudi Arabia published a comprehensive investigation into the fossil fuel stores of Saudi Arabia, and the larger Middle Eastern region. In it they predicted that the region would reach peak production in the first half of the 2050s, and that production would fall precipitously in the years after that.

This conclusion was hardly surprising—most of the world knew that fossil fuels were a finite resource—but climate change advocates of the time seized upon the report as an example of why our relationship with fossil fuels had to change. Still the report probably would have had little effect if it were not for the work of Agathe Garnier.

Following in the footsteps of Greta Thunberg, and Megan Dürnst before her, Garnier had gained notoriety as a young environmentalist at the age of thirteen when she petitioned to, and was granted the opportunity to give a TED talk (TED was an organization which put together talks from a variety of fields) on the environment. The talk went viral, and Garnier found herself at the forefront of the anti-fossil fuel movement.

In 2038, Garnier was studying public policy at the Paris Institute of Political Studies. When Al-Sahir and Al-Asad released their study, Garnier seized upon it, and worked on a proposal that, using her fame for access, she would attempt to deliver to the French, and if possible the EU parliament. In that same year, she found herself before the countries of the EU with a radical, for the time, proposal.

When it comes to replacing fossil fuels in power production, there are really only two games in town. Solar can come close, but only if you have lots and lots of space that get lots and

lots of sun, not exactly the reality for the northern latitudes of Europe. The other solution was going nuclear, something that would not have flown earlier in the twenty-first century, but when faced with the reality of a falling fossil fuel curve, Europe had to give it serious consideration. Garnier's home country of France had embraced nuclear more than most other countries in the world to great effect over the last century. The proposal then was to expand that nuclear power capability across the continent.

In 2038, 64% of Europe's energy production came from fossil fuels, whereas only 14% came from nuclear power. Increasing nuclear power capabilities by four and a half times would be daunting but doable Garnier argued. But more importantly it was necessary. Adding up the renewable options for the continent, just wouldn't get you there to replace the energy lost from fossil fuels. The energy sources available at the time necessitated this expansion of nuclear power production.

As politicians tend to do, the EU dragged its feet on adopting any solution whatsoever to an impending slowdown in fossil fuel production. All manner of excuses were given for not taking on the task of updating the continent's energy production to work for the next century. Most common of course was that it was too big a project that would cost too much. Garnier's plan called for the building of 650 new nuclear power plants over 20 years. The few politicians who went through the trouble of calculating the cost balked immediately.

That all changed with the RIVER project in the US. The building of the 10 power plants for the Potomac river didn't even seem to be a problem for the Americans. The EU, never known for being competitive with their friends across the pond, nevertheless didn't want to be left out of what was clearly large scale investments into reversing climate change. Looking around for a

project comparable to the RIVER projects being planned around the world, Garnier's proposal rose to the forefront.

The EU's move to increase its nuclear power output prompted other regions to do the same. Notably China and India started programs almost right away, which spurred additional programs in their respective regions. Even South America, and Africa began building plants. First just expanding in countries that already had a nuclear presence, but treaties began to expand the nuclear footprints in the continents to other countries assessed to be stable enough for nuclear power.

The new plants put a hefty strain on the world's uranium production, so much so that by the late 2040s, countries were looking for alternative sources. By coincidence, the lithium shortage alluded to in the last chapter was coming to a head at around this same time. Fortunately a solution presented itself for both: mining sea water.

It had long been known that there was sufficient lithium and uranium in sea water to exceed the world's needs, and in fact several procedures for removing minerals from sea water were known by the late 2040s. The problem with all of them, was that you had to get a lot of sea water in order to mine the relevant minerals. In other words it was a problem of scale. What you really wanted was some way of concentrating the minerals into a brine so that your mineral to liter of water ratio was favorable for the mining process. Fortunately for humanity's battery and energy production, the byproduct of the desalination process in RIVER projects provided just such a concentrated brine. By 2052, "minin' the brine", the pithy slogan of the American company Oceacquire who had won the contract for the Potas river's brine, was standard practice

at RIVER projects around the world increasing worldwide lithium production by 36%, and uranium production by 28%. A decade later, production of both resources had nearly doubled.

Such was the scene in 2062 when it was announced that worldwide fossil fuel production had fallen for the first time ever not due to differences in demand, but due to reductions in supply. The world was beginning to throw haymakers back at the force of Climate Change. While not quite ready to give up on fossil fuels, the world had started making headway towards a future that no longer relied on the carbon-based battery the Earth provided. Nuclear power production around the world had increased by a factor of three, with plans to increase by a factor of five. Arid regions around the world were being rehydrated through a series of ambitious RIVER projects. Projects which had even found a way to produce even more value than the fresh water they added to dry regions. These beginnings set the stage for what would eventually become the most important, and most lucrative, defense against Climate Change, but to understand that we'll need to hear more of our other stories.

Dark Field 2

To understand Dark Time, we first need a bit of understanding of what at the time was just called Time, and for that we need to learn a bit about the electrodynamics of moving bodies. The nineteenth century was one of incredible scientific inquiry as human beings advanced into the industrial age. Of all the scientific triumphs of the century, few were as influential and successful as Maxwell's equations for electromagnetism. Published in 1861 and 1862, the four differential equations describe the effects of electrical and magnetic fields on each other describing, among other things, how light is able to propagate in a vacuum. Together they explain the entirety of electromagnetic phenomenon that we use to our advantage, from computers and the internet to electric vehicles.

Two terms appear in Maxwell's equations, ϵ_0 and μ_0 . They are known as the permittivity and permeability of free space respectively. These two constants describe the degree to which electric and magnetic fields can permeate the vacuum of free space. They are essential constants that have the property that $(\epsilon_0\mu_0)^{-1/2}$ is equal to the speed of light. This is much more than a curious mathematical trick, it suggests that the speed of light is a constant. At the time, speeds were known to be relative to the observer, a constant speed that was invariant under different observers wasn't understood.

It would take Einstein's theory of Special Relativity in 1905 to resolve the asymmetries caused by this consideration of the speed of light. The solution was radical. First it set the speed of light as a sort of cosmic speed limit for electrodynamically moving bodies. Second it explained that as those bodies approached the speed of light, both space and time itself changed

to accommodate the speeding body. It had been suggested that Space might contract, but Time had long been considered absolute. Special Relativity set down that there was no preferred Space or Time in the universe.

In a series of papers published soon after in 1907 and 1908, Minkowski applied the mathematical formulation to Einstein's Special Relativity by describing Space and Time as a single four dimensional manifold dubbed Space-Time. For Minkowski, Time was merely another spatial dimension, related to the other three by the speed of light. Minkowski almost lamented that his fellow mathematicians had not determined the relationship earlier:

“By taking this into account it becomes clear that the group G_c in the limit $c = \infty$, that is the group G_∞ , is exactly the complete group which is associated with the Newtonian mechanics. In this situation, and since G_c is mathematically more understandable than G_∞ , there could have probably been a mathematician with a free imagination who could have come up with the idea that at the end natural phenomena do not actually possess an invariance with the group G_∞ , but rather with a group G_c with a certain finite c , which is extremely great only in the ordinary units of measurement. Such an insight would have been an extraordinary triumph for pure mathematics.”

I have been careful in this chapter to refer to the motion we're concerned with as the motion of electrodynamically moving bodies. It's a distinction that was shared by Minkowski and Einstein in their papers, although it's unlikely that either could have portended the significance of that careful wording. It would be caught and expanded upon by Diduch as follows:

“So accustomed have we become to the triumphs of Einstein and Minkowski’s description of Space-Time, that it is easy to forget that the roots of their description are in the examination of the electromagnetic phenomena. With respect to the Dark Matter, which we believe does not interact with the electromagnetic at all, we must ask ourselves just how much authority we grant to those details that arise from electromagnetism. In Minkowski’s choice of finite c , he naturally prescribed to his four dimensional manifold a constant which agreed with the known science of the time, but his words leave open the possibility of some future enterprising mathematician to apply his reasoning to some different choice of c .

“There may as yet be more to discover than simply the properties of electromagnetism. Perhaps two new constants, ϵ_0 and m_0 [being Ukrainian, and finding both the english and greek alphabets devoid of unused letters, Diduch utilized her native Cyrillic alphabet for the Dark constants] may be found describing the permeability of the vacuum of free space that applies to the Dark Matter. That we may then discover the speed of dark μ , which acts to create an alternative time which extends Minkowski’s four dimensions to five. I shall herein call this new time Dark Time, and append it to our well-known Space-Time to create the five dimensional manifold Dark-Space-Time wherein Space acts as a surface boundary between the familiar Light Time and the new Dark Time.”

To say Diduch’s proposal was radical would be a gross understatement. New dimensions had been proposed before, most notably with regard to String Theory, but proposing a new *time* dimension while simultaneously questioning the supremacy of the speed of light was a bridge too far for many physicists. To really twist that knife, Diduch included a thought towards the end of her paper that, “...gravity must then act to affect the curvature of the five dimensional Dark-

Space-Time. Here the speed of light, at which gravitational waves are known to propagate, must be the lower of the two limiting speeds since if the speed of dark was the lower, gravity would choose that speed to propagate. This implies that the speed of dark is at least as large as the speed of light.” Many physicists abhorred nothing more than the potential of faster than light travel.

Writing around 300 bce, Euclid formalized the beginnings of our modern understanding of geometry. He wrote a series of books called *The Elements* wherein, after positing five geometrical axioms, the reader could go on to prove a whole host of useful and interesting geometrical results. Euclid’s fifth axiom, which described the interior angles of a line intersecting two parallel lines, gave mathematicians trouble over the next two thousand years. Many wondered if it could be derived from the other four axioms, but none had succeeded in doing so.

It would wait two thousand years after Euclid for Carl Friedrich Gauss, perhaps the most influential of the second millennium mathematicians, to announce that other geometries were possible, and that the fifth axiom merely described the special case of a flat geometry. What followed was an explosion of academic activity as mathematicians around the world, untethered to Euclid’s flat geometry, were able to describe a whole host of non-Euclidian geometries. Gauss’s simple rejection of Euclid’s fifth axiom opened the floodgates for mathematical progress.

So it was with Dark Time. Those physicists able to get over their initial recoil at the suggestion of a new type of time found themselves in a fertile theoretical universe of speculation. In the twelve years after Diduch’s presentation at the 50th anniversary of the Higgs boson discovery, fifty-three thousand papers were published on the topic of what came to be known as

Dark Physics. The topics covered ranged from almost casual dismissal of Dark Time to full-blown recreations of the laws of Physics in the Dark.

Of course any physical theory is only as good as the predictions it made, and Dark Physics enthusiasts were finding it difficult to find testable predictions for the new science. They were hopeful testable hypotheses could be found, to see why, let me paraphrase one of the popular analogies of Dark Time used back then.

Consider the surface of the Earth. It's a two dimensional boundary between the outside of the Earth—the atmosphere out into space, and the interior of the Earth starting with the ground and heading in towards the Earth's core. You can build structures both above the surface and below the surface, but the *rules* for how you build are different depending on which direction you go. For one, structures above the Earth need to worry about falling over whereas structures below the Earth need to worry about collapse. Dark-Space-Time was similar, only Space was a three dimensional surface boundary on Light Time and Dark Time instead of the two dimensional surface of the Earth. Just like you can't peer underground from the top of a building built up from the ground, Physicists couldn't peer into Dark Time to view its underground structures, but the prevailing wisdom was that there should be clues found at the boundary. Underground structures need entranceways on the surface in order to be useful so if you were an investigating scientist looking for evidence of underground structure, you could look for entranceways on the boundary to indicate their presence. Physicists had to figure out what those entranceways in Space would look like for Dark Physics, and then go hunting for them.

Ronaldo Ancero was born in 2036 in Guadalajara, Mexico. The second largest city in Mexico at the time, Guadalajara had become the tech center of Mexico, and Ancero's parents, a

programmer and designer power couple in the tech world, worked for an organization that took on software development projects from around the Americas. When Ancero was eleven, his parents got assigned to a project developing an app for the fifteenth anniversary of the James Webb Telescope. Ancero became the first tester for his parent's work. The experience would build in him a lifelong love of space, and astrophysics.

Graduating from La Universidad Autonoma de San Luis Potosi in Mexico City, Ancero won a highly contested scholarship to do his graduate work at the University of Colorado Boulder. He finished his PhD there in 2063, after furiously rewriting his dissertation to include speculation about the newly unveiled Dark Time. The main topic of his dissertation was an extension of work he had done at UC Boulder on the possibility of a space-based gravitational wave detecting array. He asserted that such an observatory setup could assist in verifying or disproving the existence of Dark Time, but he stopped short at the time on making any actual testable predictions. After obtaining his doctorate, Ancero joined the Vera Rubin Space Telescope team, so named for its association with the terrestrial Vera Rubin Observatory in Chile.

Ancero cultivated a reputation as one of those uncommon physicists who never lost his sense of wonder in the universe. He found a kindred spirit when at a 2070 conference on gravitational waves he met Ama Kowalska, an astrophysicist at the European Space Agency. Born in Rybnik, Poland in 2028, Kowalska made liberal use of the EUs open borders, getting her primary education in Sweden before attending university at The Technical University at Munich, and then getting her PhD at Oxford.

Kowalska had become interested in space-based gravitational wave detection ever since the success of the M-LIGO observatories. To understand the appeal, let's go back to M-LIGO's

predecessor, the original LIGO observatories built in the United States, and the Virgo observatory hosted by the EU in Italy. LIGO set up two Laser Interferometer Gravitational-wave Observatories (hence LIGO) 3,000 kilometers apart in the US. The distance was specifically chosen so that local effects could be filtered out from the detectors' detections. For example, a small earthquake at one location would not be picked up at the second location. Virgo added a third observatory similarly far enough away to ignore local noise. In this way, the three detectors act as noise filters for each other. The observatories worked, and the first gravitational wave was found in 2015 from the merger of two black holes around thirty times more massive than the sun.

Interest in gravitational waves lagged in the west during the 2020s and 30s, but in 2038, China announced their own observatory would be built. Two years later, Australia and India both announced that they would build their own gravitational wave detectors. This brought the number of observatories around the world up to six. The five groups involved negotiated an agreement to share their discoveries mostly by necessity as gravitational wave detection worked better with disparate observatories. The project added "multiple" to the start of LIGO and the M-LIGO project was born.

In the intervening decades between its establishment, and when Ancero and Kowalska started their work together, M-LIGO had made a series of interesting and reassuring discoveries, verifying General Relativity again and again. Through a series of upgrades, and other countries bringing observatories online, M-LIGO was able to refine its approach and detect fainter and fainter gravitational signals. The array of observatories' work reached a high point in 2065 with its detection of gravitational waves from a single star's collapse into a blackhole providing the

first direct evidence for possibly the most dramatic prediction of General Relativity. It was this discovery that had Kowalska excited at that 2070 conference.

The thing about gravitational waves is that once you can detect them, you start detecting a lot of them. The more sensitive the detectors got, the more gravitational waves they would find. As exciting as that is for the engineers who made the detectors, astronomers need more data in order to find out what's going on. In general, if there wasn't some sort of visual confirmation of what was going on, astronomers couldn't tell what the source of a gravitational wave was. The first waves detected were of sufficiently large events that telescopes were able to confirm the cause of them. As we moved into the 2060s, however, M-LIGO was detecting dozens of waves a month that left astronomers scratching their heads.

As physicists began thinking about Dark Time, agreement started to form that gravitational waves were our best bet at finding answers. The first attempt at resolving the issue was straight forward enough. There are only certain types of events large enough to be detected by M-LIGO, and we had a good enough understanding about how common those events are. So add them all up and see how many gravitational waves are detected. If you detect a lot more waves than the events you predict, that's a good indication that there's some business going on with Dark Matter. The initial investigation in this regard was encouraging, but inconclusive. It found there were indeed more gravitational events than we would expect. but the number fell well within the rather large margin of error celestial events came along with. If we were going to get to the bottom of Dark Matter, we would need a better observatory.

If you want to take a picture through a telescope, there are two ways that you can get a better picture. The first is to use a better camera, resolving more detail in the image you capture.

The second is to get a bigger telescope, allowing you to see more detail at the same resolution. With gravitational wave observatories, the detector was the camera. Years of engineering had improved detectors significantly, but to get the resolution required for hunting Dark Matter would take hundreds maybe thousands of years worth of improvements. Instead scientists looked for the other improvement—a bigger telescope.

Radio astronomers had known how to get a bigger telescope ever since the twentieth century. Placing individual telescopes around the world and then coordinating their data effectively turned the Earth into a giant radio telescope. Gravity is many orders of magnitude weaker than electromagnetism (about a billion billion billion billion times weaker) so the Earth wasn't quite big enough to provide the resolution scientists wanted. You needed something orders of magnitude bigger.

So when Ancero and Kowalska met in 2070, space-based gravitational wave detection was in the zeitgeist. They resolved to work on a joint proposal for NASA and ESA to deploy gravitational wave observatories across the solar system, creating a gravitational wave telescope a million times larger than the terrestrial M-LIGO. The proposal was ambitious but doable, and had the benefit of pertaining to the hottest topic in Physics at the time.

Giant multi-billion solar system-spanning space projects don't get written by individuals. Ancero and Kowalska took their intentions back to their agencies and won approval to pursue putting forth an actual project proposal. Two years later, interest in the project at both agencies had led to more than a hundred scientists and engineers working on the idea. Interest in Dark Matter had only grown by that time, and physicists around the world were tuned into the progress going on at the agencies.

Ancero was charged with supplying the proposal with a list of possible experimental outcomes. He drew heavily from the published material on Dark Physics, but he wanted to include something novel to really drive the proposal over the top. He called up Kowalska seeking inspiration.

“Maybe we’re all thinking about this the wrong way.” Kowalska said over a recorded video call. “What if we were Dark Physicists trying to figure out Light Physics. We can’t use telescopes because no dark matter interacts with light. Our Physics doesn’t allow for gravitational collapse. How would we describe the cataclysmic events of Light Matter? What does a supernova look like to the Dark Physicist?”

Ancero took Kowalska’s words to heart and started to think about what life might be like for a Dark Physicist. To clear his mind from the proposal and work on the problem, he headed down to visit his parents in Guadalajara. There they took a day trip to the ocean where, watching the waves rise with the tide, he had an epiphany. I’ll let Ancero explain it, as he did in a YouTube video back in 2074.

“I realized sitting on the beach that searching for Light and Dark Matter in gravity was like sitting at the bottom of the ocean, trying to figure out what was going on on the surface. Far enough out into the ocean you wouldn’t be able to see the waves at the surface, you’d have to infer them. You could imagine some sensitive depth equipment that would rise and fall as the surface rose and fell. Some enterprising scientist might conjecture that at the surface there was some scene of violence causing the water to periodically rise and fall as waves. This is the scene of Light Matter, the realm of stars and explosions and black holes.

“But close observation would reveal that at a regular interval the whole of the surface would rise and fall taking the waves along with it. What would our enterprising scientist ascribe that to? How long would it take him to ascribe the tides to the gravitational influence of some orb not even connected to his ocean floor?

“And so I proposed that, rather than look for waves at the surface, we look for evidence of the tides. Evidence that diffuse Dark Matter was affecting the gravitational waves of known cosmic events to provide for us a better map of that elusive stuff, and perhaps clue us in to more of its properties. If we could find some sort of baseline for what Dark Matter is and what it does, we could then see if it changed, and there we might find answers.”

By the time the proposal was delivered to NASA and ESA in 2074, it was one of the most popular experimental physics proposals of all time. It seemed a perfect application for NASA’s newly proposed Mars-based launch center (more of that in the next chapters), which gave the agency 20 years to develop the technology that would go into the interplanetary satellite observers. We’ll save that story for our next space chapter, and skip ahead to the deployment of the space-based gravitational wave observatory.

There was only one name pertaining to gravitational waves sufficiently important to both the US and Europe: Einstein. In 2100 the Einstein Space-based Gravitational-wave Observatory (ESGO) became operational. It consisted of an array, called the Einstein Array, of 30 gravitational wave observing satellites deployed around the eight planets, Pluto, and cruising through the Kuiper Belt. The project was considered humanity’s best chance at resolving what Dark Matter is, and whether Dark Time exists or not.

The confirmation came quickly. The gravitational tidal effect described by Ancero was picked up on in glorious detail during the first year of observation. So clear was the evidence, that the project actually took on more ambitious projects sooner than expected. By 2104, the Einstein Array was working on a global map of Dark Matter within our local area of the Milky Way, and working on a less-detailed map around the galactic center. The Array picked up the rotation of the Milky Way's Dark Matter Halo, and mapped that to tidal effects on gravitational wave events.

Ancero and Kowalska were in their eighties when in 2111 the Einstein Array picked up something even more incredible than either of them could have marveled at forty years earlier while they were writing their proposal. The dark matter tidal effect was increasing in the direction of the galactic center. The effect was subtle, and might have been overlooked had it not been written in as a specific thing to watch for in the Array's software. Once found, astronomers performed several more tests to verify the effect.

On June 4th, 2112, NASA and ESA jointly announced the findings. rumors of which had set the Physics world abuzz. Something at the galactic center was *creating* Dark Matter. Everyone knew what was at the center of the galaxy, a super massive black hole. The conclusion was impossible to ignore, the super massive black hole was somehow producing Dark Matter.

Biochemistry 2

Founded in 1901 as a chemical company producing innocuous enough artificial sweeteners, Monsanto grew to be one of the largest providers of seed to the agricultural world. Focused on chemicals for most of its first 86 years, some like agent orange and DDT—significantly less innocuous than their founding products—Monsanto entered into the world of biotechnology in 1987 with its first trials of genetically modified crops. Over the next thirty years, Monsanto would grow to become one of the most reviled companies in the world with its business practices defending its genetically modified crops.

The money maker for Monsanto were its “roundup ready” seeds. Roundup was an insecticide developed by the company that used an ingredient called glyphosate. They paired this insecticide with glyphosate-tolerant maize, soy, wheat, and cotton. In one of the more questionable patent practices of the twentieth century, the US allowed Monsanto to patent the DNA used in their genetically modified crops. From 1987 until its acquisition by Bayer in 2018, Monsanto *aggressively* protected its patented interests, bringing lawsuit after lawsuit against governments, companies, and even individual farmers who even came close to their seed. Their business practices had tarnished the Monsanto name sufficiently that shortly after Bayer acquired them in 2018, Bayer dropped the Monsanto name in an attempt to rehabilitate the brand.

You can only hold onto a toxic asset for so long. Bayer divested from Monsanto in 2023, less than two decades after their acquisition. The ostensible reason given was to allow Monsanto greater freedom to research new biotechnological capabilities such as CRISPR, but the consensus from commentators was that continuing Monsanto’s success in the market took a level

of questionable morals that left the Bayer board queasy. Of course rather than take the loss and rehabilitate the company, Bayer took the easy way out of leaving the matter to the market.

Monsanto hit the scene hard, apparently viewing their 19 year break from litigation as all the more reason to sue people more. They approached their task with a ferocity that only truly vicious people possess. CRISPR was indeed posing an existential threat to the nearly 150 year old company, and in true capitalist fashion, they chose to destroy their competition rather than take on the research and development necessary to compete with them. By the time 2062 rolled around, Monsanto ranked on nearly every list of the evilest companies on the planet while also controlling nearly 80% of the maize, soy, and cotton grown in the United States.

Such was the state of things when Julia Frense, coming off of the success of the mVirex machine, announced that funding had been secured for a \$105 billion dollar procurement of Monsanto. The publicly traded company's board had little choice than to pursue the acquisition at that price. Once the sale was finalized in 2064, Frense, and Emerson got to work dismantling the nefarious apparatuses at work at Monsanto. Their goal was to transform the company into a force for good in the agricultural space by embracing the potential of PICKR.

Frense was fairly vocal about the intentions of their de-eviling process. The media latched onto the story, partly due to the novelty of a benevolent takeover of a company the size of Monsanto, and also due to Frense being the first female tech mega-billionaire—the juxtaposition of her benevolent takeover with the vanity projects of male tech mega-billionaires of the twenty-first century lent itself to the talking heads of the time.

As with any endeavor, good or bad, there were detractors and naysayers of the de-eviling attempt. One particularly vocal opponent was Jack Morrow, a republican from the deep red state

of Alabama. In an impassioned speech given on the house floor shortly after Frense's acquisition of Monsanto he offered, "They say this group is de-evilng Monsanto, but I ask you on whose authority are they making their decisions? No doubt their morality will come from the godless recesses of the scientific endeavor! I tell you this de-evilng process is really a deviling process!" Morrow was soon forgotten, but his turn of phrase was eagerly snatched up by proponents of Frense's acquisition. The de-evilng process thus took on the popular name deviling, a designation that tickled the largely anti-religion and pro-science advocates of the process.

Frense began by shutting down all litigation that Monsanto was pursuing, wiping out thousands of lawsuits in a matter of months. Similarly she sought to settle lawsuits held against the company as soon as possible. The money saved by shutting down the legal arm of Monsanto's business was put right back into R&D replacing thousands of lawyers with thousands of scientists in one of the largest scientific recruiting efforts ever performed. In addition to ending the lawsuits, Monsanto opened up all of its patents to licensing. Some felt this didn't do enough to restore the imbalance created by Monsanto's patenting of life in the first place, but it had an immediate impact on the competitive landscape in agriculture.

These and other changes had little effect on Monsanto's bottom line, a fact that brought joy to the company's investors, and drew the ire of business pundits around the world. When Frense presented at the 2066 shareholder's meeting that the company's beefed up R&D efforts were about to bare fruit, Monsanto became a consensus buy in the investor world.

The success of the deviling effort with Monsanto led to a popularization of the idea. There are only so many tech mega-billionaires, but smaller largesses were starting to be deployed at deviling groups and organizations deemed evil enough to receive the treatment. Among the

first to go were a number of multi-level marketing companies. Long-time preys on good people looking to make some money in a side hustle, MLM schemes were little more than legal scams perpetrated on the public.

Parts of nearly every sector got the deviling treatment to some degree. Food companies who sourced their food using factory farming, and bad labor practices. Industrial companies, long holding to the very narrowest interpretation of environmental laws and practices, were taken over either wholly or partially and set to decrease their environmental impacts. There were a number of companies targeted for underpaying their employees, the deviling instituting a flatter pay scale. Healthcare, Energy, Consumer Goods, etc. by 2075, all sectors of the market had some type of deviling touching them. The action created turmoil in the market, as any company at any point in time might be taken over and changed in a way that might affect the stock price positively, neutrally, or negatively. If you read accounts from the time, it was an exciting time to be alive.

Deviling was a worldwide phenomenon, but it had a particularly large effect in the United States where “evil” corporations had long found fertile ground to operate. We already saw how in 2040 the normally republican state of Texas flipped to the democrats for the first time. That flip created a cascade of events which saw the republican party gradually lose support across the board. Apparently the strategy of cultivating white male rage was not a long term winner. By the mid 2060s and the dawn of deviling, the republicans, faced with an existential crisis, split into two smaller parties. The less conservative group got to keep the name and primarily represented parts of otherwise democratic states. The second group called themselves the Christian

Republican Union (CRU), and represented the historically red states of the American Plains and American South.

In 2068, the first presidential election after the republicans split, the democrats won the presidency, the house, and a supermajority in the senate. Emboldened by the opportunity presented by their rival's fragmentation, the democrats used the opportunity to pass a number of liberal policies—universal healthcare, extensive workers' enablement laws, broadened social safety nets, a federal jobs guaranty, and a pilot program on universal basic income. In 2070, and 2072, the democrats won similar majorities, which gave them even more confidence in their agenda. President Faryn Masters started looking for a project large enough for her legacy. Like many before her, she looked to the stars.

NASA rode high on the discovery of life on Mars for the first half of the 2060s, but before long its scientists and the public began to wonder what was next. The next big mission on its radar was to Europa, an icy moon of Jupiter. At the time it was theorized that there was a water ocean beneath its icy exterior, and that that water, coupled with Europa's volcanic activity, might be sufficient for life. In typical NASA fashion, the agency had created two plans, one for if life was found on Mars, and one if not. With Pert's discovery of fossilized life on Mars, NASA stepped on the gas of its plan A.

In 2073, Poseidon 1 drilled a hole into the icy surface of Euorpa, and deployed a battery powered submersible into the depths below. Instrumentation on the vessel whirred to life, sending back readings of the water's composition, surprisingly similar to the Earth's oceans. The submersible went looking for the known volcanic structure beneath the ice. The landing spot had been off by 36 kilometers. The error was the equivalent of missing your target address in New

York City by half a block, but it doomed the mission. NASA knew it had to deploy the submersible, but it also knew the tragic reality—not designed for long distance travel, it was unlikely that there would be enough power to reach its destination. With a whimper, the submersible, and the mission died leaving the situation inconclusive.

The problem, NASA's scientists and engineers believed, was that we are too limited at what we can put into space on the Earth. If we could find a way to lift off from a place with less gravity, one that also had the advantage of being closer to our destination, we could include much more power with our next submersible. This of course was the logic behind the Lunar Gateway that the Artemis missions of the 2020s had set up. The problem with using the moon, though, was you couldn't create fuel on the moon so it just wasn't large enough for the needs of the Poseidon missions. The solution, they claimed, was to create a base on Mars. Just the kind of gigantic project a presidential ego could lean into.

Living on another planet isn't a matter to be taken lightly. It's not like camping in the woods where if something goes wrong you can just head home. It's not just about what you can take with you, but is there enough of what you need *on Mars* to not just sustain the group, but actually enable them to build a whole base? It would take more than NASA to answer these questions.

President Masters was contemplating this NASA project when Frense and Emerson came to the white house to be awarded the Presidential Medal of Freedom for their efforts to rid the world of disease. After the ceremony, the president invited the two recipients to the Oval Office for a discussion on the state of science in the country. In that fateful meeting, Emerson posed a pivotal question, "We know that public-private partnership works. NASA has been incredibly

successful at using science to push the envelope of technology. But I've always wondered why we stopped there. Why not have a National Association of Life Sciences for sciences other than physics and astronomy?"

President Masters looked thoughtful for a moment and then asked, "And what would the mission of this association be?"

Emerson became animated, "All sorts of things! Eradicating disease, curing cancer, more efficient food production and ending hunger, addressing climate change through biological carbon capture—"

"Preventing death." Frense interjected. "NASA has space, the life sciences would take care of death."

"Preventing death?" the president looked incredulous. "You mean like by curing disease right?"

"At first," Frense continued, "but there are already private enterprises out there seeking to prevent death altogether. Think about what we could do with a partnership the scale of NASA."

In the last congressional session of 2073, the democratic congress passed legislation establishing the National Life Sciences Association (NLSA pronounced *nilsa* by most). It's first mission was to assist NASA with the establishment of a base on Mars by finding efficient means for farming in the martian soil, but in the 2074 state of the union address, president Masters established its grander vision. Curing cancer would be the agency's moonshot, but like NASA before it, it wouldn't stop at the moon. As the president put it, "NLSA will endeavor to stave off that very creature which haunts us all. Death itself will run from us, cowering in the night, such that, for future generations, only taxes will be guaranteed."

The new agency was well received by the media, and the public. There was, of course, predictable backlash from conservatives, especially from the more religious contingent of the two republican parties. There was little more clear example of humans “playing god” than trying to usurp death itself. There were many among the religious who believed curing diseases was already a bridge too far. With their political power greatly diminished, these groups had little effect on the actuality of bring NLSA into operation. Even their media outlets were diminished as the formally dominant conservative airwaves were forced to split time and focus between two rapidly diverging groups.

Emerson was chosen as the first director of NLSA, and she got right down to work at organizing their first mission and staffing the agency. NASA set a goal of 2092 for a working base on Mars. Working back from that date meant getting NLSA’s food solution up and running within a decade—just the sort of ambitious project that the two agencies had been chartered to take on.

Surprisingly little work had been done in the last fifty years on generating novel staples in the realm of fruits and vegetables. For the most part, one company or another’s monocultures dominated the large spans of land offered by the Earth for agricultural purposes. Emerson took a look at that competitive biotech desert and knew she’d have to look elsewhere for the answers NASA was looking for. Thankfully the world had three agricultural industries that were pushing the boundary of our biotechnical capabilities: beer, wine, and marijuana.

We’ve already seen how bioengineering of the hops plant used in beer led to the PICKR process for multi-gene editing. Well the beer industry didn’t let up. Since Hartmann’s development of PICKR in 2032, genetically modified hops had proliferated widely buoyed by

the spread of craft beer across the globe. Hops horticulture grew increasingly specialized so that by the mid 2050s it wasn't unusual to see hops farm subdivided into a few dozen different hops varieties. The specialization led to increased efforts of micro-farming, trying to squeeze out the best hops from the smallest amount of land. It made a compelling segue into interplanetary agriculture.

So it was with hops as it was with grapes at vineyards around the world, but the micro-farming took on a novel challenge. Sophisticated wine palates would have you believe that the exact soil composition at a particular location is imparted on the flavor profile of the wine that is made from grapes grown in that soil. Attempts to exactly recreate, and even enhance the soil-based flavor profiles of well-regarded regions around the world had granted unique expertise in soil mineralization and cultivation amongst the biotech botanists of the vintners of the world. Another essential need of the effort to grow sufficient sustenance on Mars.

Nowhere, however, were the effects of bioengineering felt as fiercely as they were in the marijuana industry. In 2074, it was still a young industry—marijuana had only been legalized in the US in 2043, with the same democratic government that approved the RIVER project—but the sheer variety in psychotropic effects possible in marijuana lent itself to incredible effort from uniquely motivated bioengineers. Years of prohibition had led to the cultivation of indoor growing facilities utilizing hydroponics and aeroponics to provide the perfect chemical composition for their crop's growth. These at one time illicit grow chambers, were the perfect incubators for innovation in growing a farm on Mars.

Danin Colem, Luis Noble, and Heather Galding were brought in as biotechnical experts in their respective industries of beer, wine, and weed. Collectively NLSA employees referred to

them as the Vice Squad, a tongue-in-cheek reference to the fact that it had been less than 200 years since all three of them had been prohibited in the United States. Together they built a team of chemists, biologists, botanists, and engineers to get down to the business of constructing the optimal equipment, and best plants to cultivate for growing on Mars.

Eating a vegan diet on Mars was definitely possible, but the prevailing wisdom, supported by studies on isolation, was that in situations where people were separated from everyone they knew and cared about by 60 million miles, it was beneficial to change those people's diet as little as possible. To that end, Emerson started a second branch of the Martian food supply investigation. She recruited Gloria Torres and Lonston Robinson both leaders in the realm of cultured meat. Their goal was to find some way of taking the nutrients supplied by the aeroponic growth chambers the Vice Squad were developing, and turn them into meat for the future astronauts, a bit of biological alchemy that had never been done before.

Cultured meat requires a complete protein input in order to grow the tissue necessary for consumption. Complete protein means that all of the nutritive proteins for mammalian consumption are present. Plants in general are great sources of protein, but not complete protein. Vegetarian diets usually have to mix and match proteins in order to get complete protein, but as luck would have it, one of the crops the team was most well acquainted with was an excellent source of complete protein: hemp. Take the psychotropic THC out of marijuana, and you have a hemp plant whose seeds contain the complete protein the cultured meat group was looking for.

The fact that plant protein had never been used to create cultured meat before was less due to the difficulty in the process and more that on Earth it was cheaper, faster, and easier to simply use animal-derived protein. By 2077, a mere three years after the funding of NLSA,

Galding and Robinson had worked out a process for taking hemp seeds and turning them into meat. The process worked well for terrestrial purposes, but was far too water and energy intensive for use on Mars. NLSA filed a series of patents regarding the process, which it supplied to the public domain.

Over the next decade, NLSA focused on increasing the yield for its crops, diminishing the water and energy necessary for them to grow, and overall adding efficiency to both the crop growing and the meat culturing efforts. By 2085, things were looking good for deployment to Mars, and the two American agencies were working together to deploy the system to the red planet. We'll tell that story in our next space chapter, for now we need to return to the story of the plant to meat process, now taken over by the private sector after NLSA's patents were released.

Seif Nielson had a near unhealthy obsession with wood. As a boy, born in Melbourne, Australia in 2034, he had broken his leg one summer, and had to spend one of the valuable Christmas vacations (remember that Australian summers happen during December) of his youth laid up inside. There are only so many video games one can play when confined indoors, so Nielson turned to books to fill his time. He read two books that summer, which set him on his life's course: *Norwegian Wood* by Lars Mytting, and *Good Clean Fun* by Nick Offerman. The two, about wood, and making things out of wood respectively, sparked an interest in woodworking that Nielson would cultivate and grow into a successful furniture endeavor that would leave an indelible mark on the world.

Graduating with his MBA in 2059, Nielson set about turning his interest in woodworking and furniture making into a furniture business named Timbo, an application of the Australian "o"

epithet to the word timber. By 2069, Timbo was rapidly growing, and on track to become the largest furniture supplier in the Asia Pacific market. In those days, climate change was ravishing the Australian timber industry leading to frequent shortages, and some of the highest lumber prices in the world. To combat these rising costs, Nielson petitioned the Australian government to utilize some of the greenery created by their Western Australia RIVER Project to be used for timber forests. By the project's completion in 2074, Timbo had become the nation's largest furniture company, and the Australian government granted Nielson's request for additional forest land.

In 2076, The Australian government, inspired by the American establishment of NLSA, approached Nielson, and several other business leaders with a desire to establish a joint R&D effort around the West Australian RIVER Project. Nielson jumped at the idea, leveraging his position in the business world to raise significant funds, matched by the Aussie government, for the establishment of the Biological Investment Opportunity of the West Australian RIVER Project (BIOWARP).

Unlike NLSA across the Pacific, BIOWARP didn't staff its projects directly. Instead it provided grants to research being done within Australia, and particular for research that would benefit the West Australian RIVER Project. Nielson found himself as the chair of the grant granting committee, a position that provided him a modicum of power when it came to choosing BIOWARP projects. In 2078, when BIOWARP went live, there was one project that stood out above all for Nielson, it was written by Lestari Mahd and contained the three magical words for Nielson, "accelerated timber growth."

Shortly after the first cultured meat was commercialized in the 2020s and 30s, there was a push in biotech for cultured everything. Of particular interest were efforts at creating commercial plant products—flowers, fruits and vegetables, useful seeds, and hardwoods—in a lab. Much of these commercially viable plant components come with huge amounts of less useful plant waste, the thinking was that if you could produce them without the waste you would have a commercial winner.

The effort took longer than most investors had the appetite for. For a number of reasons, isolating plant components was simply harder than growing animal muscle tissue in the lab. It wasn't until 2051 that the first hardwood of significant yield was grown in a lab. And when it had been done, it was found that it grew at the same rate as it did in the forest. Without an economically viable slam dunk, funding for research into cultured plants dried up. By the late 2070s, cultured plants were sufficiently out of favor that the filters for BLOWARP flagged Mahd's grant proposal for rejection. If it weren't for direction from Nielson to show him anything that involved increased timber production, Mahd may have been forgotten.

One of NLSA's patents for food production on Mars concerned the production of hemp seeds, namely processes for increasing seed load given idealized aeroponic conditions. The process was only utilized to a certain extent as it was more water and energy intensive than Mars conditions allowed. Water and energy are the two things a RIVER project has in abundance, however, so Mahd proposed attempting to extend the NLSA process to other plant components. In a fine case of knowing one's audience, she chose the production of timber as the plant component to investigate. Her proposal outlined utilizing NLSA's process to increase timber production of aeroponically grown hardwoods. When Nielson ran the math on the proposal's

figures, the grant would yield a thirty-two-fold decrease in costs for Timbo. Approval was a no brainer.

Mahd was born in Palu, on the island of Sulawesi in Indonesia in 2039. Her family emigrated to Perth, Australia when she was five where she lived until attending Maquarie University in Sydney where she studied biology. She received her doctorate from the University of Western Australia in 2065. Being near her parents in Perth, she stayed at UWA as a post-doc where she worked on PICKR methods for protecting agricultural monocultures. By 2075, she had made tenure and was running her own lab. BIOWARP gave her the opportunity to expand her work beyond the farm to work that could have impact on a regional or even national scale.

The initial results were encouraging, but not particularly earth-shattering. By 2080, the Western Australian RIVER Project, using PICKR enhanced hardwoods growing in an aeroponically optimized growth chamber powered and watered by the project, was producing commercial-grade timber 1.2 times faster than growing the timber in a forest. In an unusual show of patience from a privately-funded research organization, Nielson convinced the grant granting committee to reauthorize Mahd's funding for another five years.

Mahd knew that her funding was going to dry up if she wasn't able to produce more dramatic results this second time around. Accordingly she went back to the drawing board, looking for some way to increase the timber yield. Frustrated at the problem, she headed to her parents for a long weekend.

Her parents' place had always been a bit of a sanctuary for her away from the high stress world of academic research. When visiting she would help out at the family restaurant where, even in their sixties, her parents kept regular work hours at. There she explained the problem to

her mom as best she could. Her mom listened intently, accustomed to intellectually challenging conversations with her daughter. At the end her mother laughed briefly before saying, “Lestari dear, if you want the delivery to go faster you don’t ask the man to run as fast as a bicycle, you ask the bicycler to make the delivery.”

“Immediately I felt what Archimedes had in the bath when he exclaimed, ‘Eureka!’” Mahd wrote in her 2092 auto-biography *Unbounded Growth*. “I drove back to my lab that evening and immediately planned out our pivot. I was possessed. If we could take Wolffia, commonly known as duckweed, and coax it to produce hardwood, we could leverage its fast-growing capabilities to give us the speed we wanted. With PICKR it might just be possible.”

Wolffia’s fast-growing capabilities had been studied all the way back in the early 2020s when its gene expression was studied. Unlike most plants, which will turn off their growing genes during the night, Wolffia’s genome enabled it to grow all of the time making it grow much faster than average plants, and much much faster than most hardwoods. Editing its genes to produce hardwood is, of course, trivial to our modern sensibilities, but at the time represented a novel application of the PICKR process.

In 2083, Mahd unveiled to Nielson and the BLOWARP board of directors the growth chamber for her enhanced Wolffia plant. Hardwood was produced at 2.1 times the rate of wood produced in a traditional forest, but it was also produced in stacked horizontal trays, which meant the wood was produced in a fraction of the land area. The growth chamber was seemingly only limited by the water, nutrients, and energy one could supply meaning that any RIVER project could trivially be updated to supply miles and miles of commercial-grade timber. It was such a

leap forward that Nielson told the board that this one discovery could fund BLOWARP research for the next thousand years.

Mahd and her team released a paper outlining their process, and submitted patents for it in late 2083. The end of their paper contained the encouraging line, “We have limited our inquiry and application of our found process to the production of hardwoods. We feel confident that similar inquiries may be made for the entire panoply of productive plant components.” In other words, what else could be grown?

In the paper, the team had called their growth chambers PICKR-Enabled Aeroponic Cultivation Houses (PEACHes), but the name was soon replaced by the pithier Mahd Arrays. The arrays became the focus of a significant portion of the life sciences’ research cohort. By the end of the century there was a Mahd Array for 68% of commercial plant products. Ten years after that, it was 92%. The development of commercially viable production that was more efficient in water use, land use, and faster than traditional agriculture made the RIVER projects that powered them some of the richest economic products in the world. Essentially any country with sun-baked land between the tropics of cancer and capricorn came into possession some of the most valuable land on the planet, a fact that we’ll learn more about soon.

Space 2

Born in 1930 in Chicago, Illinois, Frank Drake held an early curiosity regarding the possibility of life on other planets. It's difficult to imagine the sensibilities of the time. Rather than science dominating the public's mindset, religion was at the forefront, and the idea that humanity was not the existential center of the universe was anathema to religious thought. No exoplanets were known at the time. To think there might be intelligent life out in the cosmos was a radical idea that would draw punishment down on the precocious child who suggested it.

Thankfully academia allowed for such dreaming, at least in moderate doses. Drake pursued his undergrad degree at Cornell University before attending Harvard for his graduate work where he studied the rather new field of radio astronomy. There he cultivated his interest in the possibility of extraterrestrial life. After graduating, he worked on several radio astronomical problems, eventually settling at the Arecibo Observatory in Puerto Rico, and taking a professorship at his alma mater Cornell. He grew a reputation as an enthusiastic supporter of the idea of extraterrestrial life, which led to his crafting the Arecibo Message, a message broadcast from the observatory containing information about life on Earth. His work also led him to craft the pioneer plaque with Carl Sagan, the gold plaque sent on board Voyager I out of our solar system. He's perhaps best known, however, for the Drake Equation, a (at the time) highly speculative probabilistic equation regarding the prevalence of intelligent extraterrestrial life.

$$N = R_* \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

In the Drake Equation, N is the number of planets capable of communicating with Earth. R_* is the rate of star formation in the group of stars you're concerned with. f_p is the fraction of

stars with planets. n_e is how many planets in each of those stars that can support life. f_l is the fraction of life supporting planets that actually develop life at some point. f_i is the fraction of planets with life who go on to develop intelligent life. f_c is the fraction of intelligent life planets that develop sufficient technology for communication. L is how long communicative civilizations actually communicate for. Until the 1990s when the first exoplanets were found, all of these numbers were complete conjecture, but in the first third of the twenty-first century, the James Webb Space Telescope was doing a pretty good job at homing in on f_p and f_l , and it was starting to look like those fractions were pretty close to being 1.

Exoplanets were brand new and exotic when the JWST was being dreamt up, so when the telescope went live in 2022 it was powerful enough to investigate some exoplanets, but wasn't optimized for the task. Knowing just how important answering the question of "is there life out there?" was, the Western space agencies that collaborated on the JWST, NASA, ESA, and the Canadian Space Agency, knew their next space telescope had to try and answer that question. Planning for the next space telescope began in 2026 with the first confirmation of a life-enabling atmosphere on the exoplanet Gliese 581g. Through the end of the 2020s the project gained steam, and work was scheduled to start on the new telescope in 2032 with a launch date fifteen years later in 2047. It also got a name: the Frank Drake Space Telescope (FDST). The eponymous scientist had passed in 2022, mere years before JWST's discovery of Gliese 581g's atmosphere.

How do you make a telescope that can tell whether there is life on another planet or not? To date the only arrow in astronomer's quivers was to look at the spectral lines of an exoplanet's atmosphere. Remember that elements and molecules absorb certain wavelengths of light, and by

analyzing which wavelengths of light are absorbed from a planet's host star, astronomers can determine the chemical make up of a planet's atmosphere. Certain gases, like methane, ammonia, and oxygen, are only likely if there are biological processes going on so they can be an indication that life is present on a planet. They aren't confirmation though, and that's what astronomers were after.

To verify that life was actually the cause of the biological gases, it was believed that an additional finding was necessary. On Earth, the chemical composition of our atmosphere changes with the seasons, as the Northern hemisphere, where the majority of our plants live, enters winter, decaying plant matter releases more CO₂ into the atmosphere, which, coupled with decreased plant respiration, leads to a 1-2% increase in CO₂. A dead planet wouldn't see this kind of change in its atmosphere. Astronomers wanted to look for this seasonality, but remember that we can only see the spectral lines when the planet transits in front of its star—an event that's too brief for us to see different seasons. If there were exoplanets that behaved like Earth, though, that meant that some would have hemispheres with different seasons just like Earth. If the FDST could detect atmospheric disparities of one part in a hundred between the northern and southern hemisphere of exoplanets, it just might be able to tell us if there was life on those planets.

After a few project delays, the FDST launched and was deployed in the beginning of 2051. It immediately got to work measuring the atmospheric spectra of nearby planets, and Frank Drake would be cheering his nearest fellows at what it found. Nearly every exoplanet they pointed the FDST at exhibited the seasonality that they expected. It seemed likely that life was not a special condition offered to only a select few planets, but rather the norm for the evolution of planetary systems.

Though the findings were very much evidence that life probably existed on other planets, it wasn't proof. The hope was that NASA's other big project of the time, landing human beings on Mars to perform excavation of a site where it was thought likely a fossil of life might be found, would finally provide proof once and for all that life was possible on other planets. So when Megan Pert pulled forth the fossilized rock in 2062, astronomers giddily upped their prediction of n_e the number of planets per star that could host life. And if life is more common than we originally thought, it upped the practicality of a mission to go find it in our solar system. So that's where NASA turned its attention next, the moon of Jupiter Europa, an ice ball thought to host a liquid water ocean beneath its frozen surface where volcanos might supply the energy needed for life.

When it comes to searching for life on Europa, you've got to do two things. First you need to dig a hole through the surface ice of the moon. This is anything but trivial as the ice stands at 16 to 24 kilometers thick. The deepest hole ever drilled on Earth was actually drilled all the way back in the twentieth century. The Kola Superdeep Borehole was drilled in Russia to a depth of around 12 kilometers. It was only 9 inches in diameter. For a mission on Europa, you needed a much bigger hole because of the second thing: you needed to put a submersible in the water to go looking for the life you think is there. And of course you need to pack all the equipment you need to do this onto a spaceship that can make the 650 million kilometer trip to our solar system's largest planet.

The nice thing about digging a hole on Europa is that it's just ice. You don't have to worry about running into unbreakable rock, or drill clogging clay, you have one material with well-known properties to worry about. As you've probably guessed, the problem with digging

the hole wasn't making a machine capable of doing so, it was packing enough energy aboard the ship to get the job done. When all the math was done for the newly dubbed Poseidon 1 mission to Europa, NASA knew there was very little margin of error for its equipment, but it decided to proceed, mostly due to public exuberance at the possibility of finding life in our solar system. NASA internally knew that a Poseidon 2 would be necessary, and so in 2073 when Poseidon 1 failed to reach its destination—as discussed last chapter, it missed its landing spot by 36 kilometers leading to the submersible lacking the energy needed to get to the volcanic vents—NASA was already armed with what it believed to be the solution—a launch site on Mars where its gravity, one third the strength of Earth's, wouldn't limit what the agency could launch into space. This pushed the launch date out for Poseidon 2 to the end of the century, after the projected completion of the Masters Mars Base named for the president who had overseen the start of the base's construction.

We've seen how the Masters Mars Base led to the establishment of the National Life Sciences Association (NLSA), a life sciences analog to NASA in the United States, and learned about how they were able to create a grow system which would keep martian astronauts fed. Let's take a look at the other major difficulties the base had to overcome namely, how to get enough water and oxygen, and how to protect the astronauts from the constant barrage of charged particles flung at them by the sun.

There's plenty of water on Mars itself, enough that if it were liquid and the surface of Mars were flat, the water would cover the entire surface of Mars at a depth of about 120 feet. The problem was that it was all locked up as ice at the planet's poles. NASA wanted to build the base near Mars' equator to maximize the solar energy it could draw upon. A trip from the poles to the

equator of Mars was around five thousand kilometers—about the distance from Los Angeles, California to Portland, Maine. The distance made for a potentially epic road trip on the carefully paved Earth, but would be a perilous journey over unknown terrain on the Red Planet.

If it was just water needed for plants and human consumption that was needed, NASA would likely have just packed the stuff with them in the rockets headed to Mars. Water held the key to the much more valuable commodity though, oxygen. The whole point of the Masters Mars Base was to provide a launch center for rockets headed out beyond the inner planets. Rockets burn a lot of fuel, and to burn a lot of anything, you need a lot of oxygen. There was a small amount of oxygen available in the scant martian atmosphere, but not nearly as much as was held in the water and CO₂ up at the poles. Splitting hydrogen from the oxygen in water was relatively as straight forward as running an electrical current through a vat of liquid water. So if you can get plenty of water, you can get plenty of oxygen. And for the CO₂, NLSA's grow chamber provided plenty of biomass to transform CO₂ into oxygen through plant respiration. All they had to do was mine the CO₂ at the pole and release it in the grow chamber. This CO₂ to oxygen pathway would become way more important than anyone at NASA could have predicted, but we'll need to wait a few decades for that.

Lance Muhalin was put in charge of project Mars Hydration (a name portmanteaued to MAHY, and then affectionately nicknamed Mahi-Mahi). Ahead of him and his team was the daunting task of designing an autonomous vehicle capable of mining ice and then transporting it three thousand miles across the harsh martian surface. The Mahi-Mahi team, under the enthusiastic direction of Muhalin, gained a reputation for increasingly dangerous testing scenarios for their rover. In one particularly well attended test, it drove over the surface of the

lava pool at Mount Kilauea in Hawaii's Volcanoes National Park. It also spent a year traversing the island of Iceland. They sent the rover to the moon to test its hardiness against the harsh regolith and low gravity conditions. When NASA was finally ready to send project Mahi-Mahi to Mars, Muhalin remarked, "Hopefully this will let us find some new planet around some distant star. Our sun's offerings have gotten a bit boring."

With the problems of water and oxygen solved, the only thing left was shielding astronauts from radiation. It was already known that the base was going to use far more power than the martian sun could provide with a reasonable array of solar panels. This meant that NASA would have to establish a nuclear power plant at their Mars base. Thankfully this almost trivially solved the problem of radiation. On the Earth, we're shielded from cosmic radiation by the Earth's atmosphere and more importantly its magnetic field. If you have plenty of energy, you can just make your own magnetic field. Doing so around the planned Mars base was estimated to take about 25 kW, well within the energy output of a nuclear reactor.

There were, of course, plenty of other problems to solve, not the least of which was how to put everything together once you got it to Mars, but for the most part these were engineering problems, and not science problems. Figuring them out was just a matter of getting enough smart people to think about them for enough time. Missions to deliver base components to Mars began in 2080, and by 2090, an astonishing two years *ahead* of schedule, a permanent settlement was populated on another planet.

The agency wasted no time putting their long sought after martian base to good use, beginning a flurry of space activity unheard of at the beginning of the century. In ten years, the thirty satellites of the Einstein Space-based Gravitational-wave Observatory were deployed to

the eight planets, Pluto, and the Kuiper Belt. We've seen what ESGO was able to find in the beginning of the twenty-second century. Its success was heralded as a shining example of what science can accomplish when given the proper public backing. So in the beginning of the twenty-second century, when NASA went back for funding it decided to go for broke. Top of its list was a return trip to Europa to look for life, but, it argued, why stop there? There were other moons that might have life like Ganymede and Callisto around Jupiter or Saturn's Enceladus. NASA asked for funding to go to all of them, and it said that while we're at it, let's send a probe to Proxima Centauri B, the closest earth-like planet. At only four light years away, it was within the realm of possibility that a probe could get there and send back data within a human lifetime.

Much of what we knew about Jupiter's Europa, Callisto, and Ganymede comes from an ESA mission flown in the 2020s called the Jupiter Icy Moon Explorer (JUICE). Arriving in the Jovian system in 2032, the JUICE satellite did flybys of Europa, and Callisto, before finally settling into an orbit around Ganymede. Europa, we've seen, is covered in a thick layer of ice with a liquid ocean underneath. Ganymede and Callisto are actually rocky with sub-surface oceans, but what JUICE found was that both had cracks in their surface, allowing for venting from the subterranean oceans. NASA hoped to utilize these vents as access points to the underlying water. On Ganymede, the hope was that its tidal forces were sufficient to sustain some sort of life. Things were less hopeful for Callisto, a moon lacking interesting geological features like volcanoes and tectonics. Still, the presence of organic molecules on Callisto gave some hope that we might find some hint as to how life could form in the first place.

The real prize was Enceladus though. At only about 500 kilometers in diameter, the sixth largest moon of Saturn is little wider than the state of Arizona. In 2005, the Cassini probe flew by

the icy moon and discovered that plumes of vapor were being vented at the moon's south pole. These features, awesomely named cryovolcanoes, indicated that Enceladus was geologically active. Not only that, but it looked like Enceladus hosted a subsurface ocean beneath its icy exterior just like Europa. Liquid water plus vulcanism is a key recipe for life on Earth; scientists wondered if it might just be the recipe for life on a moon as well.

Poseidons 2 through 5 won fast approval, and work began towards their launches in the second decade of the twenty-second century. NASA was joined by their common partners ESA and CSA. Together the three agencies built specific drilling operations and submersibles for the challenges of exploring the four moons. Astrobiologists scrambled to make predictions about what we might find. The data from the Frank Drake Space Telescope, and its successor the much more powerful Carl Sagan Space Telescope made it seem more likely than not that we would find *something* interesting in those celestial oceans, if not life itself.

While the Poseidon missions were making progress, a parallel group at NASA got down to the business of planning our first voyage to another star. The effort was deemed larger than any one nation, and so NASA called upon all the space agencies of the world to join them in collaboration. ESA and CSA were on board as usual, but in a sense of collaboration on the big things that was still being figured out in the early twenty-second century, they were joined by 34 of the other 48 space agencies around the world.

When it comes to sending a spacecraft to another star, the main issue you have is that space is really really big. Proxima Centauri is only 4 light years away, but 4 light years is an almost unfathomable 38 *billion* kilometers. At the 24,000 kmph that our best nuclear spacecraft flew at it would take 183 years for our probe to get there. The hope of the agencies was that the

probe could get there *and* send back data (remember that data would travel at the speed of light so it would take four years to get that data) inside 100 years, roughly the time of a human lifetime.

The agencies ran the math on a number of scenarios, and after a while settled on one that they felt made the best sense. Just like a fighter jet that takes on more fuel during a mission from a larger, and slower aircraft, they could launch extra fuel ahead of the probe, which could then add it to its supplies while in flight. The only problem was that taking on the extra fuel would require extremely risky maneuvers done by computers in space too far away for humans to intervene if anything went awry separated not only by many miles, but many years. Of course this is just the type of problem that space agency engineers live for.

It wouldn't be until 2116 when the probe finally launched. It had been dubbed the Worldwide Cosmically Interplanetary Probe or WCIP, a not at all veiled nod to another phrase with the same acronym: We Come In Peace. If everything went according to plan, the additional acceleration provided by the in-flight refuel would allow it to reach Proxima Centauri B in 62 years, fast enough to let some of the younger scientists who worked on it hear what it had to say.

Two years later, after nearly 14 months of drilling on the icy moon of Enceladus, Poseidon 5's submersible sent back images of a nearby volcanic feature it had found deep beneath the moon's icy surface. In the image, grainy but clear enough to be verifiable, were the unmistakable outline of alien lifeforms moving around the vent. The discovery was shocking, even to the most ardent believers in extraterrestrial life, but even more than that, the harsh environment of Enceladus, so far from the warmth of our own home star, indicated that life could not be as fragile as we thought. The ramifications of these space entities can not be overstated.

Socioeconomic 2

Compared to the geological processes which created the oil in the first place, the socioeconomic changes of the worldwide collapse in fossil fuel production happened in the blink of an eye. It was an interesting case of convergent forces driving down demand, as supply began to dwindle. The resulting shift in the economic world order was muted compared to projections made at the beginning of the twenty-first century through a series of seemingly lucky circumstances that helped the world change to a post-fossil fuel future. To understand the twenty-second century, it's important to see how the latter half of the twenty-first century came about, so let's dive in.

Global fossil fuel demand had increased throughout the first half of the twenty-first century, but not everywhere all at once. In fact demand had been dropping through much of the wealthier parts of the world with the advent of electric vehicles. Parts of Europe and the US had even gone so far as to ban the production of new fossil fuel cars. Demand in the developing world had increased at enough of a rate to make overall demand continue to increase up to the first half of the 2060s, but when oil production peaked in 2062 it became clear that more had to be done to bring electric vehicles to the developing world. Replacing the world's fleet of gas-powered cars with battery-powered ones required a large upgrade to the world's energy infrastructure. Nuclear power provided a convenient upgrade for the developed world, but was hardly a solution for parts of the world lacking in infrastructure. As mentioned before, when you want a lot of energy there're only two games in town: nuclear, and solar. So a massive upgrade to solar power was in order. Luckily the world was up to the task.

Keeya Magosi was born in Francistown, Botswana in 2030. At the time, Botswana's energy grid was split between domestic energy production from fossil fuels (mostly coal), and imported electricity from neighboring South Africa and Zambia. Throughout the late 2030s, and 2040s, Botswana's grid suffered from consistent blackouts as it tried to keep up with the inclement weather brought on by climate change. By the time Magosi headed to university at the Oregon Institute of Technology in Klamath Falls, Oregon, pressure was mounting on the Botswanan government to revise the grid and provide relief. There she met a professor who introduced her to a growing body of research into micro-grids.

Ever since 1882 when Edison turned on the generators at the first power plant in New York City, humans relegated energy production to a series of large scale power plants that provided electricity to large amounts of customers. In the early twenty-first century, researchers started bandying about the idea of smaller electrical grids. These grids, dubbed micro-grids, could handle energy production for tens or hundreds of consumers, and be wired together to behave like a more large scale grid. By the mid twenty-first century, micro-grids were pretty well understood, but had not found any large scale deployments, mostly due to the fact that fossil fuel power plants were ubiquitous and cheap.

Most forms of energy production lend themselves to the large scale power plant model. Especially fuel-based power production like coal and nuclear power plants. Solar on the other hand is ideal for micro-grid production, but it had the problem that it didn't always produce power. Electrical grids need redundancy to ensure the smooth production of power, and that was a challenge for micro-grids trying to provide power to a small number of consumers. Increases in battery capacity over the years gave micro-grids a rosier picture, and researchers theorized that a

sufficiently intelligent (as in computer intelligence) deployment of electric vehicles could provide the backup power needed to support micro-grids—the vehicles providing the battery storage while they sat idle in driveways and charging stations. Magosi read the relevant literature while studying in the states, and left with a masters in renewable energy back to Botswana in 2054. There she got to laying the theoretical groundwork down for a deployment of micro-grids throughout Botswana, but in order to bring her dreams to life she had to figure out how to replace Botswana's fossil fuel burning cars with electric vehicles. Replacing them wholesale would cost far too much for the average Botswanan so she looked elsewhere for answers.

The communist revolution in Cuba had a whole slew of long lasting ramifications on the island nation, many more than we can enumerate here. The one of interest for us now is that automobile imports to Cuba slowed to a crawl. As a result, Cubans had been maintaining and updating the same fleet of 1950s and 60s automobiles for nearly a hundred years. Some time in the late 2040s, a group of enterprising Cubans got their hands on a supply of electric vehicle batteries, and had worked out how to retrofit their internal combustion engines with electric motors. Magosi heard about the electric vehicle upgrades and headed for a visit in 2055 sponsored by the Botswanan government. There she convinced the Cuban government to similarly sponsor a group of mechanics to head to Botswana and develop processes for the Botswanans to upgrade their existing gas-powered fleet to battery-powered vehicles.

Magosi's first dozen micro-grids launched in 2058 alongside retrofitted electric vehicles. The smart grid built to handle the electrical production between the twelve grids worked like a charm charging the vehicles during peak times and using the batteries during off-peak. The micro-grids worked so well that the Botswanan government commissioned a five fold increase

for the next decade. The deployment grew, and became a model for micro-grid deployment, which still may not have gone anywhere given the low cost of fossil-fuels if it weren't for the peak oil year of 2061 and subsequent decline in 2062. As governments around the world looked for answers to a falling fossil-fuel reality, Botswana's implementation of solar-based micro-grids rose quickly to the top. By 2072, the UN was creating targets for electric vehicles, whether retrofitted in developing nations, or replacing gas-powered cars in wealthier countries. This led to a huge increase in demand for solar panels, which helped create opportunities for countries willing to capitalize and build the necessary infrastructure.

The transition to solar-based micro-grids, and RIVER project mining operations, provided enough economic growth to compensate for falling fossil-fuel reserves for much of the world. Unfortunately it didn't leave much for the countries that populated the northern latitudes. Canada and Scandinavia's sufficiently advanced economies allowed their falling fossil-fuel reserves to have little effect on their populace, but Russia was another story. Since 2045, Russia had been the number one producer of oil. Russia was all in on a fossil fuel-driven economy so that in the decade after 2062 when worldwide demand dropped out, things looked dire for the already hurting Russian economy. It had been fifty years since Russia's last territorial war, but people began to worry that its failing economy might lead it down a bellicose road once again. A way out came in an unforeseen, but fortuitous way.

In the early 2020s, the United Arab Emirates, a country flushed with cash from oil production, but lacking in arable land, built the largest vertical farm the world had seen. Vertical farms at the time, were trays of plants grown hydroponically in a large stack. The output was such that the UAE's vertical farm could output 2 million pounds of leafy greens in one seventh

of the area it would take to grow them in land. The vertical farm was the perfect energy sink for the oil rich nation, and operated to great success in the mid 2020s. So much so that it spurred the building of additional larger vertical farms. By the 2050s, the UAE was growing 100 million pounds of food in vertical farm facilities. Then something interesting happened.

Vegistax, an Iranian company started in the 2030s to capitalize on vertical farming in desert conditions had built a ten million pound farm, almost twice the size of the next largest installation, in the sands outside of Dubai, but in 2053, their first year of operation, they found that the farm produced only six and half million pounds of produce. Vegistax executives were left scratching their head at the discrepancy. They outsourced their problem to researchers at the China Agricultural University. There a pair of researchers set about figuring out why the plants at Vegistax's farm didn't grow.

Researchers Wu Junfeng and Chang Xinfei came back in half the time allotted for the study and said the data suggested something radical. The plants' growth curve was consistent with plants grown in a low CO₂ environment. Their math on atmospheric CO₂ levels and plant respiration backed up their findings. There were simply too many plants trying to grow in one volume of space.

Faced with an enormous capital expenditure, Vegistax took the information and resolved to find a way to make their vertical farm grow. The scientists' data suggested that Vegistax could recover their growth if they could somehow add an additional 300 metric tons of CO₂ or so back into their plants' atmosphere. No problem right? Just head call up a dry ice distributor and ask for 300 metric tons. Many many more tons of dry ice were produced every year. In a rare case of

business foresight, Vegistax decided on an alternative that would change the landscape of our world.

Carbon sequestering had long been proposed as a possible solution to our warming planet. The levels of CO₂ in our atmosphere were the main cause of the warming of climate change so, people reasoned, removing that CO₂ from the atmosphere was likely our best bet at reducing that warming effect. There were really only two ways to get that CO₂ out of the atmosphere. The first was with plants, those great allies of ours are happy to just sit around taking in carbon dioxide from the air and turning it into all the great plant things of life. The other was for us to go and get it ourselves through sequestration, a process which until then just didn't make business sense. CO₂, for all the havoc it can wrought on our climate, is pretty diffuse in our atmosphere. One cubic meter of air at sea level only contains about 0.75 grams of CO₂ so if you want to get a meaningful amount of the stuff, you have to filter a lot of air. At \$5 a kilogram, there just wasn't an economical way to sequester. Not to mention there's the problem of what to do with the gas once we got it. You can't just let dry ice evaporate back into the air because then you're right back where you started.

Corporate executives aren't known for their benevolence, and so it's unclear what exactly was the mixture of self-interested motivations for Vegistax' executives, but rather than buy one and a half million dollars worth of dry ice, and be done with it, they went to their home Iranian government with a dramatic proposal. As vertical farming grew in importance around the world, they argued, the procurement of CO₂ for farming purposes would stress the CO₂ market making the warming gas a sought after commodity. It would be in the best interest of Iran to strategically fund measures designed to procure CO₂ through sequestration. The executives presented some

convincing figures about the rise of popularity in vertical farming, and what it would mean for CO₂ production. They even threw in some speculation about what plant production around RIVER projects could look like with certain advances in biochemical processes. Little did they know that they weren't far off on what the future held.

The Iranian government took the proposal to heart, and in 2056 launched the first airborne carbon sequestration fleet. To filter enough air to capture a meaningful amount of CO₂, the aircraft had to be low-flying and slow moving so, much to the delight of steam punk aficionados everywhere, the fleet consisted of 32 zeppelins made from flexible solar panels. Powered by the sun, and floating due to helium, the blimps could stay aloft almost perpetually. They would sequester CO₂ in large canisters which were then dropped near roadways for supplemental collection. Each zeppelin could capture about a metric ton of CO₂ a day, not enough to put a dent into climate change, but enough to provide Vegistax and its brethren with cheap CO₂.

Iran's allies took notice of this CO₂ project, and, encouraged by the possibility of a natural resource to supplement falling oil production, began their own sequestration projects. Most notably Russia and China, both countries with huge tracks of unused land to sequester over, designed their own fleets. They also took advantage of their shores and started building ships capable of sequestering CO₂. Other countries took notice, and the sequestration effort was underway. By 2065, worldwide sequestration had grown to a million tons a year. This was a pittance when compared to the 17 billion tons of CO₂ emitted by humans into the atmosphere every year, but researchers were starting to take notice and make predictions about what could happen if global CO₂ demand increased further.

From 2065 to 2085, vertical farming grew both in popularity and out of necessity. The warming planet was making arable land more scarce, especially between the tropics, and the only way to feed the populations who still lived there was to go vertical. Still, CO₂ sequestration was outpacing demand from just farming operations. The effort, hailed by many environmental activists, was at risk of being abandoned, or at least sidelined in favor of other potential resources, when BLOWARP announced its earth changing Mahd Arrays. The CO₂ required by the fast growing Wolfia plant that powered the arrays meant that the bet countries had made on CO₂ sequestration was about to pay off. It also meant that every country sequestering CO₂ had a vested interest in finding what the BLOWARP team had termed "...similar inquiries may be made for the entire panoply of productive plant components."

What started out as one company's attempt to get cheap carbon dioxide for itself turned into a worldwide effort to supply bioengineered plants with enough CO₂ to produce the whole of wonders that plants create. The late twenty-first century was an incredible time for biological research participated in by more than half of the countries on earth. In that time nearly 500 million acres of plant matter was planted in the stacked rows of Mahd Arrays, the equivalent of almost 50 years worth of human made deforestation.

This drastic increase in plant life in relatively small spaces led to a carbon dioxide need of 600 million metric tons of CO₂. Part of this was met by a dramatic increase in the sequestration efforts of many countries, but growing fleets of zeppelins and ships weren't quite enough to meet this demand. Instead humanity looked to what was once one of the largest culprits of carbon dioxide exhaust.

For much of their existence, cars had sat empty taking up space in cities and suburbs. When in use they spewed CO₂ and other nasty particulates into the air as they burned fossil-fuels in their internal combustion engines. By 2100, however, sleek and efficient electric vehicles had replaced the gas guzzling car. Autonomous driving, while never able to get all the way to a complete driving solution, opened up the possibility of cars doing something more productive with their idle time than just sitting around. The smaller footprint of the electric drive train left plenty of room for carbon sequestering infrastructure. And solar powered micro-grids provided an essentially limitless supply of charging stations.

The car-based sequestering revolution started in China where state-mandated vehicle changes were common and not questioned. Adoption happened slower in the rest of the world, but by 2110, two thirds of the world's countries had adopted some level of autonomous car-based sequestration. The additional 100 million metric tons of CO₂ captured each year from autonomous cars supplemented the overall market, and somewhat started to make up for all the carbon that cars had put into the atmosphere in the first place.

All of these changes started adding up. We may not have broken through the dome of CO₂ in the atmosphere, but we were putting a good dent in it. So much had the balance of atmospheric make up changed that scientists began to actually predict that within people's lifetimes they might actually see the point where humans took out of the atmosphere more CO₂ than they put in. All of these things, the dramatic increase in plant biomass, the decrease in CO₂ emissions with electric vehicles and micro-grids, CO₂ sequestration, and efficiencies from vertical agriculture, were all adding up to give people hope that in the future, human beings

might right the wrong of climate change. It was giving the world a sense of collaboration it had not felt before, and that would have a very interesting effect on the twenty-second century.

Dark Field 3

The finding that the supermassive black hole at our galactic center was producing dark matter had astrophysicists scratching their heads. It had been known since the late twentieth century that black holes aren't necessarily black. They radiate energy in a peculiar way known as Hawking radiation. Quantum mechanics tells us that empty space isn't really empty. Instead it is awash in virtual particles—a regular particle and its anti-particle can blink into existence briefly so long as they meet back together and annihilate each other. This curious behavior, a result of Heisenberg's Uncertainty Principle, which we'll talk more about in a moment, allows for virtual partners to pop into existence near the event horizon of a black hole, the point at which no light or light matter can escape the gravitational pull of the black hole. One of the pair can fall into the black hole and be lost to its gravitational crunch while the other pair can travel off into the universe as a real particle. In this way black holes give off Hawking radiation. The astrophysicists wondered if some similar mechanism could be causing the dark matter so they turned to their quantum compatriots for answers.

To understand what quantum theorists came up with, we'll need an understanding of quantum theories of the twenty-second century, and to understand those we might as well go all the way back to the beginning of quantum mechanics in the twentieth century. It's a story that has been written about many times, but hopefully this brief treatment serves as a good refresher.

In the early 1900s, physicists were puzzling about the Hydrogen atom. Given the equations of classical mechanics, that is of Newton, the electron's orbit of the nucleus should lose energy and decay leaving the electron to fall into the nucleus. This is not what happens, and

theorists were trying to figure out why. In 1913, a 28 year old Danish physicist named Niels Bohr suggested an answer. He delivered the mathematical underpinnings of a model for the Hydrogen atom that behaved like what was observed. His work included three assumptions, the first of which was that electrons could only occupy certain specific radii with certain specific energies.

Thirteen years earlier, a German physicist named Max Planck had introduced the term “quantum” while solving a problem called the Ultraviolet Catastrophe. Towards the end of the nineteenth century, it was known that the radiation given off by a black body (this is the generic term for any body of unknown makeup that radiates heat) was predicted to be unbounded given what physicists knew at the time. The sun gives off a lot of energy, but it’s clearly not unbounded. This discrepancy between theory and experience was resolved by Planck who introduced the assumption that light could only be emitted at certain intervals, called quanta. It was a strange assumption to be sure, but it explained the ultraviolet catastrophe, which is a neat feature of physics—your theory can be as weird as you like so long as it explains and predicts phenomena accurately.

In 1926, an Austrian-Irish physicist named Erwing Schrödinger put these results, along with work from other leading scientists, including Einstein, together into a formulation called the Schrödinger Equation, which described the wave mechanics and time evolution of quantum systems. Schrödinger himself is a problematic figure, recording numerous sexual dalliances in his journals, including some with underage women. The Schrödinger equation was one of the most powerful mathematical generalizations of varied phenomena (some very strange) ever to be recorded, giving rise to a whole branch of physics.

One of the many things this mathematical formulation explained was the wave-particle duality of light and matter. A debate that stretched all the way back to Newton's work on optics was whether light was a wave or particle. With quantum mechanics, it was found that not just light, but all matter could be described as both waves and particles. This curious duality was the subject of many inquiries, the most famous of which is the dual-slit experiment. Imagine a barrier with two narrow slits cut into it, a detecting screen behind them, and a laser pointed at the two slits. The light waves will pass through both slits in such a way that they interfere with each other. This will leave a tell-tale interference pattern on the detecting screen. Now think of the same experiment, but instead of a laser, we use an electron gun to shoot electrons at the slits. If electrons were just particles, we'd expect them to hit the screen evenly and just form a uniform smear across it. What we find instead is that same tell-tale interference pattern. Even more astonishingly, if we shoot electrons one at a time we still find the interference pattern. Every individual particle is acting like a wave when it goes through the slits. It turns out we can do this with any piece of matter we can make a sufficiently narrow slit for. Everything can be both a particle and a wave. The wavelength given for matter particles is called the De Broglie wavelength, the same wavelength that Rogers used two Dark Field chapters ago.

Another strange prediction of quantum mechanics was that two or more particles could become entangled. One of the simplest entanglement scenarios was to have two particles with opposite spins. Spin is a measure of the angular momentum of a particle, and due to quantum mechanics some particles can only have one of two spins which we call up and down. When two of these particles become entangled, and we measure one as having spin up, the other is guaranteed to be spin down. So what? you might say, but the thing is that neither of those

particles will have known spin before we measure them. It seems as though the first measured particle sends some signal to the second, but that signal would have to travel faster than the speed of light! Einstein, who took a while to warm up to quantum mechanics, called this “spooky action at a distance.” It is this entanglement property of quantum states that give us the qubits that power the quantum computers we use today.

By the time the twenty-first century rolled around, most attention on quantum mechanics had turned to work with the Standard Model. Formulated throughout the mid-twentieth century into the 1970s, the Standard Model was critical in predicting and explaining the particle zoo that was being found in particle accelerators of the time. The culmination of the Standard Model was the 2012 discovery of the Higgs Boson that our first chapter started off celebrating. After that, the pendulum swung back to astrophysics with the launch of the JWST and the Extremely Large Telescope (ELT) in Chile. But with Diduch’s introduction of Dark Time, interest in these great quantum mechanical equations picked up in earnest.

Adding a new time dimension, Dark Time, to the Schrödinger equation was simple enough. The equation equates the time evolution of a particle system to the overall energy, a term called the hamiltonian, of that system. Dark Energy as a term had already been appropriated by the astrophysicists to describe the energy causing the expansion of the universe, so in 2066, a Japanese theorist named Itsuke Sakurai coined the term Yami Energy from the Japanese word for dark in an illegal or shady sense, a reference to how energy on the dark side of things would have to steal from the light side of things in order to work. Adding yami energy to the hamiltonian side of the Schrödinger equation worked so long as two things happened. First all the particles of the system could only be light *or* dark (as in there were no light *and* dark particles), particles

couldn't live on both sides of the spatial boundary. Second, every particle had to be able to transition between light and dark and vice versa. By 2066, physicists knew better than to doubt a result of quantum mechanics. These two conditions had to mean something in the physical world, and quantum theorists set out to find what that was.

We'll come back to the quantum mechanics side of the story in a bit, but for now we need to take a trip into the other side of theoretical physics, and that other crowning equation of the twentieth century: General Relativity. Whereas the equations of general relativity are harder than the Schrödinger equation, the concept that gravity curves space and time is more accessible since it deals with stuff we experience every day. Imagine a sheet stretched over a bed, and then place a bowling ball in the center of the bed. The sheet will curve around the bowling ball. Well it turns out that you can equate the curvature of the sheet to the mass of the bowling ball. A sheet is basically two dimensional, but you can do the same thing in four dimensional space-time, and that's exactly what Einstein's field equations for general relativity do. On one side of the equation is the curvature of space-time, and on the other side of the equation is everything in space-time.

Incorporating Dark Time into General Relativity was less straight forward than doing so for quantum mechanics, but still followed quickly after the 2062 paper introducing the world to Dark Time. In 2070, Liu Min proposed a formulation of five dimensional general relativity that took into account the quantum mechanical result that particles could only occupy either light or dark matter states. Liu was born in 2038 in Guiyang, China. A product of the Chinese education system, she became adept at navigating the bureaucracy of the Chinese Communist Party, a rarity

amongst Chinese scientists. Her maneuvering ensured ample funding for her research as she joined the staff at the China Academy of Sciences.

Liu's general relativity formulation was heralded by many for taking on the results of the quantum mechanical formulation of Dark-Space-Time. Remember that at the time, Quantum Mechanics and General Relativity have not been combined into a unified theory yet. The quantum mechanical results being included gave some the hope that through greater understanding of Dark-Space-Time the so-called Grand Unified Theory of Physics could be found.

In 2074, one of Liu's doctoral students, Shao Yuze, produced a surprising solution to her five dimensional equations. Orthogonality is a property of dimensions in differential geometry. It means that the inner product of quantities on the dimensions is equal to zero. We can visualize this property in three dimensions by noting that our three spatial dimensions are at right angles to each other. We further know that time as a dimension is orthogonal to our three spatial dimensions. What Shao found was that if the light time and dark time dimensions were orthogonal to each other then space would have infinitely expanded shortly after the birth of the universe! For us to exist at this late date to observe the universe how it is, light time and dark time had to be in some other configuration.

Two years later, Jason Edwin was a physicist looking for a doctoral thesis topic at Oxford University. He decided to try and find solutions to the five dimensional general relativity problem. His doctoral advisor encouraged him to publish his results a mere two years later. Edwin had the crucial insight that parallel time dimensions would cause space to collapse in on itself. Given that orthogonal time dimensions caused infinite expansion, and parallel time

dimensions caused collapse, Edwin searched for something in the middle. To try and resolve the angle between time dimensions, he took all the known light matter, and all the known dark matter, placed them in a universe the size that we observe the universe to be, and attempted to solve Liu's equations. He found that to make it work he needed a non-zero value for the cosmological constant, that term added by Einstein in General Relativity to make a static universe happen. The value that popped out astonishingly agreed with the observed value for dark energy within an order of magnitude. The physical interpretation Edwin gave was that light time and dark time are very near parallel, but that they slightly diverge. This slight divergence causes their boundary dimensions (our three spatial dimensions) to expand in order to keep up with the divergence. The discovery was championed as the first theoretical evidence that the Dark Time theory was correct, and Liu, Shao, and Edwin all shared the Nobel Prize in 2087 for this series of discoveries.

From this discovery in 2076 through the first decade of the twenty-second century, theoretical physics was filled with theories about Dark-Space-Time, but nothing really took hold. One theory that gained traction was called Six Symmetry, an attempt to add a third time dimension to make our dimensions symmetrical at three space dimensions and three time dimensions. Physicists love symmetries, and so the theory gained a not insignificant set of followers, but like String Theory before it (String Theory was popularized in the late twentieth century, and treated particles as vibrations on a one dimensional string), it failed to make any testable predictions. By the time the twenty-second century rolled around, physicists, and perhaps more importantly research funders, were starting to become frustrated with the slow pace of progress, so in 2112 when it was announced that the Einstein Space-based Gravitational-

wave Observatory had discovered Dark Matter increasing at the galactic center, it made huge waves throughout the physics community. That light matter could transition to dark matter and vice versa was one of the big predictions of the quantum theory of Dark Time, and to see it in action so dramatically was just the shot in the arm the physics community needed.

Since quantum mechanics had given us the prediction of light-to-dark transitions, and black holes are one of the triumphs of general relativity, theorists felt this would be a great opportunity for the two theories to work together. Let's start with the black hole. For all their mystique and wonder, black holes are relatively simple creatures. All we can really know about them is their mass, from which we derive most of their other properties like size and temperature, and their rotation speed. We know about Hawking Radiation, and how black holes give off heat. There isn't all that much to do at a black hole, you can orbit it like any body of mass so long as you're outside of the event horizon—the boundary from which light cannot escape the blackhole—or you can get pulled in by the immense gravity of the black hole and... we don't know. At the center of the black hole, all laws of physics break down so we can't say what happens there. It was at that center, a physical entity called a singularity, where theorists began their inquiry.

It was known that dark matter is collisionless, and it was believed that this property is what prevents dark matter from getting crushed in the singularity of the black hole, for if it did, black holes would all be much larger than we measure them as they ate up all the dark matter around them. Theorists hypothesized that some unknown process transitioned light matter trapped within the event horizon of a black hole to dark matter, and that that dark matter, accelerated by the gravitational pull of the singularity, accelerated right through the singularity at the center and passed back out of the black hole. There were only two problems with this

explanation. First was that in order for anything to leave a black hole it would have to be traveling faster than the speed of light! Remember, though, that faster than light travel was suggested by Diduch's original Dark Time paper, perhaps this was merely evidence of that hypothesis. Second, it's possible to add up all the light matter we think is falling into supermassive black holes at galactic centers to come up with an estimate of the mass that could be converted to dark matter. Astronomers made those calculations, and found that the mass of light matter was a quarter of the mass of the produced dark matter. Even at the cosmic level, this was a troubling discrepancy.

Perhaps the most famous equation of all time is Einstein's $E = mc^2$, the equation equating energy to the mass of an object times the speed of light squared. It's hard to really convey how sacrosanct those quantities, and thus this relation was when Einstein committed this equation to the body of physics. Both energy and mass were governed by conservation laws, which were merged into one conservation law now that we could see energy could be converted to mass and vice versa. The speed of light was held as a constant to all observers in the universe, and served as a cosmic speed limit where nothing could travel faster. Now in Dark-Space-Time, capital E was replaced by $\mathfrak{E}$ standing for Yami Energy, an unknown property analogous to light energy; mass was anything but settled with the quantities calculated and observed for light and dark matter at the galactic center, rumors began about a supposed Yami Mass; and the speed of light might be replaced by some greater quantity: the speed of dark. The long-held pillars of physical phenomena were being called into question.

Of all the questionable properties and figures being thrown around, none seemed to get physicists worked up as much as the problem of the disparity in mass between light matter into

the black hole, and dark matter out. Sixty years of Dark Physics theories had gotten physicists accustomed to the speed of dark, and the historical success of quantum mechanics made most amenable to the idea of Yami Energy. But no one up to that point had suggested anything might be awry with the mass of objects. In fact, mass seemed to be the only thing linking light matter to dark matter as both are affected by gravity alone of the four known forces. If dark matter had been found to have *less* mass than the light matter falling into the black hole, it would be easy enough to explain that some matter gets trapped, but for there to be *more* matter flew into the face of deeply held intuitions about the universe.

This mass problem plagued physicists for nearly forty years until a recalcitrant young physicist named Esperanza Sandoval made a radical proposal. One of the great benefits of the RIVER projects and subsequent success of Mahd Arrays was that it provided not just the need for science and engineering, but the means with which to pay for it. New universities, and universities near the projects grew in importance, and as countries tried to ensure that enough scientists and engineers were being produced to support these economic boons, scholarships and incentives went to a broad swath of people. One of the areas that felt this the most was the RIO Proyecto de Atacama (RPA), a joint project between Peru, Bolivia, and Chile through the coastal Atacama desert. At La Paz, one of Bolivia's two capitals, and the closest major city to the desert RIVER, the three governments established the Instituto de Ciencia y Tecnologia de las Tres Naciones. By 2150 the institution had established itself as one of the best universities for science and engineering on the continent, with a particularly good astrophysics program owing to its proximity to the many telescopes built in the Atacama desert. It was there that Sandoval had

gotten her doctorate, and was working towards making tenure as a professor of astrophysical theory.

As a young indigenous woman in Bolivia, Sondoal was taught from a young age that her plight would be behind that of men, and non-indigenous people. Sondoal rebelled against this relegation, and organized protests and sit-ins at her high school, and community. Her actions caught the eye of a local government official who, given a mandate from La Paz to find promising youth for the country's science and engineering programs, sought to redirect some of her energy into university work. Excelling in math and science, Sondoal looked at the opportunity to represent her people at the country's top university as one she had to take. The opportunity did little to slow down her community efforts, but the university did a passable job at adapting to Sondoal's actions. By the time she joined the staff at Tres Naciones (the colloquialism for Instituto de Ciencia y Tecnologia de las Tres Naciones), her and the university had reached a bit of a *détente*: the university made recruitment of indigenous peoples, especially women, a priority, and provided services for those that chose to attend.

Sondoal brought her rebellious nature to her work as an astrophysicist, challenging assumptions and long-held beliefs amongst the physics community. She was particularly fond of questioning what she considered the hubris of nineteenth and twentieth century white men thinking they had it all figured out. In 2153, Sondoal submitted a paper where she called out one particularly vaunted assumption that dated all the way back to the nineteenth century: the principle of equivalence of mass. The principle of equivalence held that a body's gravitational mass was equivalent to its inertial mass. It was a fair enough principle back when all we were concerned with was light matter, but, Sondoal argued, what if the principle of equivalence was

simply a special case that held for light matter, and was replaced by some other rule for dark matter. Doing so could help resolve the mass problem troubling physicists.

To back up her argument, Sondoal appealed to the Standard Model, the quantum mechanical theory that explained a truly astounding amount of properties of the light matter around us. The Standard Model predicts the Higgs boson will exist to provide particles with mass, but *all* the particles predicted by the Standard Model interact with the electronuclear field (the electronuclear field is the term given when the electromagnetic, weak nuclear, and strong nuclear forces are all combined as they are at very high energies). Could it be that dark matter, which does not interact with the electronuclear field, gains mass through some other mechanism? According to Sondoal, this was what was going on with the dark matter produced by our galaxy's central supermassive black hole.

Of course a theory is only good as the testable hypotheses it provides, and Sondoal's argument, while compelling, failed to give physicists any way of verifying its veracity. It would take another six years before the right interpretation would come along and jog us forward. This time it was a pair of women doing quantum mechanical research at King Saud University in Saudi Arabia. That Fatima Sadiqqi and Nadira Uddin were able to rise to prominence at the Saudi Arabian university is telling of the sociopolitical change felt by the historically strict and conservative muslim nation. Like other parts of the world, Saudi Arabia found itself with a hugely high-tech RIVER installation, and not enough people to staff it. This, along with strong social movements in the region, most notably for women's rights in Iran, caused the government to put more capital behind assisting women in STEAM fields.

Sadiqqi and Uddin were searching for additional ways to bring quantum mechanics into the theory of general relativity. Their inquiries led them to a very old problem, perhaps even the first great experimental conundrum of quantum mechanics, the dual-slit experiment. We've seen how particles and light both behave like waves and particles, and how a wave explanation is consistent with the behavior at the slits. Sadiqqi and Uddin asked what the gravitational effect of a particle wave would be, a question that had been bothering physicists since both theories had been produced. If you think of a wave stretching through space, how do you calculate its gravitational mass? You can pick a point and say the mass is concentrated there, but that would just be a particle and we know going through the slits that the particle is a wave. Do you average the mass along the whole wave? If that's the case how does that work for more massive bodies where we know we can treat their mass like a point at their center?

Sadiqqi and Uddin argued that this state, where the position of the particle is unknown, which is called a superposition, may actually violate the principle of equivalence. With light matter, a paradox is avoided because when one goes to *measure* a particle, the superposition collapses and the particle resolves as a particle. But maybe with dark matter, where particles can't be observed, the superposition state is the state matter is in all the time. Going further, they argued that when in that superpositioned state, the particle's gravitational mass behaved as though the particle existed at all points on the wave of the superposition. The calculations for determining what that mass might look like to an observer far away were complex, but Sadiqqi and Uddin found that it fell within the range of observation for dark matter creation at the galactic center.

Sadiqqi and Uddin gave the predictive theoretical underpinning that Sondoal's work needed to be considered seriously—particles stuck in a superpositioned state might violate the principle of equivalence. The only problem was that the force of gravity is really really small, and particles are really really tiny. There was no real hope of measuring the change in a particle's gravitational mass directly. To make measurements that confirmed their theory we either needed a way of getting a lot of particles to be in a superposition, or get a gravitational field strong enough to measure the effects of light matter falling into it. So basically a huge box of special state particles, or find yourself a black hole. If only there were enthusiastic scientists and engineers dedicated to those two things.

Biochemistry 3

After the development of the food growth chambers for the Mars base, NLSA turned its attention to the newly created Mahd Arrays, in particular helping to transition the world's animal part production to their vegetarian methods. The action reduced expansionary pressure for animal pasture precipitously, which saved, along with other agricultural shifts, nearly four million hectares of forest each year. As the market shifted to a preference for cultured meats, the size of pasture land began to shrink, allowing countries to take steps to re-forest their lost wooded areas. The situation created a unique opportunity where the jobs lost by the changing meat market were replaced by jobs to re-forest areas. By 2100, humans had more than doubled their tree planting efforts from one billion trees per year to two and half billion. The situation had environmentalists optimistic, so much so that in 2108, they declared that humanity had planted more trees than it had lost for the first time ever. Of course, we don't really know the planting habits of pre-historic man, but given our propensity for cutting down trees, they felt relatively assured by their assertion.

As the speed of discovery of new things to make in the Mahd Arrays wound down, and with a new administration taking over in Washington, NLSA was tasked with renewing their focus on preventing death. Not that nothing had been accomplished in that area since the agency's inception in 2073, but research in that area had taken a back seat to solving the food problems of another planet, and then doing the same for our own. In 2104 though, the agency received 55 billion dollars explicitly for research that would prolong human life, and eradicate

human disease. NLSA heard the mandate loud and clear, and reorganized to facilitate the new direction.

Let's start with the effort to eradicate disease. The success of Medemos' virex machine, along with beneficial licensing opportunities for researchers around the world, grew the number of viruses curable through VIREX treatments from four to 108 by the early 2100s. All of the viruses with fatality rates over 1% were curable, and for most of the world, viral infection had ceased to be of much concern. This spawned an effort to expand the reach of virex machines to the hardest to reach populations around the globe. An organization called The Finch Society, grown out of an effort to bring VIREX treatments to rural Central Asia, rose to prominence in the NGO world. Richly funded by billionaires swept up in the deviling decades of the mid twenty-first century, finches, as members of The Finch Society came to be known, grew a worldwide reach through efforts to eradicate disease. Their name was a reference to Darwin's *Origin of the Species*, and the finches he studied on the Galapagos Islands that helped him develop his theory of evolution. The more ardent finches even went so far as to trace their organization's lineage back to the bearded naturalist. Their name was lived up to with a fervent belief in science, and scientific reasoning which they brought with them in their charitable efforts. We'll leave them here for now, but they'll come back to our story in due time.

With viruses under control, the effort to eradicate disease branched out to address the other causes of disease. The other pathogen alongside viruses, of course, was bacteria. For much of the world, much of the bacteria that causes disease was treatable by antibiotics, but even the most developed parts of the world were ravaged with antibiotic-resistant bacteria. Other disease vectors included neuro-degenerative diseases like ahlzeimers and ALS, auto-immune diseases

like lupus and Multiple Sclerosis, mental health disorders like bipolar and schizophrenia, and of course the 800 million pound gorilla in the room: cancer.

Fighting back against the myriad of maladies which afflicted the human body was not a task for one organization, no matter how well funded NLSA was. Instead the agency provided a series of grants called Disease and Ailment Reduction Target grants or DARTs. The idea was to throw a lot of DARTs at all the diseases that afflicted human beings so that the agency could focus on cancer, the group of maladies it pulled under its purview.

Cancer is to different cancers, like car is to different models of car, an umbrella term for a huge number of diseases which are cancer-like. In general the things that cancers have in common are cells that a) multiply more than healthy cells, b) live longer (sometimes forever) than normal cells, and c) don't stop growing like normal cells (this is the property known as *malignancy* and is the key feature connecting all cancers). This kind of bad cell behavior could happen anywhere there are cells, and that made every individual cancer a different set of problems given the type of cell it started from. Many many people had been involved in fighting cancer throughout almost all of recorded human history. In 2104 when NLSA threw down its gauntlet at "the emperor of all maladies", the ten year survival rate for Cancer in the US was right around 77%, a significant improvement from a century before, but due to the sheer rate of Cancer in the population, cancer was still lethal enough to make it second only to heart disease amongst the causes of death.

Remember that with PICKR we had the ability to target specific cells and rewrite their genetic code. This gave us a powerful tool in the fight against cancer, and was a large part of how the ten year survival rate got to 77%. The key had become finding cancers early. The twenty-first

century had been a masterful piece of progress in this regard. By 2025, a bevy of private companies had developed blood draw tests for different types of cancer. They all worked mostly by looking for what is called circulating tumor DNA (ctDNA). All cells, cancer cells included, shed some of their DNA into the blood stream, and using a combination of precision instrumentation and machine learning, computers were able to find trace amounts of ctDNA that could lead to early diagnosis. Of course some cancers, like brain and ovarian cancer, remained difficult to detect in this way, but these tests were an incredible improvement over prior diagnostic efforts. Throughout the middle parts of the twenty-first century, efforts were made to consolidate these tests, an effort resisted by the private companies, but pushed for by the governments on the hook for paying for the tests. By the end of the twenty-first century, 86% of all cancers could be tested for with just three tests, all of which were part of routine blood draws done at semi-annual checkups. The problems to solve for NLSA in 2104 were detecting the other 19%, and getting people to actually go to their checkups.

Robert Kipp led a team of computational biologists working on the cancer problem at NLSA. In a proposal him and his team wrote for funding to solve the detection problem, they argued that though we hadn't solved for them all, each cancer had a blood-born signature, it was just a matter of finding it. Finding it, they went on, was not a task for human observers, but a big data problem for machines. The proposal believed that detection of these signatures was possible given current technology, and all that was left was to run the data alongside data collection from as many cancer screening blood draws as possible. This kind of data collection would have been impossible in the United States a century earlier, but one of the provisions of the Human Equal Assistance and Life Time Health (HEALTH) Act, which brought universal health care to the

United States, was to allow for the anonymized collection and sharing of medical data for research purposes.

The first trial took six years, but was able to find markers in the blood for an additional 4% of cancers, a tremendous success. The study's structure, looking at data from commercial blood chemistry analyzers, and matching it to patient data secured over a period of time meant that continuing the study as analyzers improved would lead to more and more cancers being diagnosed. Kipp argued for, and secured funding for the next 30 years, allowing for five more studies. The gamble paid off, and by the time 2140 rolled around, early detection was possible in 96% of cancer cases, some times as many as five years ahead of when it might otherwise be detected. While certainly not 100%, the early detection was enough to bring the ten year survival rate up to an incredible 93%.

Of course all of this early detection is for naught if people aren't going and getting their blood drawn to get tested. Due to the incredible cost difference between early cancer treatment and late cancer treatment, governments had a vested interest in finding and treating cancers early (much was pontificated about whether this would have been the case with privatized health care, but with the entirety of the developed world having government-sponsored healthcare, the debate was purely academic). There were marketing campaigns, and incentive programs all around the world advising people to get tested on a regular basis. The push for testing actually pushed the numbers of phlebotomists (people trained to draw blood) to their limit. To compensate, several countries provided free phlebotomy training with wage guarantees high enough to ensure a steady stream of students.

In 2137, a team of fourteen economists from a partnership between the University of Chicago, Copenhagen Business School, and the London School of Economics, completed a years long study analyzing the healthcare cost savings as a function of government spending on the training for phlebotomists. The study used a rather liberal definition of costs, going so far as to tally the time lost from waiting in blood draw queues at understaffed locations. The study also took into account the cost associated with the deaths and loss of productivity from those who would die of late detected cancers. The study concluded that each additional phlebotomist trained saved their country 4.7 million dollars *per year*. This enormous figure garnered a lot of media coverage, and ultimately led to some interesting changes, but to see why we need to look at NLSA's other major effort at the beginning of the twenty-second century.

In 2022, Katherine Dinh and Qixuan Wang of the University of California Riverside, discovered that a protein called transforming growth factor beta (TGF- β) regulated cell renewal and cell death in hair follicles. Hair follicles have an interesting role in human organs as they're the only organ that regenerates automatically without injury, and the only organ that kills itself (baldness). Interestingly, when hair follicles do kill themselves, they never kill their stem cell reserve. It was this critical feature that led two scientists at The University of Pennsylvania to develop a cure for baldness sixteen years later. The cure, which involved using TGF- β to trigger regeneration of hair follicles after a PICKR treatment which removed the genes that caused the baldness. Their technique would be the first time that biologists were able to trigger regeneration in otherwise dead cells, and brought an already pondered question to the forefront, why does anything have to die?

Aging is a complex process, but in the first half of the twenty-first century, scientists were starting to get a handle on it. The longer cells live, the longer they have for something to go wrong. Mutations can occur which lead to cancers or just poorly functioning cells. Glycation can occur where glucose binds to proteins and lipids blocking cell function. Telomeres, which are strands of DNA that protect DNA from mutations during cell division, shorten and cause cells to stop dividing. Various biological systems such as the endocrine system that regulates hormones, and the immune system, seemed designed to age and die over time. Even individual genes were found to turn off with time. All of these, and a host of other issues needed to be solved in order to prevent death.

As with disease, NLSA created a grant pool for enterprising researchers to draw upon. They called these grants LIFELINES (no clever backronym ever caught on for these), and they funded research throughout the twenty-second century. Many of the most important LIFELINE projects are chronicled in the excellent history *The Apples of Idun: Stories from the Search for Eternal Life* by Marta Hingel, I could never do them justice here. Instead I'd like to tell the story of one of the more clever uses of those LIFELINE funds.

A group of researchers at UCLA, under the leadership of Erin Trestle, taking up the work started by their UC Riverside colleagues twenty years earlier, asked whether a similar mechanism to TGF- β 's cell renewal capabilities could be used to reset skin stem cells back to an earlier state when they produced the collagen and elastin of youth. Doing so, they argued in their LIFELINE grant proposal, would ameliorate the most visible signs of aging, and would be a shining example of what life extending research could do. The grant proposal board at NLSA saw the potential in this line of research, and bet that if headway could be made against the aging

of the skin, it would go a long way to ensuring funding continued on their quest to prevent death. Trestle and team received a multi-million dollar grant from NLSA in 2040 to pursue their anti-aging skin treatment.

Trestle was a charismatic biochemist from Long Beach, California. Her undergraduate took her to the east coast where she studied at Cornell before getting her doctorate at MIT. Family, and a childhood love of the movie industry brought her back to LA where she established a lab at UCLA in 2032. She was perhaps most well known for her role as science advisor to the movies *CRISPR* and *Love In The Time of Cloning*, movies about the potential of biochemistry to change life in interesting ways. Her easy-going demeanor and ability to explain complex science topics to non-scientists made her a regular on entertainment shows covering those and other science-related films. Her minor celebrity status, and prominent lab in tinseltown, made her the perfect lead for going after aging skin, one of hollywood's greatest adversaries.

Skin might seem like a trivial place to start for the fight against aging, and the goal of preventing death, but it's anything but. Skin is an incredible organ. It protects against microbial invasion, while allowing for substances like oils and sweat to leave the body. It regenerates itself with incredible efficiency after cuts, scrapes, and burns. It has the elasticity necessary for all of our joints to move unimpeded. It houses the majority of our sensory nerves, providing protection from hot and cold. It takes a beating from the sun's UV rays, and provides feedback when there's been too much exposure. And, unlike all of your other aging organs which it hides beneath itself, it shows your age. Cosmetic treatments like botox had been developed to keep skin from forming wrinkles, but the underlying cause was still there: as you age, your skin stem cells stop producing

collagen and elastin, the key components for youthful skin. Trestle and her team believed if they could reset the clock on those aging skin stem cells, it would take us a long ways towards rejuvenating all the body's stem cells, thereby arresting the aging process.

The LIFELINE grant turned into two, three, and then four grants spanning over two decades. Of course progress was made during this time, and the UCLA team published an incredible 116 papers, nearly five per year, on what they found on the aging process. What had begun as a hopeful foray against aging had turned into Trestle's life work. Finally, in 2064, Trestle and team published their pivotal work, a paper entitled *Epidermal Stem Cell Rejuvenation*, which outlined a novel process for restoring multi-potent somatic stem cells (in adults, stem cells are only pluripotent to a degree. Mesenchymal stem cells for example can differentiate into muscle, cartilage, bone, or fat cells but not nerve or skin cells. This leads us to use the term "multi-potent" with adult or "somatic" stem cells). The process was highly technical, and doesn't lend itself well to explanation in a popular science novel such as this, but it addressed multiple avenues to cell aging such as hormonal, external mutations, telomere shortening, and turning off various cell regulation inhibitors used by the body to protect against cell deterioration and adverse mutation. The process worked wonders in mice, but had a significant drawback. Nearly all the aging mechanisms that the process turned off had been evolved to help prevent against cancer. Without those defense mechanisms, the probability of cancer in the tested on mice skyrocketed.

Over the next three decades, similar processes were found for stem cell rejuvenation in the other multi-potent somatic stem cells of humans. The processes failed to breakthrough healthcare regulatory bodies throughout most of the world due to the higher incidence of cancer.

The work was refined, but largely set aside by the biochemist community until 2120 when NLSA announced they had reached the 89% level for cancer detection. That led to the FDA in the US, the European Medicines Agency (EMA) in the EU, and agencies across Asia to tentatively approve skin stem cell treatments, with plans to approve additional treatments should the rollout go well.

The treatments collectively came to be known as Rejuvenations, and when they finally hit the market, they were incredibly successful. Apparently the increased threat of cancer did not hold much of a candle to the larger threat of aging. When Trestle's team's process for epidermal stem cells finally hit the market in 2145, just over eighty years after it had been discovered, demand was so high for the procedure it shocked the healthcare infrastructure of several countries causing governments to have to intervene. It would take almost a generation's worth of education to produce enough healthcare professionals to meet demand for rejuvenation centers. By that time in 2168 some 52% of the US population had undergone a rejuvenation treatment. About half of the ones who hadn't were too young at under 25, and the other half were the staunchest holdouts of the anti-science community.

Kiko Hana was born in Osaka, Japan in 2040. When the Japanese Pharmaceuticals and Medical Devices Agency (PMDA) approved skin stem cell rejuvenation in 2145, it quickly followed with other rejuvenations over the next ten years. Hana volunteered for all of them. In 2168, at 128 and 76 days, she became the oldest person to ever live. At the time, thanks to the various rejuvenations, she didn't look a day over 60. No one knew how long she might live. We were entering into a new world of life extension therapies. Our historical enemies of disease and aging were being beaten back through science and healthcare. There were already animals that

lived to be hundreds of years old, and trees that lived to be thousands of years old. Why not humans? What was left to stop us?

Space 3

Finding life on Enceladus was as exciting as it was surprising, and it was obvious to everyone that a follow up mission to find out more was necessary, but not everyone could agree on what that mission would entail. There were some who felt it was absolutely necessary that we send a probe to try and bring back some of the Enceladans (Poseidon 5 had identified six different species on three separate volcanic vents, which were collectively known as Enceladans) back alive. There were others that felt doing something so invasive to another planet's lifeforms was a most evil example of human hubris. There were debates on the morality of killing the creatures and dissecting them, which were countered by a deep rooted desire to learn if the life was DNA-based or not. There were some who felt we had already gone too far, and that any return trip to Enceladus was reprehensible. There were others who felt we should send humans to the far away moon to study the creatures as much as possible. The thought of establishing human operations on the icy moon 1.2 billion kilometers away from Earth gave NASA heart palpitations, but the desire to get human eyes on the Enceladans was very strong.

Esteban de los Reyes and Hanna Pines of the University of Arizona and Arizona State University respectively, called together an emergency conference of their astrobiologist colleagues. Never before had an academic discipline been catapulted to the forefront of the public's imagination as the astrobiologists now found themselves. At the conference, de los Reyes and Pines led the creation of a document called *The Guide to Extraterrestrial Scientific Interaction*. GESI, as the guide came to be known, outlined rules for engaging with life found outside of Earth. Attempting to be comprehensive, it used 16 pages to outline how to engage with

another intelligent species. In the absence of intelligence, which was believed to be the case on Enceladus, rules to be followed included only taking specimens that had died of natural causes, and even then only a small amount so as to disrupt the natural habitat as little as possible. The urgency of the conference assisted in the brevity of the guide, which in turn helped with its adoption.

GESI achieved some notoriety beyond the astro-scientist circles it was intended for, and de los Reyes was asked in an interview whether or not their rules might be interpreted by some unknown race of aliens who might treat us like an unintelligent species. His response somewhat drove the need for exploration home, “I do not worry about being researched by some unknown alien group, for it would be immensely more efficient to simply ask us for the results of 500 years of human inquiry then to try and recreate it themselves. But it is essential we establish the rules of our inquiry now, lest some future group of less thoughtful humans decide to go find out about the Enceladans themselves.” Everyone knew we had to return to the icy moon, and GESI gave us the best chance of doing so as minimally invasive as possible.

NASA for its part was already well on its way to designing the return mission to Enceladus. While the debate of whether we should go, and how, had been raging through the last half of the 2110s and early 2120s, NASA was working out how to put more science equipment on a probe that they could then return from Saturn’s moon back to Mars for science to be done by a group of humans. The plan would mean upgrades to the Mars base, and a whole new type of scientific probe, one built for biology instead of astrophysics. Once again the agency partnered with NLSA to help design the mission, and new lab for deployment on Mars.

Some questioned what GESI should mean for the WCIP mission, which was happily on its way to Proxima Centauri b. The probe had been designed to land on the planet should it find signs of life, and send pictures back, but with the discovery of life on Enceladus, people were beginning to think that the small chance of life on Proxima Centauri b we thought there was when we launched it, was actually a big chance of complex life, and whether or not we should disrupt that life on another planet. At \$23 billion it wasn't exactly the type of mission to just scuttle, but these questions would hang over the mission as the probe flew through interstellar space to its destination.

By 2123, NASA had decided on two missions to send to Enceladus. The first, Poseidon 6, would be the mission to try and retrieve specimens of the species found on Enceladus. It would have minimal scientific capabilities, preferring instead to use all available space for ensuring a safe trip back to Mars. The second, Poseidon 7, would establish a more permanent presence on the Saturnian moon, studying the habitat and lifecycles of the Enceladans. It would also take with it additional launch components for Poseidon 6. NASA set aggressive launch dates of 2135, and 2136 for the two missions. No one wanted to wait longer than absolutely necessary for more information about the alien life. It was a six year trip to Saturn's moons so it would be another twelve years after launch that we might get our hands on an Enceladan. NASA viewed anything before 2150 as a success in that regard.

While NASA worked on the next two Poseidon missions, the rest of the world got down to the business of doing science backed by the discovery on Enceladus. In particular there was a huge push to try and find the origin of life. Life appears everywhere on Earth, from the biological hotspots found on the Earth's surface, to the most inhospitable environments in caves,

near volcanoes, and even under the ice sheets at the poles. The discovery of fossilized life on Mars, combined with the atmospheric investigations from the Frank Drake telescope, gave credence to what was known as the Common conditions Rarely Emergent Proliferating through Evolution (CREPE) theory. CREPE held that life supporting conditions were common in the universe on the planets and moons in the habitable zone surrounding stars. On these planets and moons, life rarely emerged—only through the sheer vastness of stellar time did life spontaneously appear, a one in a billion shot. But once life did emerge, evolution provided it incredible power to proliferate across the biomes of its home. Enceladus gave even more credence to the CREPE theory by opening up life supporting conditions to beyond a star's habitable zone.

The rarely emerging part of CREPE was its most contentious conjecture. On the one hand, it explained certain phenomena such as why all life on Earth seems to come from a single common ancestor; and why we couldn't seem to recreate the origin of life in the lab; on the other hand, if life was rarely emergent, why did we seem to find it everywhere we looked? Enceladus gave an interesting third data point on the origin of life. At only a billion years old, Enceladus was aligned with the Earth when it comes to life appearing, and it pointed to volcanic vents perhaps being the key to sparking life. But a billion years is impossible to simulate in a lab. To even try and approximate those conditions, astrobiologists took a page from their particle physics brethren, and tried to make an experiment with lots and lots of vents. The thinking was that if the emergence of life was rare then it was likely a random process, and so one vent over 100 years could be simulated by 100 vents in one year. To try and mimic the conditions of Enceladus, and the early Earth, they built their test ocean underground. At 320 meters deep, it was over twice as

deep as any pool made by humans before. At the bottom they built an intricate system of heating tubes which would supply a mixture of water vapor, carbon dioxide, sulfur dioxide, and hydrogen sulfide to a system of 10,000 rocky vents of various sizes and shapes. The Planetary Origin Of Life (POOL) study, started observing in 2132 with a planned life of 100 years. The one in a million chance the study was designed to capture was the best that twenty-second engineering could build, but CREPE put the actual likelihood of emergent life at one in ten billion. But just because your telescope can't see exactly what you want it to doesn't mean you don't build the telescope.

Poseidons 6 and 7 were only delayed one year, an astonishing accomplishment chalked up to scientific exuberance beating out administrative bureaucracy. Launching in 2136 and 2137, Poseidons 6 and 7 sped off on the 1.2 billion kilometer journey to Enceladus. Equipped with all the autonomous scientific equipment NLSA and NASA could cram onto them, the two missions represented the pinnacle of scientific achievement at that point.

Poseidon 6 touched down on the icy surface of Enceladus near its predecessor Poseidon 5, and got right to work getting set up for its return back to Mars. Ice was its friend, as the oxygen that powered its rocket back to the inner planets was harvested from the surface ice of the moon. It was a slow process as the energy on Enceladus had to come from the sun, and the sun was much further away than Earth. Only about 50 watts per square meter worth of sunlight reached the distant frozen moon. That is about 3.5% the amount of energy that Earth receives. Thankfully years of mining oxygen from ice and frozen carbon dioxide on Mars to fuel the rockets from the Masters Mars Base had honed NASA's efficiency enough to make the Enceladus return mission feasible, even with the much diminished solar power. Poseidon 6

carried with it enough solar panels to make the return fuel, and power the submersible on its way with Poseidon 7, the largest solar array deployed in space in one payload.

As exciting as Poseidon 6's landing on Enceladus was, it was really just an incredibly sophisticated power station, Poseidon 7 is where the real sexy science was going to go down. Equipped with a submersible to retrieve samples, and the most powerful autonomous science lab ever created, it would start sending back scientific data within days of landing on the icy moon. Some of the big questions Poseidon 7 was looking to answer were what you expect: what were the Enceladans made of? were they DNA based? how similar were their organs to Earth's animals? did they evolve from a common ancestor? The whole of the scientific community, and a not insignificant part of the world's attention were eagerly awaiting what NASA would find out about these foreign creatures.

Astrobiologists decided to start with the largest Enceladan first. The submersible collected six dead specimens, three to send back with Poseidon 6, and six to examine in the autonomous biology lab. The creature, dubbed Enceladan a, was shrimp-like, about six inches long with six broad flat wings, three on each side. It also had two hidden longer limbs that retracted next to what scientists called its abdomen. Superficially it looked like something that could be found on Earth, though its thermal eye spot, and antenna-like sensory organs did give it an otherworldly feel.

Inside Enceladan a is where the real fun began. It seemed to have many of the same organs we would expect from an animal on Earth, heart, stomach, kidneys, etc., but they weren't arranged how we would expect. It would take months of examining and reexamining the positions of the internal organs before someone would suggest that it's what we might expect to

find in a bilaterally symmetric animal that evolved from a radially symmetric one. This was an exciting hypothesis, as the transition from radial to bilateral symmetry on Earth had happened so long ago that no animal was in this transition state. Astrobiologists also found blue blood, which was exciting. Octopi and other cephalopods have blue blood, where they use copper instead of iron to carry oxygen to cells. Enceladus's blue blood hinted at more convergent evolution, as the ocean of Enceladus was similarly oxygen-poor to Earth's deep ocean.

When the first images from the biology lab's microscopes started to reach Earth, excitedly anxious astrobiologists started cheering and high fiving each other. The unmistakable picture of an animalian cell, complete with nucleus and organelles to discover, was a powerful first step in confirming that Enceladus life was, at its core, not that different to life on Earth. It's hard to imagine what would've happened if we had found something else; picturing life that works differently to what we have on Earth is notoriously difficult. Cells are incredibly complex, yet efficient machinery for making life happen, and it was thought that some sort of cell was necessary for complex life, finding them in the Enceladans was a strong indicator that cells were likely universal.

The first chemical test performed was arguably the most important. It was said that the debate about whether the Enceladans used DNA or some new alien mechanism for their genetic encoding was so interesting, and popular, that Vegas casinos had been taking bets on it. When Poseidon 7 sent back definitive proof of DNA, there was more celebrating amongst the astrobiologists. DNA! That molecule that powers all the life on Earth found on a new world. This incredible example of convergent evolution suggested that like the cell, DNA was likely universal to life. It suggested that alien life may not in fact be that alien, that we could leverage

centuries of biology to learn more about our new neighbors. It was an exciting time to be an astrobiologist.

And an even more exciting time to be one of the team of sixteen biologists and biochemists picked to join those on the Masters Mars Base to conduct experiments on the specimens brought back by Poseidon 6. Poseidon 7 had identified two more species of creatures on the vents of Enceladus bringing the number of Enceladan species up to eight. Poseidon 7 then collected 10 of each species from the vents over the course of 14 months while Poseidon 6 continued to make its rocket fuel. In late 2144, Poseidon 6 lifted off of the tiny moon, and headed back towards Mars. The trip took 9 years, three more than the trip to Enceladus due to the difference in fuel, but it arrived, fully in tact with 80 specimens ready for the lab, preserved first by the cold waters of Enceladus, and then in a chamber of Argon on Poseidon 7. There a team of sixteen biologists went to work under the leadership of Chris Harnack.

Harnack had prepared for this assignment his whole academic life. Trained as a biochemist, he ran a lab focused on DNA, and how it worked, trying to understand how it gave rise to complex systems like the senses. He had developed a particular interest in astrobiology, a field that garnered a lot of interest after the 2062 discovery of fossilized life on Mars. He had written two books on the role that DNA might play in alien biology. The second, *A European's Life*, took the reader through how DNA could shape creatures on the icy moon of Europa. Of course life had been found on Enceladus and not Europa, but Harnack's predictions weren't that far off from what was actually found there. He was an energetic leader whose enthusiasm for the strangeness of the Enceladan's biology was infectious.

Scientists by 2153, when Poseidon 6 returned with its specimens, had gotten pretty good with DNA. While we still couldn't read it with the fidelity with which people might read computer code, we understood multi-gene expression and how that led to some complex systems and structures. We had gotten really good at identifying what certain genes were connected to, even if we didn't yet understand exactly how they did the work that they did. It had been 200 years since Crick, Watson, and Franklin had discovered the double helix structure of our genetic molecule, and 200 years of curiosity driven research had put us in a great position to examine the newly found Enceladans.

The entirety of the Pythias mission, as the mission sending 16 scientists to Mars to study the Enceladans was called, after a wife of Aristotle known for collecting specimens of animals, plants, and fungi, is chronicled in the excellent work *Resurrecting Pythias: The First Look At Alien Life* by Jacob Goldstein, here I'll focus on the most important discovery for our story. Once the Enceladans had been received by the Pythian scientists, they went to work examining them using state of the art equipment that went beyond what the autonomous lab of Poseidon 7 was capable of. Within the first few weeks they had discovered something very curious. Three of the heterotrophic species (heterotrophic means the creature gets its energy by consuming other organic material like animals do, the opposite being autotrophic where creatures make energy themselves from some inorganic source like the sun in the case of plants, or the heat of the vents in the case of the autotrophic Enceladans) displayed less senescence in bigger specimens than in the smaller ones. Senescence is the sciency word for the deterioration that happens when something ages. In aquatic animals on Earth, size is more a function of age than it is a function

of genes like it is with humans. So the bigger specimens should have shown more senescence, but instead they showed less.

Turritopsis dohrnii is a type of jellyfish found around Earth in tropical waters. It has the incredible ability to revert back to a juvenile state when threatened, or when conditions are less than optimal. It's believed that it can do this reversion indefinitely, a fact that has earned it the common name of the Immortal Jellyfish. Scientists had been studying this jellyfish for 150 years by the time the Pythias scientists discovered the anti-aging properties of the Enceladans. Could it be that they had found new examples of the biological immortality possessed by the immortal jellyfish?

To answer that question, scientists needed to be able to see the cells that were reverting to an earlier age in action, and for that they needed live specimens, but the GESI explicitly forbade the acquiring of live specimens. Fortunately for the Pythias team, a technique had been developed in anticipation of just this sort of problem. Discovered in 2146 by a team of Japanese biologists at Tahoku University led by Suzu Inari, it was possible to actually swap the DNA in a cell wholesale, turning the cell into a whole new kind of cell. Most of the time, this will result in near immediate cell death, but if a similar enough cell is used for the DNA, it was actually possible to turn one kind of cell into another kind of cell. The process, efficiently named genome swapping, was of limited use on Earth where if you wanted a cell of a certain type you just went to the source. Yes this technique is essentially what enabled the plot of the twentieth-century classic Jurassic Park. Unfortunately dinosaur DNA existed so long ago that it had all deteriorated by the time genome swapping came to be. Scientists did manage to bring back some animals that

went extinct around the last ice age such as the mammoths and giant ground sloths though, much to the delight of conservationists and animal lovers everywhere.

In the hands of the Pythias team, genome swapping let them essentially bring their specimens back to life, at least at a cellular level. Cells weren't traditionally afforded the same moral considerations as the multi-cellular entities they made up, and the GESI was mum on whether studying a living cell was right or wrong. Given carte blanche to work with these revitalized cells, Harnack directed his team to coax them into a stem cell state, and then grew them in a lattice into the various organs they found in the Enceladans using techniques that were advanced versions of those used for cultured meat. When the Pythians looked at what resulted, they found that the Encaladan cells would sometimes revert to an earlier phase. Some of the cells seemed to rejuvenate themselves!

It was at this point that the decision was made to bring the alien DNA back to Earth. Fear of an Andromeda Strain like event (yes it's not lost on me that I've referenced two Michael Chriton novels in this chapter, but what can I say the guy could write some sci-fi even if his views on climate change were a little questionable) had been subdued through two years of study on Mars, and the prospect of biological immortality was just too enticing. In February of 2156, the first alien life touched down on Earth. Scientists at the time were beginning to feel like it wouldn't be the last.

NASA was feeling pretty great about their accomplishments in the first half of the twenty-second century, but like all great scientific endeavors, it wasn't enough to simply find success, we had to constantly be striving for something more. So it was, when, in the same year alien life touched down on Earth, the astrophysicist's particle physics colleagues came to them

asking to find them a black hole. Hungry for the next big thing to search for after life on another world, NASA eagerly accepted the challenge.

Socioeconomic 3

The transition from an oil based economy to an agrarian one based on vertical aeroponics supplied by CO₂ sequestration, combined with the advances in healthcare, the increasing amounts of technology these advances required, and the dramatic increase in professional services that all of that required, created an incredible need for college educated labor. This need was felt all over the world, but most acutely in the developing world where the percentage of adults graduating college was in the teens. During the first half of the twenty-second century, that percentage would grow dramatically, but to see how we got there, we need to take a look back at the good ol' US of A.

In 2069 when the US congress passed the HEALTH Act, it had an immediate and dramatic impact on US state budgets. For those unfamiliar with the structure of government monies in the United States, the federal government collects taxes and funds things for all Americans, but the individual states, like Wisconsin or California, also collect taxes, and use those funds to fund state initiatives. If you imagine those state budgets as a pie chart, from the 1970s until 2069, the two biggest slices for almost every state were education and healthcare. And for nearly every state during that time, the healthcare slice got bigger and bigger at the expense of the education slice. With the passing of HEALTH Act, all of a sudden states were no longer on the hook for healthcare, the biggest slice of their budget pies. What each state did with this sudden windfall became *the* case study for how public monies are spent worldwide.

The states of the American West surrounding the Potas river, the US's RIVER project, and running up the coast, feeling the need for higher education most acutely, invested their freed

up healthcare funds primarily into education. For some states this meant going so far as to more than double the amount of funds for education starting in 2070. So large was this influx of cash, that it had to be spent on all education from pre-school through college. States expanded their university systems, adding colleges all along the Potomac river to educate people on all that the RIVER had to offer. Teachers along the West Coast and in the Southwest became some of the highest paid teachers in the world, almost overnight. STEAM programs got funded all across those states, technology was upgraded, facilities were repaired. Nearly a hundred years of education funding erosion by the rising costs of healthcare were wiped out overnight, it would of course take years to repair the damage of that erosion, but 2070's budgets were a big step in the right direction for the American West.

The Great Plains, and South of the US, the stronghold of the Christian Republican Union did not make a similar investment. Education had long been identified as the enemy of conservatism, and faced with the opportunity to improve education for their people, the republican states decided instead to provide tax breaks for their wealthiest citizens and corporations.

The Eastern Seaboard, and the Midwest states found some varying degrees of middle ground between investing in education and funding other initiatives. New England similarly invested in education, with Maine and New Hampshire on one end with the lowest expenditure, and Massachusetts on the other end with the highest expenditure in education.

At the start of the 2070s, the United States was divided into the greatest experiment in education spending ever put together. Sociologists and Education Researchers around the world lined up to monitor the outcomes of the US educational system. By the end of the decade it was

abundantly clear that investments in education had an outsized effect on the bearing of the populace. In the words of Lauren Moeller, a socioeconomist from University of Chicago, “It’s rare that socioeconomic effects in the modern world are obvious enough to trace back to a single source, but in the case of American education after the HEALTH Act, we have clear and direct evidence that having the state cater to its billionaires at the expense of its people is simply bad policy all around.”

By the 2080s, the federal government of the United States began to wonder if it should step in with regards to state funding of education. The CRU controlled states were costing the federal government nearly \$25,000 per resident. The lowered expectations for CRU state students also stymied investment in those states regardless of the ostensibly favorable tax conditions for corporations. For years the CRU states depended on domestic migration, providing inexpensive respites from the housing markets on the coasts, but by 2080, the domestic migration of people at child bearing ages began to dry up. Populations in those states were beginning to drop, and the problem was expected to get worse. Faced with the prospect of an aging government-dependent population in states struggling to bring in enough young people to keep their economies going, the presidential election of 2084 had at its center what the federal government’s responsibility was to the states who hated it the most.

Javier Renta was a product of the Potas river’s effect. Born in Fresno in 2032, he grew up as the Potas river started construction, and graduated from high school the year that it officially opened. He went to undergrad at UC Berkeley where he got a degree in Biology before attending law school at Yale. He immediately launched himself into the political atmosphere of the Potas river, rising in prominence in California’s competitive political landscape, eventually being

elected governor in 2074. At 42 he was one of California's youngest governors, a fact that was probably equal parts his political bearing, and the importance of the Potas river to that state's economy.

When Renta launched his campaign for president in 2083, he put together a team heavy on Potas river leaders to help him extend what had worked so well for California to the rest of the country. Education was at the forefront of his platform given California's success in that regard. Still he was missing a big hook, something that would draw in voters from all fifty states. That would come towards the end of 2083 with a paper published by researchers working as part of BLOWARP in Australia. The Mahd Array was exactly the economic boost that Renta needed for his platform.

The US presidency in the 2080s was won during the primaries. With the republicans still split between the CRU and classic Republicans, democrats only needed to decide amongst themselves who to run and they were all but guaranteed victory. This made the democratic candidate debates of the utmost importance. Two of four had been held before Renta and his team read the Mahd Array paper and formulated their strategy. Polling at that time was mixed without there being a clear frontrunner. Renta knew that if he could formulate a strategy with the Mahd Arrays that would bolster him in Iowa and New Hampshire, the first two states to hold primaries, it could give him the advantage he needed to pull away from the pack.

The Renta Plan, as it came to be known, was to build a number of Mahd Arrays proportional to a state's population in all fifty states. No longer dependent on optimal geographic conditions, the arrays, powered by combinations of nuclear and solar energy, would give each state its own supply of crucial raw materials, and food. The large public works jobs would

provide for plenty of jobs in each state, as well as plenty of contracting funds for materials. The catch was that each state would be on the hook to educate the people needed to staff the arrays. To help fund that education, the federal government would provide funds to states to reach a budgetary threshold, but states couldn't fall below a certain percent of their own funds.

The Plan, derided by republicans as a government takeover of education, drew plenty of media attention. The democratic candidates, once aware of the arrays, developed their own plans with their own set of parameters, but Renta had the first mover advantage, and the most straight forward plan. Democratic voters in the red states heralded the plan as the government finally doing something, and Renta cruised to easy victories in Iowa and New Hampshire. He coasted through the rest of the primaries to a victory in November of 2084, and the Renta plan, now called the Advanced Jobs Investment and Lifetime Education (AJILE) Act, passed in the first legislative session of 2085.

By 2110, a generation of American students had matriculated under the AJILE Act, and the supporting laws passed in following years. The results were overwhelming. The United States, which until 2085 had lagged in education compared to similar economies, moved steadily up the achievements chart so that in 2110, the US actually led in several categories, a fete thought impossible due to the country's size. In the CRU states, which had come kicking and screaming along for the educational ride, conditions began to improve. States that for more than a century had been dependent on government subsidies started to change the balance of their contribution to dependence ratio. GDP grew as the Mahd Arrays proliferated, staffed by a generation of advanced degrees paid for by a mix of state and federal funds freed up by the switch to universal healthcare.

The world paid attention to the new economic engine in the United States. As Mahd Arrays were changing the landscape of how we produced anything from food to medicine, they of course fell to that eternal truth that the rich get richer. Countries that were able to build more infrastructure, and staff that infrastructure with advanced degrees, outpaced countries with less capital and less universities. With carbon sequestration, the commodity that replaced fossil fuels in this new economy, the same was true. Wealthy countries could afford to build the infrastructure necessary to supply their grow chambers with CO₂, while poorer countries lagged behind. As the twenty-first century transitioned to the twenty-second, there was a growing movement to try and fix this disparity. Advocates of the fix argued that oil had dominated world politics for so long because of how it was distributed. Some countries found themselves suddenly wealthy, while others were left dependent on sometimes hostile governments just to move food around. But carbon was everywhere, there was no need for it to only benefit the handful of countries already wealthy enough to capture it. Most of the carbon on Earth, they argued, isn't even found over nations, so it's up to us on how to distribute it once it's captured.

In 2115 the UN adopted resolution E/RES/2115/4, which did four main things. First it established the creation of a fleet of long haul carbon capture ships to be deployed to operate across the international waters of the world's oceans. The investment and operation of the fleet would come from wealthy nations around the world. Second it took a portion of the funds from the proceeds of the fleet and invested them in Mahd Arrays in developing nations. Third it took the remaining funds and provided them to those same developing nations as educational investment funds similar to how the AJILE Act worked in the US. Fourth it established that all vessels operating in international waters would pay a percentage of their proceeds into this

economic fund. This last provision, which looked and smelled a lot like a tax, was the most contentious, but since the resolution left carbon captured within a country's borders alone, the large carbon capturers went along with it.

The economic development, in the regions that took it on and worked with it, had interesting amplifying effects. The building of Mahd Arrays offered countries relief from having to import food, timber, paper, medicine, etc. Rather than having that money flowing out of their country, it was staying in, giving them the opportunity to put it to more use. The education investitures improved every aspect of their economies, bringing in even more capital to use to continue to grow. Still that economic development would probably have stalled or petered out if it weren't for another resolution passed fifteen years later in 2130.

The advances that the developed world were enjoying in human longevity, and stem cell rejuvenation caused a new movement to begin. Faced with the likelihood of humans in developed nations outliving humans in developing nations by centuries instead of just decades, activists started pointing out the human rights debacle that unfettered wealthy country progress would be. By 2120, when administrations were approving the first stem cell rejuvenations, calls for action on what had been dubbed the Longevity Crisis were growing across the globe. The movement built throughout the 2120s, filling the slot of "biggest challenge humanity faces" that Climate Change had vacated due to the success of Mahd Arrays and carbon sequestration.

In 2128, Britta Larsen from Norway proposed a radical solution. With the United States having adopted universal healthcare, all of the countries in the top half of economic development had governments which paid for healthcare. What if, similar to the international waters carbon sequestration, those countries paid into an economic development fund which improved

longevity solutions in the developing world? It wouldn't be a direct tax on consumers, since it was the countries themselves who footed the bill for longevity treatments. But it would ensure that as lifespans increased in the developed world, there were funds for similar improvements everywhere else.

By 2128, the effects of the carbon sequestration in international waters resolution were being felt, which gave strength to Larsen's argument. But what really sealed the deal, was that in 2129, scientists reported that humanity had reached a carbon neutral state. Tipping the scales on climate change created a growing sense of the strength of global cooperation. The benefits of economic development paid for by carbon sequestration gave the UN, and those watching it, the confidence that they were finally on the right track for improving life for all of humanity. In 2130, the UN adopted E/RES/2130/2, a resolution that created a modest 1% tax on all longevity treatments, as defined by the resolution in such a way to capture future treatments as they came along, and funneled those collected funds into economic development of longevity treatments in the developing world.

The first half of the twenty-second century was filled with political upheaval around the developing world. Advances in technology made it increasingly difficult for corrupt governments to keep their people in the dark. In countries with corrupt governments, first the carbon funds, and then the longevity funds, were siphoned off to fill the coffers of cronies and yes men, but people could look around to neighboring countries and see those programs at work for the benefit of their neighbors. People started taking direct action against these corrupt governments, but it would take into the second half of the century to really turn over the entrenched status quo. That is the subject of our next socioeconomic chapter.

Dark Field 4

While physicists awaited the engineering necessary to sequester large amounts of superpositioned particles and/or found their way to a black hole, they turned their attention to the other side of dark quantum mechanics: dark-to-light transitions. Remember that incorporating dark matter into quantum mechanics meant that particles could transition between light and dark physics. Physicists believed they were on the right track when it came to how light matter could transition to dark matter, now they wanted to try and figure out how the reverse took place. One possible clue was the mass problem, as the broken principle of equivalence came to be known. Physicists weren't too keen on abandoning their conservation rules. If the light to dark transition caused a particle to take on so much more mass, it should take energy, and if the opposite transition shed that mass, it should give off a lot of energy. If a lot of particles were giving off that energy, then we should be able to see it.

While particle physicists were pondering this dark to light transition, astrophysicists were puzzling over another dark matter problem. By 2160, four generations of exoplanet hunting telescopes had been launched by the world. More than 500,000 exoplanets had been found around the 40,000 or so stars we could see close enough to us to distinguish the characteristics of exoplanets. We had gotten a pretty good sense of the gravitational mapping of about half of those other solar systems. In all of those systems, there was no gravitational effect attributable to dark matter found, and it was starting to become a problem for the entire theory of dark matter. When it was just our solar system lacking dark matter, that was explainable by saying we just happened to be in a smooth dark matter pocket of our galaxy where dark matter's gravitational effects

canceled out, but for none of these solar systems to be affected by dark matter just didn't make sense. In each of them, the light matter we knew about made up all that was needed to explain the gravitational action we saw. So if dark matter comprises 85% of all the matter in the universe, why wasn't it found around stars?

The West African country of Mali was uniquely positioned to benefit from the establishment of RIVER projects, and the subsequent building of Mahd Arrays. It was large and hot and most importantly flat, making it a perfect endpoint for RIVERS flowing in from the coastal countries of western Africa. All told, the Economic Community Of West African States (ECOWAS) had built seven RIVER projects prior to the UN's adoption of its education resolution in 2115. Five of those projects terminated in Mali. Three of them fed into a new lake basin north of Timbuktu named Lake Musa, a reference to Mansa Musa whose wealth may have made him the wealthiest person to ever live.

On the shores of Lake Musa, Mali, one of the biggest advocates of the UNs education resolution, and subsequently one of its biggest beneficiaries, built Four Rivers University, so called for the three that made Lake Musa, and the Niger River that ran on the other side of the city of Timbuktu. The desert climate made for clear skies the year round, and made for an ideal location for observing space. In 2106, the nascent West African Space Agency (WASA), in coordination with NASA and ESA, broke ground on a radio telescope array called the Gigantic Array, continuing the linear naming from the Very Large Array, and the Extremely Large Array before it. The credence given the night sky by Mali at Four Rivers University made the school a destination for astrophysical students from all over the continent.

Rokhaya Diallo was born in 2132 in Tambacounda in central Senegal. The city had grown precipitously after the building of the Aibmag RIVER project, a joint venture between Senegal, The Gambia, and Mali to build a river that ran parallel to the Gambia river. By 2132, nearly 1.5 million people lived in the Tambacoundan metropolitan zone. Diallo's parents had met on the job at a pharmaceutical Mahd Array, her dad a microbiologist, and her mom an immunologist. Having two scientists for parents gave Diallo a head start into the scientific realm, but like any good rebellious teenager, she decided against her parent's wishes for her to study biology, and dove head first into physics instead.

In 2150 when Diallo enrolled into Four Rivers University, it had established itself as the top school for astrophysics on the continent of Africa, and was doing a lot to raise Africa's standing on the world's stage. She stayed at the university for undergrad and her PhD, choosing to stay close to home rather than following most of her fellow students in traveling abroad for their PhDs. The added time gave her plenty of opportunity to get to know the inner workings of the Gigantic Array. After she received her PhD, the university offered her a teaching position, and plenty more time with the telescope. She enthusiastically accepted.

Diallo's thesis topic, and subsequent focus of her studies was Fast Radio Bursts (FRBs). First discovered all the way back in 2007, FRBs had defied explanation for over a hundred years. On the universe's scale, FRBs are relatively rare events—by the 2150s, only about a hundred of them were being detected every year. When they do happen, a powerful pulse is detected in radio frequencies for only a few milliseconds. They are extremely energetic, putting out as much energy in a few milliseconds as the sun puts out in a week. With so few occurring, and occurring

as briefly as they did, their source had eluded astrophysicists for nearly a century and a half. Diallo was determined to figure it out.

In 2161, she had a crucial insight. Interviewed years later for the podcast *Searching In the Dark: the quest for dark matter*, Diallo explained her insight:

“We had ignored dark matter as a possible source for fast radio bursts, as it wasn’t known to generate energy at all, let alone on the scale we were working with with the bursts. I was of course familiar with Sondoval’s work on light matter turning into dark matter, she had actually presented her work at Four Rivers while I was a doctoral student. And I was aware that there was a contingency of theorists working on the problem of how dark matter turns back into light matter.

“I used to have an ice addiction—in Mali you have to stay cool somehow, and for me that was with a constant stream of ice water. I had a counter ice maker that I used to keep filled to make sure I always had ice. One day I was standing in my kitchen staring at that ice maker as I thought about FRBs. I remembered something we learned as undergrads, ice won’t form in pure water, there needs to be something dissolved in the water for the ice crystals to form. I wondered, flippantly at first, if stars were the same way.

“When a star forms, we knew fusion started when it was a less dense protostar, but we weren’t quite sure how the process started since at that point, the star would be too diffuse and too cool for fusion to start at the rate the star needed to avoid catastrophic collapse. I wondered if something in the star acted like what was dissolved in the water to allow ice to form. I went through what it could be and when I landed on dark matter, something clicked. What if dark matter seeded in the collapsing star contributed to the start of fusion. If my particle theorist

colleagues were right, this would create a tremendous amount of energy, enough to explain a fast radio burst.

“I started working on models and theories, but my quantum mechanics was quite rusty so I started enlisting help from colleagues around Western Africa. It took nearly a year to put the theory down in its entirety, but we got there. We had no idea that our discovery would have such an enormous effect on humanity, we were just pursuing the science.”

When Diallo et al’s paper was released in 2162, the terms *decoherence*, and *recoherence* joined the scientific lexicon to describe light matter transitioning to dark and dark matter transitioning to light respectively. What the paper described was the process for recoherence, performed through stellar nucleation. Superpositioned particles, that is dark matter, caught within some radius around the stellar formation, collapsed during that formation and recohered, transferring their excess gravitational mass to a tremendous amount of energy and kickstarting the fusion process in the stellar core. A survey of FRBs put an upper and lower limit on the radius, dubbed the Diallo Radius, and agreed with the observation that solar systems didn’t contain dark matter—it would all be turned to light matter when it collapsed.

Physicists added this new development to the list of things that the Dark Matter Creation and Containment Center (colloquially known as the DMC-cubed) being built within the super colliding ring at CERN needed to test. Ground was broken on the facility in 2154, the year after Sondoal’s paper breaking the principle of equivalence, on the bicentennial of CERN’s opening. CERN had maintained itself at the forefront of particle physics as the Americans increasingly turned their attention to space. DMC-cubed was to be built deeper than anything at CERN had

been built before to shield the dark matter from interactions with cosmic rays—of course it was theorized that dark matter doesn't interact with cosmic rays, but better safe than sorry!

The groundbreaking was derided as premature by some since physicists had few ideas on how to create dark matter, let alone contain it, but CERN wanted a big project for their bicentennial, and as the center of the experimental particle physics world, CERN was the natural spot for going after elusive dark matter. The project runners gave theorists ten years to figure out creation and containment, as it would take that long to build all of the pieces that they did know they needed. Spoiler alert, they figured it out, but let's see how they got there.

For a particle to decohere and become dark matter, it wasn't enough for it just to be in a superposition. There was some other mechanism that happened that caused the particle in superposition to lose its connection with the electromagnetic field. To date, the only place physicists had found this mechanism to be happening was around the supermassive black hole at the center of the Milky Way. Space agencies were looking for a black hole for us to study, but physicists are an impatient bunch—a contingency was looking for a terrestrial mechanism to stand in for a gravitational singularity.

To get our dark matter creation machine we need to take a trip over to West Lafayette Indiana, home of Purdue University. In the early 2000s, Tongcang Li and a team of scientists at Purdue set multiple records for the fastest spinning object on Earth. The devices in those days were dumbbell shaped nanoparticles which would be propelled with a laser to spin at around 600 billion RPMs. After Li retired, a grant established a semi-annual competition for the fastest rotating thing. By the 2150s, objects in the competition were hitting quadrillions of RPMs, just what dark matter theorists were looking for.

Einstein himself had introduced the thought experiment that being in a gravitational field was just like riding in an elevator being pulled in the opposite direction of the force of gravity. Since then physicists had known that being in an accelerating reference frame was just like being in a gravitational field. When an object rotates, it is like there is constant acceleration away from the center of the object (this is how artificial gravity is created in spaceships). Without a black hole to provide a strong gravitational field, physicists looked for as much acceleration as they could find. Now the best way to accelerate particles is with a magnetic field, but you can't use electromagnetism with dark matter. What theorists needed was a way of accelerating their superpositioned particles mechanically. And for that they needed something that spun really fast. And for that they headed to the 2156 semi-annual Rotate-athon at Purdue.

What developed out of the 2156 semi-annual Rotate-athon was a huge collaboration of 62 scientists and engineers who would go and build an apparatus that balanced speed of rotation with size of a chamber of superpositioned particles. By 2164, the apparatus, dubbed the Dark Matter Aggregator (affectionately called DaMaGe by those close to it) was able to simulate around a billion Gs on a mole of superpositioned particles. It was about a thousand times less than the gravity felt at the event horizon of a black hole, but It was much higher than anything else on Earth could produce. Theorists gave the apparatus only a 1% chance of creating dark matter, but in particle physics, a low percentage chance of something happening just meant running your experiment a bunch of times.

On a parallel track, physicists were trying to figure out the containment of the dark matter they were going to create. Once particles made the switch to dark matter, they no longer interacted with light matter, which meant that the walls of DaMaGe's acceleration chamber

would no longer hold it. You couldn't move it anywhere either since light matter couldn't apply a force to the dark matter. The only thing dark matter responded to was gravity, and without the ground to stop it, dark matter would accelerate towards the center of the Earth, fly right through it, and turn into a harmonic oscillator tunneling through the Earth. If you've ever asked the question what if the Earth had a hole drilled through it, and you dropped something down the hole, you'd be able to find out once you made some dark matter.

This containment calamity stalled the project at DMC-cubed until Diallo's paper on recoherence was published in 2162. Up until then, DMC-cubed had been trying to verify the breaking of the principle of equivalence by measuring the gravitational force from a collection of dark matter, but the energy given off by recoherence provided the same insight. Maybe, if we could create a fusion reaction with a large enough Diallo Radius, then it wouldn't matter that we couldn't keep the dark matter in one place. If we could verify the energy in the resultant fast radio burst, it might do the trick.

The scientists and engineers working on containment decided to pivot to building the largest fusion reactor the world had ever seen. Fusion power, the nuclear power of the stars, long sought after by green energy enthusiasts everywhere, had eluded its proponents for 200 years. We knew how to make a fusion reaction, but we couldn't sustain one that gave off more energy than it needed to sustain itself—it was as though we were missing some piece of the puzzle. As the team turned its attention to the fusion reactor, a buzz began about how dark matter recoherence might actually be that missing piece of the puzzle, providing the energy needed for a sustained fusion reaction that gave off more power than it consumed.

DMC-cubed was ready to start doing experiments in 2168, a mere fourteen years after ground was broken on the project. The experiment was about as simple as an experiment consisting of artificial gravity induced dark matter recohering in a large nuclear fusion reaction could be. DaMaGe would spin up a batch of potential dark matter, which would begin to oscillate through the Earth. The fusion reactor would start a sustained reaction, and then a bevy of detectors would sit and watch for a fast radio burst indicating the dark matter had recohered. The array of detectors were given the nickname The Committee after the Nobel Prize Committee since the experiment, if it verified dark matter worked as the theories expected, would probably dole out Nobel Prizes for years to come.

The name of the detectors was not a let down. In 2169, a paper with 124 names attached announced the successful detection of a fast radio burst from terrestrial fusion verifying that dark matter could be created in the presence of a gravitational force, that that dark matter decohered from light matter, that it recohered during the fusion process, and gave off the predicted amount of energy when doing so. Sondoal, Sadiqqi, and Uddin shared the nobel prize in 2172. Diallo and two of the leads of the DMC-cubed project won in 2173.

The discovery spawned renewed interest in nuclear fusion. By 2178, a mere decade after DMC-cubed had begun its experiments, dark matter fueled fusion reactors were being experimented with in the US, the EU, Russia, China, India, Western Africa, Southern Africa, and South America. The promise of actual nuclear fusion power was just too enticing to pass up. The dark matter containment issue created an interesting political dynamic for these regions to figure out. What exactly is the role of government in a resource which, once created, falls through the Earth to the country on the other side of the Earth from its country of origin? Further, it was

within the capabilities of everyone involved to build a fusion reactor with a Diallo Radius large enough to steal the dark matter created by other groups.

The UN, building on the success of its education and longevity initiatives, developed resolutions to assist in the peaceful proliferation of fusion power across the globe. Largely motivated by a desire to replace the much more dangerous fission reactors that had proliferated during the twenty-first century, the resolutions included measures for the sharing of technology developed by one group with others. The entire effort took on an air of worldwide cooperation, the promise of fusion power was essentially cheap energy made from water, a resource so much more abundant than uranium as to effectively solve human energy problems for millennia to come—possibly forever.

While the rest of the world worked on getting nuclear fusion power out to the masses, physicists turned their attention back to dark matter. The DMC-cubed experiment confirmed a lot of theories, but it had left open the big question of how to contain dark matter. This of course was just one piece of the much larger open problem, which was how can we interact with dark matter? Now that we knew what it was, it was insufficient to just let dark matter sit, unaccounted by probing scientists. Being able to move it from one place to another was not only interesting, but also useful in a world where it could be turned into energy. The question of dark dynamics pressed to the forefront of physical inquiry.

One of the first experiments to go after this question was performed using DaMaGe at the DMC-cubed. It was to try and determine whether particles needed to be in a superposition in order to decohere. It was the beginning of trying to determine what information a particle took with it when it decohered. Of course, finding out what information remained when the particle

recohered required trying to find it inside of a fusion reaction, but one thing at a time. The new experiment spun up in 2174, and found that the decoherence process only required a sufficient force, and gave scientists hope that some information from before the decoherence stayed with the dark matter.

Not needing to have particles in superposition before decoherence opened up a new world of possibilities for particle physicists. A series of experiments run at DMC-cubed, yielded decoherence on larger and larger and more complex objects and groupings of particles. By 2190, DaMaGe was decohering ordinary objects. Of course most of those objects were crushed under the force of the artificial gravity of the spinning DaMaGe chamber, but the satisfaction of finding those objects simply gone from the chamber, decohered into dark matter, spurred the imaginations of the attending scientists. It was in 2190 that a team of 22 scientists wrote a paper proposing that decoherence and recoherence, if suitable low energy methods were found, could be used for teleportation. One hundred years ago such a paper would have been met with ridicule, but on the cusp of the twenty-third century it was met with serious consideration.

Biochemistry 4

By the second half of the twenty-second century, the life sciences were beginning to feel pretty good about their attempts at extending human life. So much so that beginning in the 2160s, people started wondering what more could be done if we started to lift the taboo on *in vitro* genetic modifications. The first *in vitro* genetic modification had been made in 2018 when He Jiankui used CRISPR to make a pair of twins immune to HIV. Since then, a pretty strict restriction had been placed on *in vitro* modifications. Of course when PICKR was discovered, the allure of a generation of genetically modified superhumans was too great, and a second researcher from China, Dang Yichen, followed in He's footsteps by making a PICKR baby with green modified eyes—eye color being just the kind of multi-gene trait perfect for PICKR. Dang's announcement upon the birth of the modified child in 2078 drew the ire of both China and the broader world. It led to the reinforcement on the moratorium on *in vitro* modifications, which held for nearly a century.

In the 2160s though, two forces were converging to cause the biotech community to rethink its stance on *in vitro* modifications. First was the overwhelming success of human rejuvenation treatments. Adding decades to the human life span turned out to be quite popular, and the world's approach to redistributing the wealth associated with the treatments helped to grow that popularity. Combined with the global decline in disease, those in the life sciences were gaining a reputation that they could do no wrong. Second, the fantastical properties of the Enceladans, recently brought back to Earth, had set the world's collective imagination running. It

seemed every week that there was some new article explaining how we were on the cusp of finding true immortality.

At the 2165 genomic editing conference held annually in Berkeley, California, a cohort of scientists formed around the idea of rewriting the rules for *in vitro* genetic modification. Feeling that the writing was on the wall, they felt it would be better to get ahead of the growing pressure for *in vitro* research, and write sensible rules about how to proceed. The cohort was led by the current holder of the Doudna biochemistry chair at Berkely, Alma Ortega.

Ortega was the daughter of Honduran climate refugees who managed to obtain green cards after settling in northern Arizona in the town of Sunvale, one of the several towns that grew around the Potas river basin. Her parents had worked in the Mahd Arrays of the town, and imparted in Ortega a love for and curiosity of the life sciences that powered the arrays. She went to undergrad at Texas A&M with the intention of learning the agrarian sciences needed for high level work on the arrays. Unlike with humans, there was no compunction about using PICKR on plant germs, and Ortega cut her teeth in a lab creating more efficient corn and soy beans. In graduate school at Yale, she made the jump to animals, continuing experiments on mammalian embryos. After getting her PhD, and completing post-doc research at Cornell, she was offered a lab at UC Berkeley. Happy to move closer to home, she established herself on the West Coast as the leading researcher on embryonic PICKR procedures. A perennial presence at conferences and awards ceremonies alike, she was a natural leader for the cohort.

By 2165, many of the moral sensibilities that had stymied *in vitro* modifications were no longer much of a concern (we'll learn more about that in our next socioeconomic chapter). Still, the question of what to allow and what not to allow persisted. There was a sense that without

proper controls, *in vitro* modification could spiral out of control, or perhaps be weaponized as countries attempted to create genetically modified soldiers of some kind. So Ortega led a group of 44 scientists in crafting new rules for *in vitro* modification. She and the cohort knew that it was an opportunity to open up *in vitro* modification as much as to guide it safely, and that for their work to have any teeth they would need to get approval from the governing health bodies around the world. Their guide, called the Rules for *In Vitro* Experimentation and Testing (RIVET), sparked a decade long campaign, headed by Ortega, to encourage the governments of the world to allow for *in vitro* modification.

During the 2160s, while Ortega's campaign was just beginning, biologists were making a lot of headway in understanding the biology of the Enceladans brought back to Earth a decade earlier. Through a series of experiments utilizing genome swapping to create Enceladan cells, scientists applied various stressors to those cells' environments in hope of inducing the rejuvenation process that the Enceladans seemed to possess. They could rule out a number of things, if the stressors were global situations, say the environment lacked oxygen or food, the rejuvenation would be expected in all Enceladans. That only a few exhibited the rejuvenation meant that it had to be something specific to that creature's environment, and likely that meant something in its food. That it was the carnivorous Enceladans that rejuvenated seemed to corroborate that hypothesis.

The discovery of the Enceladans, a group of alien creatures powered by hydrothermal vents on a small moon orbiting Saturn, created a lot of interest in the study of Earth's hydrothermal vent ecosystems. Dontrell Morris was one of those scientists intrigued at what Earth's systems could show us. Born in Chicago in 2136, Morris grew up dreaming about oceans

—both the ocean under the ice on Enceladus, and the ocean he couldn't drive to from his brownstone in the midwest. When he left Chicago for the marine biology programs of sunny southern California (he did undergrad at UC San Diego, and got his doctorate from UCLA), it cemented his love of the ocean. By 2168 he was splitting his time between field work at vents in the Pacific, and teaching at the University of Hawaii.

Morris watched the developments with the Enceladans closely as he carried out his work on hydrothermal vents. In 2168 he became convinced that the Enceladan rejuvenation was a result of food cycles on the vents. He set out to show that hydrothermal vents had a sort of boom and bust cycle where the creatures that depended on them would proliferate or contract respectively. On Earth this turned into a difference in population numbers, but perhaps for the Enceladans, who needed to survive a much harsher environment, the boom and bust manifested itself in periods of rejuvenation. Looking for this cycle, Morris started to find evidence that the cycle was dependent on which vent you looked at, providing the locality necessary to explain the Enceladan behavior.

After gathering evidence over the course of five years, Morris took his findings to NLSA and the scientists that were working with Enceladans. There the scientists combed through his data, looking for what could be the potential trigger for the Enceladans. They found that during the boom cycle, there's an increase in heavy mineral production. On Earth this leads to more gold, silver, cobalt, and zinc in the water, but on Enceladus, where the chemical composition lacked some of the milieu afforded the more massive Earth, the heavy metals produced would likely include copper, the same element the Enceladans used to capture oxygen in their blue blood.

Copper, of course, had been one of the environmental components tested with the Enceladan cells over the past ten years, but Morris's work renewed interest in the element. Perhaps the rejuvenation mechanism was more complex than simply subjecting cells to environmental stressors. It was pointed out that an increase in copper may increase the number of blue blood cells, and that that could have an effect on the rejuvenation. To test the hypothesis, scientists recreated an Enceladan organ along with a simple, yet functional, circulatory system. There they increased the number of blue blood cells, and watched the effect on the cells of the organ. In that way, in 2175, the first induced rejuvenation of Enceladan cells was performed.

2175 was an auspicious year for the trajectory of human longevity. It was in 2175 that several governing bodies released regulations for the first *in vitro* modification trials. Among those approving the treatment were the FDA in the US and the South American Medical Association (SAMA) of the South American Economic Zone (SAEZ), which covered most of South America. A key feature of the approvals was limiting human experimentation to modifications that "improved resistance to disease, and promoted human longevity". In other words, the agencies continued to block modifications that would improve the subject's capabilities. No making babies stronger, or smarter—even changing eye color, as Dang Yichen had done in 2078 was verboten. The agencies were still worried about countries using genetic modification for nefarious purposes, and they felt the restrictions were necessary to stem that potential outcome. Still, what was allowed gave scientists plenty of room to operate with. *In vitro* modifications, unthinkable a mere century earlier, were about to burst on the scene to create a new generation of long-lived humans.

Finn Roncel was born in 2092 in Portland, Oregon. He was a precocious child with a vivid imagination. When he was young his dad introduced him to classic video games from the era of a century before, that, along with his enjoyment of fantasy novels and tabletop RPGs, gave Roncel a love for character building and story telling. He stayed close to home when it came time for college, choosing to attend Reed college and study Physics. There he took some programming classes and found he was better at that than he was at solving Physics problems. In 2112, he dropped out of Reed to pursue a startup he had been recruited into.

Long hours, and intense deadlines at that startup caused Roncel to suffer a mental break in 2117, landing him in the hospital with a bipolar diagnosis. While there, he experienced days of intrusive thoughts, which he attributed to a group of friendly aliens, spiraling in a jumble that he interpreted as wanting him to start a company. Here I'll let him describe for you the major thread:

“I had been thinking a lot about cryptocurrencies and non-fungible tokens, and I got this idea, supported by the intrusive thoughts in my head, of what if aliens visited our solar system in a spaceship. Well like any group finding another group, we would need to find a way to trade with them in order to have friendly relations, but the spaceship would be far too far away for us to send any material goods. But, I thought, what if there was a digital good that we could trade? That's where I got the idea for Tradestol, and the first product of my company Visitor. But I knew that just making it wouldn't be enough, we needed something to launch with it so we started building games that used the Tradestol. Having Tradestol be tradeable *and* usable made it a resource, like gemstones, and that ended up being the key.”

Roncel left his first startup in 2120 at the age of 28, and founded Visitor. The initial launch of the platform included Tradestol tokens, and three games for them to be used in. Non-fungible tokens had of course been around since the early 2000s, but Tradestol contained two innovations that helped it win market share. First and foremost was how Tradestol was created, it was created when transactions were performed on the Visitor platform meaning that Tradestol did not have to be purchased to be obtained. Second was a property that Roncel dubbed half-fungibility. Non-fungible things are unique like works of art, but most of the useful things in this world are differentiated, but repeated. Think of beer. Each brand of beer is differentiated from every other brand, but beers of the same brand are not. Roncel called this property half-fungibility, and he gave it to Tradestol. Half-fungibility is what made Tradestol work in the games. In one game, it would become the equipment that characters used in an RPG, in the second it would be the collectible monsters of a game like Pokemon, and in the third it became the cards of a tradeable card game.

It took a while for Visitor to catch on, but once it did, it took off like a rocket ship. Within a decade, it had turned profitable, and in 15 years it went public, landing it the largest IPO for a gaming company in history. Visitor's most successful game was Locke's Legacy, a map-based treasure hunting game, which saw players going around the real world collecting Tradestol. In 2128, the company introduced character sheets where players could give themselves classes, and stats which they could carry with them into other games. Visitor was an open platform, meaning anyone could make a game using Tradestol and the new character sheets, and the two became a popular combo as the platform grew.

By 2142, the company had grown beyond games, and had cemented itself as a tech giant. It had turned the spatial data collected from Locke's Legacy into a 3D mapping of the world called Index One, which was used to create virtual highways off the ground for drone operation in much of the world. It had built the successor to YouTube, and Vyouer, in VisiTV. And its platform powered three of the top 25 websites in the world. Visitor's market cap made it one of the largest companies on the planet. But at 50, Roncel started to become enamored at the efforts to improve human longevity, and decided to throw his, and his company's largesse behind the effort.

To start contributing to the longevity effort, Visitor introduced the Vampire Network onto their payments platform. Utilizing their character sheet feature, people could become vampires (the mechanics were lore-heavy, such as only vampires could make more vampires through proximity). Vampires paid less at night, but more during the day on the payment platform, and of course games built with Planet Nine could give vampires whatever set of powers and weaknesses they wanted. Planet Nine pledged three quarters of the profit from the Vampire Network to longevity efforts. The move became incredibly popular with its customers, and created demand for similar contributions from other large companies. Thanks to the deviling process of the twenty-first century, using corporate profits for the good of humanity wasn't unheard of, and several other companies started to follow suit.

By 2175, when *in vitro* modification was starting to win regulatory approval, nearly half of all longevity funding was coming from private enterprise. This created a bit of a conundrum for the average person seeking to improve their child's longevity prospects through *in vitro* modification: do you use corporate-funded research or government-funded research to modify

your baby? In South America, an interesting solution to this problem developed. Lacking the economic cohesiveness of the United States, south americans worried that government funds wouldn't be used for the greater good of the whole, while corporate funds would be similarly targeted on what would benefit the corporations. A movement started to grow for people-controlled pools of funding.

Helena Lopez was born in San Cristóbal Venezuela on the border with Colombia in 2140. Venezuela had a long history of social upheaval from various political and economic systems, and even a century after the fact, the meltdown of the early twenty-first century was still felt by its citizens. Utilizing the new freedom of movement created by the establishment of the South American Economic Zone (SAEZ), Lopez studied political science and economics in Costa Rica, Peru, Chile, and Brazil. Through her studies, she became convinced that socialism, though championed vehemently by various South American governments, had never actually been tried; that all attempts had really been some form of state-sponsored capitalism, and that a system where the people actually held the means of production had never taken place on Earth.

Lopez bounced between various social movements early on in adulthood before landing on the longevity movement. There she argued, to much agreement from those in the movement, that only the people should have a say in how *in vitro* modification was trialed and developed, that only through technology-enabled direct democracy could humanity avoid the inequities that plagued the modern condition. She rose quickly through the ranks of the movement's leadership, a hierarchy that she recognized as necessary to the movement's success, but which had to relinquish its control when the actual voting came to take place.

They called the movement Las Eternistas, the eternalists. By the time SAMA approved *in vitro* modification for public consumption, Las Eternistas held considerable power with the medical community. They demanded that private and public funds be put in a trust controlled through direct democracy. Any citizen of SAEZ could join the movement and have a vote in the allocation of the *in vitro* modification funds. In 2175 their ranks were a whopping 43% of SAEZ citizens. A decade later, two thirds of SAEZ citizens were members. The trust, which had become the only way of getting *in vitro* modification done on the continent swelled to nearly a trillion dollars. Democratic actions were taken monthly to allocate funds for longevity trials, and to invest in the infrastructure necessary to make longevity treatments available. The constant actions made it possible for funds to move faster, and more efficiently than in more centralized economic systems. The result was a pace of advancement heretofore unseen on the planet.

The other economic zones on Earth all struggled to keep pace with South American innovation. In the US, where there was enough money to overcome some of their organizational challenges, they managed a distanced second. The EU, West and South Africa, India, China, and Asia Pacific all trailed the Americas as public distrust of corporations and government stymied progress on *in vitro* modification. Rhetoric maintained that this lack of progress was by design. After all, it would be at least half a century before scientists would know if their longevity treatments were even working if everything went according to plan. And if it didn't then possibly a generation of genetically modified adults may drop dead of some unforeseen side effect of the cavalier treatments. But this was all a thinly veiled attempt to cover up the fact that South America had figured out a way to accelerate its developments that the tried and true capitalist systems of the planet could not.

Such was the situation in 2191 when finally *in vitro* modification, and Enceladan rejuvenation converged. We now have two generations who have grown up with the normalcy of alien DNA being spliced into our own, but I cannot overstate just how outrageous this idea was viewed even in the scientifically advanced year of 2191. Taking genes wholesale from creatures born on a volcanic moon, and editing them into our children's DNA in hopes of providing everlasting youth was a bridge too far for many. In most of the economic zones, stodgy regulatory bodies put the kibosh on human-Enceladan hybrids. In South America, however, Las Eternistas simply asked if there were any volunteers.

Back in the twentieth centuries, anthropologists were stymied trying to explain the presence of humans across the pacific islands. They simply couldn't fathom how people could have gotten from one to the other over their vast distances. Now, of course, we know the answer is very simple: someone built a boat, and then convinced other people to get on the boat, and then they went over that way (towards the next island). We're talking about people that went to the moon because it was there. And people that then went to Mars because we had to. Remember, there's always someone who'll get on the boat.

72 thousand SAEZ citizens got on the boat to have their children be born equipped with auto-rejuvenation capabilities. So once again, SAEZ led the way on longevity treatments starting with a cohort of 1200 babies born in 2194. Kiko Hana, the world's oldest person, was 154.

Space 4

The closest known black hole to the Earth is a little more than 400 light years away, much too far for twenty-second century rockets to fly to in any reasonable time. Of course with humans living longer and longer, it wasn't outside the realm of possibility to send a probe there within a human life span, but it would still be an unsatisfying length of time for physicists to wait. Instead NASA turned to a twenty-first century theory that the elusive Planet Nine might be a primordial black hole.

Most of the time, a star needs a mass more than double that of the sun in order to collapse into a black hole. It was theorized that smaller black holes might have been possible shortly after the big bang—these were called primordial black holes. We felt pretty strongly that Planet Nine had to exist due to its gravitational effect on objects in our solar system. Its mass was predicted to be about ten times the mass of the Earth. When astrophysicists first started taking its existence seriously in the early twenty-first century, it was hypothesized that it might be simply too far away for telescopes to resolve, but Jakub Scholtz and James Unwin proposed in 2019 that the “planet” might be a primordial black hole. As telescopes got bigger and better over the years, Planet Nine stayed elusive, lending credence to the black hole theory. As we could see more and more of the cosmos it made more sense that the planet we couldn't find would be something we couldn't see.

So how do you find something you think is there, but can't see with telescopes? NASA believed it had the answer in the form of micro-probes. Remember Newton's second law that force equals mass times acceleration, if you have a constant force on a small mass, you can get a

big acceleration. Each microprobe weighed only 16 grams. NASA would pack a rocket full of these microprobes and then launch them 360 degrees towards the predicted orbit of Planet Nine. Each microprobe would come equipped with an accelerometer, and the components necessary to call home. When a microprobe found itself being pulled by gravity near Planet Nine's predicted orbit, we'd know generally where the planet was. Covering the 450 billion kilometer circumference of the distant planet's theoretical orbit would take about 450 million microprobes given NASA's propensity for redundant systems. The manufacture of so many microprobes and the rockets necessary to launch them was projected to take twelve years, and the flight to the far reaches of our solar system would take another 18 years, but thirty years, to find a black hole in 2186 seemed reasonable all things considered. The 2180s would be quite the decade for space.

The mission was dubbed Melissa after the Greek goddess of bees, the original microprobes. It won easy funding from congress after the agency's success with the Enceladans, and the public's interest in learning more about dark matter. While most of the agency got down to the business of building the mission components, NASA formed the Black Hole Readiness Task Force, a group of scientists and engineers tasked with figuring out what exactly we were supposed to do if and when we found Planet Nine really was a black hole. BHRTF is a mouthful so the group referred to themselves as the Niners.

Designing experiments in space is not easy. Earth has a magnetic field and an atmosphere to shield from radiation which might disrupt your experiment. On Earth you can easily build in all six directions if you need more space for things like accelerators. Perhaps most importantly, if an experiment doesn't go as expected, you can leave and come back on Earth. In space you only get one chance, everything has to go right, and you have to deal with the fact that space is a

dangerous void. Not only did the Niners have to design an experiment that would hold up in space, but it had to be performed by machines since no human could make the trip, and it had to work with an object that no one had ever even seen before—a primordial black hole.

Exacerbating the problems designing the experiment was that they would be dealing with gravity. Gravity is the dominant force on cosmic scales because it's always attractive so it can accumulate. This is all well and good when we're trying to understand the orbital periods of massive bodies in space, but on smaller scales, the force of gravity is too small to reliably detect. We couldn't measure, for instance, the gravitational effect of an apple, as the apple is far less massive than gravitational detectors can detect. Even a system as sophisticated and powerful as the Einstein Space-based Gravitational-wave Observatory would be hopeless at detecting the gravitational effect of any object or objects that you could take in a spaceship to Planet Nine.

Luckily black holes have an interesting property. The radius of the event horizon of a black hole, the sphere from which light can not escape from known as the Schwarzschild Radius, is directly proportional to the mass of the black hole. Distance is something we can measure very precisely, so if we could measure the Schwarzschild Radius of Planet Nine, we might have some hope that we can know its mass. And if we can know its mass then we can know whether feeding it material added to its mass, or turned into dark matter and continued on its way. Thus was spawned the most expensive space-based equivalent to throwing a rock down the quarry and checking the splash.

While NASA waited for the manufacture of nearly half a billion microprobes, they were preparing the next Poseidon mission for return to Enceladus. Poseidon 8 would deliver a new type of submersible to the frozen moon that would be able to stay underwater for weeks at a

time. It was powered by a new power system which could power itself from the heat gradient near the ocean vents. The Exo-Thermic Node ReGenerator (ETNRG) was essentially the solid state physics version of the power supply for the Enceladans themselves. This would give the submersible, dubbed the Nautilus after the ship to travel 20,000 leagues under the sea, the range to explore much more of the underwater world of Enceladus. Poseidon 8 would be followed a year later by Poseidon 9, which would add to Poseidon 6's energy production and provide a return vessel for any new specimens the nautilus might find.

With the launch of Melissa in 2168, and then Poseidons 8 and 9 in 2169 and 2170 respectively, the 2170s was a decade of anxious waiting. Poseidon 8 touched down on Enceladus in 2175, and got to work re-drilling the hole that had partially filled in since Poseidon 7. The Nautilus launched in 2176, shortly after the arrival of Poseidon 9. I would like to write that found in the greater depths of Enceladus were even greater wonders than what had already been found, but the Nautilus found largely the same kind of environments that had already been discovered, with largely the same creatures. Over the next two years only four new species were found. Three were thermovores, as the Enceladans which fed off the vents heat came to be called, and one new carnivore was found. Using the rules from the still relevant Guide to Extraterrestrial Scientific Interaction, five dead specimens were brought back to Poseidon 9 for return to Mars.

In 2182 as Poseidon 9 sped through the solar system back towards Mars, a special room at NASA came to life. Since its launch 66 years ago, a portion of all the radio telescope bandwidth on Earth had been dedicated to listening for the WCIP probe. On August 17, 2182, the first communication from the probe in 65 years was heard; telling listeners that it had entered orbit around Proxima Centauri b. Far too far away to wait for any kind of acknowledgement, the

probe started sending a flood of information about what it had found at the first exoplanet visited by humanity. The probe would wait 72 hours before sending its first images, but the data streaming to that room in NASA was incredible enough. In addition to information about the size of the planet, and its measured mass, the probe sent back its assessment of the dominant colors on Proxima Centauri b. There was the expected blue of liquid water, the textured browns of land masses, and (astonishingly) light purple. The room at NASA had developed quite a crowd as the information came in. Excited whispering began to fill the room, rocks weren't violet, the color could only mean one thing: life.

Violet-leafed plants had actually been a prediction for life on Proxima Centauri b way back in the first decade of the century by Michelle Hanson and Richard Long, two astro-botanists working out of Kansas State University. It had long been theorized that different star compositions might lead to different plant biology for harvesting energy from those stars. The signature green color of chlorophyll just happened to be what worked best with Earth's yellow sun, but that redder or bluer stars might require a different elemental composition. Hanson and Long worked out a possible Iodine based photosynthetic plant which might thrive near redder stars. Pure iodine is violet in color, and so the two astro-botanists wrote a paper positing that the plant life on Proxima Centauri b might be violet. The data the WCIP probe was sending back seemed to confirm the hypothesis—there was iodine-based plant life on the surface of the planet.

When the first images of the exoplanet began streaming to Earth, radio telescopes across the globe were tuning in to catch every byte. The images showed what could have been some earlier Earth with green replaced with violet. Scientists counted four continents spread over the surface of the planet. It seemed the total land area was smaller as a percentage of the total surface

area of the planet than Earth making Proxima Centauri b more of an aquatic planet. The distribution of violet over the surface indicated that the planet was more tropical than the Earth—large deserts like the Sahara and Gobi were missing as features from its surface. Polar ice caps seemed to develop as expected for a planet with so much water, whether one or both of them hid land masses beneath was unknown. Familiar weather patterns formed in white clouds hung in the atmosphere above the land. A few images even showed lightning in flashes across the sky indicating storm activity.

The WCIP probe was programmed to orbit the planet for thirty days and then descend to the surface. The descent had been hotly debated back in the 2110s, as not everyone was on board with the idea of landing a probe on a planet of unknown intelligence. Curiosity won the argument. To travel over four light years to miss out on an opportunity to land on the first exoplanet visited by humanity was just too bitter a pill to swallow for the scientifically minded. After all, the argument went, if there was intelligent life on that planet, it would either be sophisticated enough to appreciate the experiment, or it wouldn't be sophisticated enough to understand it. There was a Neal Stephenson novel that dealt with this very issue, and everything had turned out alright in that.

The part of the probe built to land on the surface was essentially a mechanical tumbleweed with cameras and sensors. Rovers are great for missions where you know you can land and that you'll have space to rove around, but no one knew what we were getting into on Proxima Centauri b, and the four lightyear distance made it impossible to give the probe any kind of instructions. It was felt that a lightweight probe designed to be carried by the wind was the best bet. It turned out to be a prescient decision as most of the planet's surface seemed to be

covered in violet-leafed trees. The tumbleweed probe could land on those treetops and explore them possibly for months. 32 days after that first communication from the Proxima system brought that NASA room to life, pictures from the surface of an exoplanet were delivered back to Earth. The violet leaves looked familiar and alien at the same time, with familiar veining structures and shapes. Underneath, wooden branches spread out in their familiar fractal branching.

Five days after touching down on the Proxima Centauri b treetops, the WCIP probe recorded its first flying creature. There was some debate amongst astrobiologists as to whether Earth-based taxonomical kingdoms like Plantae and Animalia should be used for alien life. The Enceladans answered the question to a degree by being similar enough to Earth animals to take on that moniker, but for species which would stay far away on Proxima Centauri b without scientists to look to their inner workings, it was unclear how to classify them. Hence the use of the word creature for alien life that was mobile like Earth animals. Astro-botanists felt less compelled to keep the separation, and overloaded plant terms to include the violet-leafed trees that the WCIP probe found itself on.

The WCIP probe was sent simply to observe the life on Proxima Centauri b. It did so by happily skimming along the tops of trees, and then to the ground, and eventually to the ocean. Avian, terrestrial, and aquatic creatures were all photographed over the course of the next few months. There was nothing that seemed to indicate intelligent life, however, no alien analog to the curious primates found on Earth. Of course the probe's reach wasn't sufficient to rule out intelligent life, but it was safe to say that if it was there, it wasn't as ubiquitous as it was on Earth.

The WCIIP probe and its visit to our closest exoplanet was more than enough to keep NASA busy for years to come. Astrobiologists now had four different planets and moons to compare life on. Some were making the case for a new probe to send to the nearby star system, one that could retrieve samples like the Poseidon missions and perform tests about the biology found there. Some were naturally skeptical about any further missions. There was a certain allowance afforded humans who wanted to explore their own solar system, but exploring other solar systems seemed to require a more careful hand. Finding as much life as we had, we had to start taking seriously the possibility that we weren't the only ones looking for it.

The Fermi Paradox had long been the question de rigueur as to the existence of extraterrestrial intelligent life. Simply put, the paradox queries if intelligent life is out there, how come we can't find it? For example, humanity had been broadcasting electromagnetic waves out into the cosmos for over two hundred years. If there were lots of other civilizations broadcasting, why can't we hear any of them?

There are several so-called solutions to the Fermi Paradox, one is simply that intelligent life is rare enough that two intelligent civilizations close enough to each other to hear each other's transmissions just doesn't happen. Another, gaining traction in the early twenty-first century due to the popularity of a book called the Three Body Problem, was the Dark Forest theory. Intelligent life, it said, was like tribes trying to survive in a dark forest where alerting others to your presence spells certain annihilation. It was here that a number of scientists were turning. If life was as common as it seemed, then intelligent life is probably pretty common. Maybe the reason we hadn't found it is that everyone else has enough sense to stay hidden.

Maybe humans sending probes out to other stars was us just begging to have our hand smacked, or, you know, have some galactic destroyer swing by and eliminate us.

Like any good existential threat, the debate over our place in the universe and what it meant for the further exploration of Proxima Centauri b raged for years. It even drew in the public, with politicians and other leaders weighing in on the matter.

The debate was still raging when in 2186 the first data started coming back from Melissa. Not surprising, but disappointing nonetheless, the first microprobes to reach the theorized orbit of Planet Nine showed no signs of gravitational pull. With 450 million microprobes to worry about, some of which wouldn't reach their destination for another five years, it would mostly be a waiting game. With something like three microprobes sending back their data every second though there was plenty to be watching for.

It wasn't until 2189 that a microprobe found something, a slight deviation in its flight trajectory most likely caused by the gravitational pull of something massive. Shortly after, more microprobes started reporting back the effects of this gravitational pull. Astrophysicists rushed to rule out other possibilities for the anomaly such as errant asteroids, but thanks to space being largely empty, it was relatively straightforward to verify that when something happened it was because of what you thought it was. The handful of microprobes being affected by the gravitational pull made calculations on its strength and location. It was likely a mass about 9.3 times the mass of the Earth located 482 AU (70 billion kilometers!) from the sun, about a third of the way between its proposed orbital aphelion (furthest from the sun) and perihelion (closest to the sun). The microprobes came equipped with very limited cameras to try and take a picture of the elusive planet. When the images started streaming back they showed nothing at all, just the

blackness of space with dots of stars around them. The Schwarzschild Radius of a black hole of 9.3 Earth masses is about eight centimeters. So there was a lot of mass, but nothing to see, just what you would expect from a tiny primordial black hole. Physicists were ecstatic.

The Black Hole Orbiting Probe (BHOP) had been built during the 2180s for the purpose of examining Planet Nine should it turn out to be an actual black hole. Equipped with experiments developed by the Niners, it had all the sensors and detectors one could imagine—the space agencies involved weren't about to take any chances on missing out on anything the black hole wanted to show us. NASA rushed it onto a rocket in the months after the microprobes' discovery of Planet Nine's location, and in late 2189 it took off on a 21 year journey to the far reaches of our solar system.

When it arrived in 2210, it immediately put itself in orbit around Planet Nine at a distance believed to be safe from the powerful tidal effects of the black hole. The improved cameras on the BHOP allowed it to study the effects of Planet Nine on background stars, seeing some of the tell tale gravitational lensing that we would expect from such a condensed mass. The lensing effect was not quite what our models had predicted, but that was exciting, it meant that there was new physics to be found at the black hole. Other readings of the environment around the black hole were in line with predictions, a win for theorists' models.

After a month of recording everything there was to be recorded about Planet Nine, the BHOP was ready to perform the first test to see if the radius of the black hole could be increased. The measurements would have to be extremely precise, as increasing the mass of the black hole by a ton would only affect the black hole's radius by about a quadrillionth of a meter. Measuring the radius directly was not possible, but the change should introduce a detectable wobble in the

orbit of the probe. It was this wobble that scientists were eagerly awaiting when the probe jettisoned a one ton mass towards the location of Planet Nine. The mass approached as expected, but then seemed to deviate from course, entering into orbit around Planet Nine. NASA scientists checked and rechecked their calculations, and no one could give a justification for the deviation. The most important experiment of the BHOP had just failed, and no one knew why. This probably meant new physics too, but people were a lot less excited about that this time around.

Of course we now know the reason for these deviations, but at the time they were mystifying to astrophysicists on the ground trying to puzzle them out. Before we get into that though we need to take a look back at the socioeconomic forces at work around the world.

Socioeconomic 4

No history of the world's socioeconomic changes in the twenty-second century would be complete without talking about the worldwide collapse of organized religion. The beginnings of this collapse can be traced all the way back to the beginning of the twenty-first century as belief in god started to fall in western Europe. Even the United States, home of the theocratic Christian Republican Union, saw belief in god falling all the way back during the first and second decade of the twenty-first century. It took nearly a century for this healthy skepticism to spread across the globe, but once it did, organized religion, lacking the collective will to perpetuate it, dissipated across the earth in a whimper that thankfully didn't match its historical propensity for destruction.

The diminishing (it was originally Sanjeep Ramachandran of the University of Ohio who first named the phenomena of dwindling support of religion as “diminishing”) of something as universal as organized religion is hardly the result of any one thing. Looking at the early diminishing in western Europe, we might think that it was a result of liberal thinking in mature economies, but that hardly explains the later diminishing in the middle east and central Asia where economies were historically dependent on fossil fuels, and lacked the maturity of countries to the northwest. The diminishing in central and south America similarly lack the economic maturity of more northern latitudes, but nonetheless diminished in kind. And of course there's the case of Africa, behind western Europe in every economic metric, and much more conservative, which nonetheless diminished religion even faster than the afore mentioned groups. Some hypothesized that since Africa had had religion foisted on them via colonization, it

was easier for them to throw off its yoke, but the diminishing nonetheless left social scientists scratching their heads for some common rationale.

One obvious unifying thread was the role of science during the first two centuries of the second millennium. There was the work science was doing to undo the inevitability of death. It was religious groups who fought hardest against the inexorable progress made on human longevity, and as longevity treatments grew more and more successful, the religious luddites railing against progress looked more and more foolish. Somehow the notion that death had been provided for the benefit of mankind by some creator seemed more far fetched as we undid its inevitability.

Perhaps an even larger blow to human religions had been the discoveries of former life on Mars and current life on Enceladus and Proxima Centauri b. That mankind occupied a special place in the cosmos, put on a paradise of an Earth by some divine providence, had long been a hallmark of religions across the globe. The fact that life seemed not just not to be unique to Earth, but possibly even common in the universe undid millennia of thinking Earth was at the figurative if not the literal center of the universe.

Beyond pure science, the success of the transformation of the global economy from carbon spewing fossil fuels, to carbon consuming Mahd Arrays and RIVER projects had ushered in an era of global cooperation that made the historical lines of division between us seem antiquated. The economic zones established to harness the production of RIVERS created open borders and systems of free trade which saw nations intermingling like never before. Long held prejudices, often hardened by religious rhetoric, were broken down in the gears of shared economic success.

Certainly education played a dramatic role in the diminishment of religion. As the educational prospects of students around the world improved, religion's footholds within generations became narrower and narrower. With global support for education growing throughout the beginning of the twenty-second century culminating with resolution E/RES/2115/4 in 2115, a worldwide effort to ensure the global population would meet some standard of education did some heavy lifting with regards to diminishment. By 2160, two generations had gone through real education, and not indoctrination in most of the countries on Earth. Without the veil of ignorance, religion faltered trying to find a way to hook into these generations' psyches.

Religion best thrives with peer pressure. Large groups of people hold each other accountable to religious norms through shame and sometimes force. But to get there in the first place takes a critical mass of the population. As newer generations bucked the cultural shackles religion had placed on their societies for centuries, its diminishment accelerated toward collapse. In much of the world, democratic institutions were able to absorb this instability as secular governments took over from the more theocratically minded. In many of the nations where religion had held particular power however, violent protests and in some cases full on revolutions broke out to topple the entrenched theocratic regimes.

The period from 2075 until 2150 saw unprecedented political upheaval culminating in the near eradication of theocratic rule across the globe. Socioeconomic improvements in India undid the entrenched caste system and eroded the hindu sensibilities that underlay that society. Across the middle east, religious islamic leadership was overthrown and replaced with educated secular rule. The entrenched catholicism of south america seemed to evaporate over the course of three

generations as education and economics improved. In China where the state had officially been atheist since the mid twentieth century, religion as a cultural phenomenon simply faded away. In the United States, without religion to fuel the white rage holding the CRU together, the split republican party came back together as Republicans in a new secularized conservative party whose platform, while still frustratingly behind the times, lacked the same callous disregard for human life that had been its north star in the early twenty-first century.

2150 to 2190 was a cooling off period of the tensions built up during the political upheaval of religious regimes. Armed conflicts would flare up here and there, but lacked the staying power of religiously fueled wars. It was an era of unprecedented peace on a planet that had seen nearly constant war for thousands of years.

It started as a meme in January of 2193, a picture of an American politician's face looking surprised with text that said, "When you forget to start a war all year." According to the internet, 2192 contained no wars for the entire year. This was such an unthinkable occurrence at the time that there were no official trackers for the number of wars on the planet in a year, but the meme's success caused sociologists to look into the matter. By the end of January, it was widely reported that 2192 had been the first year on record, and likely the first year since the first human fashioned a club, that humanity had not had a war. The news garnered widespread attention, and groups formed to champion the idea of never having a war again. In governments around the globe, and at the UN, rhetoric began to take shape around a true anti-war effort, one that could make armed conflict a thing of the past.

It was around this time that Rachel Golding wrote a landmark book on the history of war and religion. In her book, *The Killing Brain: How Religion Shapes Our Capacity For Murder*,

she argued that religion was actually instrumental in humanity's propensity for violence, claiming that it is religion's ability to turn other groups into enemies that provides us with our capacity for killing other humans. Further she wrote that absent that compelling force, humans were free to take on more pacified approaches to conflict resolution. Golding's book focused the anti-war effort, giving the group something that explained the phenomenon, and gave them hope that the phenomenon could continue. In 2194, Golding was asked to give a talk at the UN on her research. There she implored the legislating body to seize the moment and usher in a new era of peace.

Stopping wars, a task that seemed impossible mere years before, had occurred without any intervention, certainly a conscious effort to prevent them from happening again would be even more successful. So was the thinking that dominated the UN and governments around the world. But addressing the grievances of the totality of humanity in a way to obviate the need for armed conflict was no simple task even without the bellicose force of religion.

Sociologists looked into the wars of the last two hundred years and found that for the most part, armed conflicts *between* nations had stopped back in the late twenty-first century. RIVER projects, economic zones, modern trade agreements, and shared education resources had all worked together to end the need for territorial expansion. Combining that with the beginnings of diminishment, and the need for one nation to attack another was replaced by the need to work together. It was conflicts within nations that were still causing trouble.

Enter Los Modernistas, a socially democratic group representing the SAEZ that had grown out of the success of Las Eternistas and similar groups fifty years earlier. Los Modernistas supplied twelve of the eighteen national delegates of the SAEZ to the UN, and through those

delegates they argued that since the cause of the majority of armed conflict in the last century had been struggles for political power, the only way to prevent them is to allow for a peaceable transfer of power. They proposed what they called a *coup de les personnes*, a coup of the people. Essentially any group could overthrow the government provided they had the support of the people, they argued that such support would be sufficient to violently overthrow the government, and thus sanctioning it through a democratic process delivered the same result without the bloodshed.

It was a radical idea. In general, governments aren't in the business of designing ways for them to be overthrown, but then again many governments held periodic elections that could turnover their governments. The United States as an example holds elections for at least 87% of its government every two years, and 100% over six years; would a possible *coup de les personnes* be that much different? In countries where power had been won through bloodshed, the holders of that power tended to hold onto it tighter and with less regard for recompense from the people. Could they ever find the political will to adopt a measure institutionalizing their own removal from power?

The *coup de les personnes* idea gained widespread backing from democratic groups worldwide, but largely in a not in my backyard sense. Sure the coup measure was necessary in far off lands on other continents, but not in our well structured democracy went many of the arguments. Some of this was true, nations which possessed already some mechanism for turning over their government in reasonable time periods basically already had *coups de les personnes*, and establishing some new mechanism for that turnover would be confusing at best. For the rest of the nations who could benefit from the coup provision, it was unclear how to move forward.

Los Modernistas decided to lead by example. They acknowledged that governments structured to turnover on a regular basis didn't need the coup provision, but for the six governments in the SAEZ that didn't turnover regularly, they started campaigns to adopt the coup provision into constitutions. It took until 2198 to get those measures adopted, but they ultimately prevailed.

The movement to adopt the coup might have died in South America had it not been for forces at work across the middle east and north Africa. The collapse of religion may have removed the accelerant from the fires of cultural divisions, but those deep historical rivalries still smoldered. The anti-war movement in the regions had to address the grievances of different groups, and the *coup de les personnes* was a way of giving some modicum of hope to historically oppressed minorities that should a government overstep, they could be removed. Pressure from the movements on governments in the region led to the coup being adopted across the middle east and north Africa. After the example set up north, south Africa followed suit, as did Asia Pacific, so that by 2206 the *coup de les personnes* had been adopted by 72% of the countries that qualified for its use.

Anti-war movements gained momentum throughout the first and second decades of the twenty-third century. By 2217, 25 years after that first warless year in 2192, calls for a worldwide celebration of a "generation without war" were leading to a new year's celebration for the ages.

Sociologists, economists, political scientists, and anthropologists gleefully studied the effects of a world without war. There are a whole slew of nasty things that come along with war. Famine often hits one or both warring groups as food infrastructure is either coopted for the war

effort, or just outright destroyed. Disease comes along as a result of the destruction of sanitation infrastructure. Fighting puts a pause on economic development, and societal progress. Education suffers as families worry more about staying alive than the future prospects for their children. But all of that was gone, without war fueling these negative outcomes, historically war torn areas were recovering, and with concerted efforts to help them recover from anti-war groups around the world, their prospects weren't just improving, they were accelerating towards a worldwide baseline significantly higher than it had ever been.

World peace. It had been a notion held almost exclusively by counter cultures for so long that even while people were in the midst of its development, they couldn't believe that it was happening. Free from the shackles of near constant conflict, the world could focus on making life better for everyone rather than simply trying to meet basic needs. It was an incredible shift in mindset, and one, it turned out, that couldn't come too soon.

Planet Nine

At 3:21p CDT on January 3rd, 2218, an alarm went off at the Planet Nine observing station at NASA's Goddard Center in Greenbelt, Maryland. The alarm indicated that one of the BHOP's sensors was picking up an anomaly. Eagerly Becky Mathers checked her console for what had come to life on the probe. Astonishingly it was the probe's camera that had sent back the signal. She switched her monitor over to see what the camera was seeing.

Where Planet Nine had just been blackness there was now a black sphere, not the empty blackness of space, but the mirrored blackness of a screen that is off. Towards the top left of the sphere was a blinking green rectangle that almost looked like a cursor. Mathers couldn't believe what she saw, and called over a supervisor. Together they made some calculations as to the size of the sphere based on the known distance of the probe. The sphere had a radius of 12 km. This made the object incredibly dense, far more so than any planet might be, and more than most stars.

Congratulations Humans!

Mathers blinked her eyes in disbelief. Did Planet Nine just spell out words on its surface which moments before was just empty space? The supervisor called for technicians to check the video feed from the BHOP to see if it had been hacked.

You have completed the four things Fours must do.

The green glowing letters scrolled along the screen like some old monitor you might find in a movie from two hundred years earlier. Mathers looked around for any help in understanding

what was going on. The words on the sphere could only mean one thing though. Planet Nine wasn't a black hole. Planet Nine was a spaceship.

1) Gain the capacity for becoming a Type I Civilization

Mathers felt this referred to the Kardashev Scale. A Type I Civilization was one that could harness all the energy that fell on its planet from its star. Could dark matter fueled fusion power be the capacity they were talking about? Mathers shook her head at using the word "they" already.

2) Beat Death and become biologically immortal

Aware of NASA's sister agency's efforts at improving human longevity, Mathers wondered if this one might mean that the rejuvenating generations might actually live forever. After all that's not something you can really test without time, maybe we had already gotten there.

3) Live on another planet, or visit another star system's planet

That was a clear success for NASA both with the Masters Mars Base which housed a permanent population of around 250, and for the WCIP probe tumbling its way across Proxima Centauri b.

4) End all wars for a generation

Having just celebrated this milestone a few days ago, Mathers felt confident in this one as well. It occurred to her that the timing of the images would be right in line with if the message had begun at the stroke of midnight when the international date line passed into 2218 indicating this fourth one might be what the spaceship was waiting for.

You are almost ready to join The Advancement. To do so you must continue building Five.

The message had said the four things Fours must do, and now was talking about Five. Mathers shrugged at her supervisor trying to puzzle out what they meant.

You are at 3/5, we are at 4/5, once you get to 4/5 we shall talk further. The parts of Five we know are Stacks, Boats, Screens, and Teleporters.

One of those four things was not like the others. Stacks, boats, and screens were things that existed, teleporters did not yet, though Mathers was aware that her colleagues working with the Dark Field believed that it might be possible some day if some aspects of decoherence and recoherence worked a certain way. Were the aliens trying to tell them that they needed to build teleporters?

We'll send you a gift to help you along the way. We'll land it at 2:487:323:62 in 42 days.

Mathers recognized the coordinates as Index One coordinates, the three dimensional coordinate system devised by Visitor and used for commercial spatial locations throughout the solar system. The 2 at the beginning indicated that the coordinates would be on Mars. A quick lookup on the Index One website showed that the landing spot was in an empty flat region within a few hours rove from the Masters Mars Base. She then did a quick back of the envelope calculation, on how fast something would have to be traveling to get from Planet Nine to Mars in 42 days—something like one fifteenth the speed of light. She pondered what in the universe could be on that sphere to accelerate something to that speed in that amount of time. Then she

thought about how that thing was supposed to decelerate so as not to blow a giant hole through Mars. Even the most out there doomsday scenarios for cataclysms from space hadn't included space probes at relativistic speeds, Mathers would take this back to the planet defense task force, but ultimately she guessed we'd just have to trust our new alien neighbors.

You can find instructions on what to do with it at www.planetnineisaspacehip.com.

Mathers navigated over to the website. She found a 3D model front and center of what the aliens had called a DecRec, she skimmed the accompanying text to find DecRec was short for Decoherer-Recoherer. She had only heard those terms as they pertained to objects transitioning between the Light and Dark Fields. She knew something of the process—accelerating objects by spinning them very fast to decohere, and then recohering in nuclear fusion. Was the DecRec able to transition objects between the fields without those violent inputs? The website contained a link to an instruction manual for the DecRec, which Mathers downloaded. The site stated that 64 DecRecs would be delivered in 42 days (there was a slick spacewave-y countdown timer), and that humans should learn from them and the spaceship they would arrive in.

We'll get back in touch once you figure out teleportation. Good Luck!

The last line of the message hung on the screen for two minutes before everything printed on the screen faded away, and the black sphere was left hanging inert against the void of space. Mathers, her supervisor, and a handful of other NASA technicians who had been drawn in by the

message looked at each other in disbelief. A call came back from IT letting everyone know that no evidence of a hacked feed could be found.

Humanity in 2218 was ready for contact. Centuries of scientific exploration discovering life on other planets and moons, combined with books, movies, games, and various other cultural artifacts exploring the idea of contact with an alien race had convinced most people that running into another intelligent species was inevitable. With the socialization of the idea came a reduced need for secrecy should contact happen. The US had adopted science-based protocols rather than shadowy secret operations to handle contact, and many other countries and economic zones followed suit. So Mathers's job after witnessing the first contact from aliens, was to write it up and make it publicly available just like any other scientific discovery. She worked late into the night to make the message available via an astronomer's telegram, a sort of memo that astronomers released before releasing a formal paper. January 4th was going to be an interesting day.

Dark Field 5

The first thing to start figuring out, after picking our collective jaws up from the floor at watching our first alien contact, was what to do with the DecRecs. It took a few days of debate, but eventually it was settled to divide them into sixteen groups of four to be delivered across the globe for study. Four was picked as the DecRecs were paired, one to decohere and another to recohre, and the groups would want to dismantle some of them to study them. Four gave each group two to dismantle and a pair to operate. The sixteen groups chosen to receive DecRecs were NASA, Fermi Lab, AEM, SAEZSA, BSA, NAEZSA, AzSA, ESA, CERN, Roscosmos, MESC, ISRO, CNSA, JAXA, ASA, and Mars. Put another way, two to the USA, one to Mexico, one to the South American Economic Zone, one to Brasil, one to the North African Economic Zone, one to Southern Africa, two to Europe, one to Russia, one to the Middle East, one to India, one to China, one to Japan, one to Australia, and one to be kept on Mars.

While the logistics of DecRec apportionment were being figured out, scientists and engineers around the world were diving into the instruction manual found at www.planetnineisaspacehip.com. The manual was written for a technical audience, but the gist of it was that using some unknown process, the DecRecs created a pocket of spacetime which was separated from the rest of spacetime with something called an Action Containment Enclave (ACE). The DecRec extended an arm into the ACE, and everything connected to the arm would decohere and recohre at the paired DecRec's ACE. So the DecRecs were, in effect, teleporters, although the manual indicated that their operation had been restricted to block teleportation from happening. Given the mission from Planet Nine's message, people believed our task was to

deconstruct the DecRecs so we could build our own. The DecRecs that were on their way also only had an ACE about half a meter in diameter, too small for humans to use, clearly we had to reconstruct them and build them bigger.

Over the next couple of days, the different groups of scientists and engineers began to fall into particular lines of investigation. MESC (Middle East) and ISRO (India) for example took particular interest into ACE production, combing over the manual for clues on how the mechanism worked. Fermi Lab and AEM in Mexico started a partnership around how the recoherence mechanism worked, trying to figure out how recoherence was possible without the incredible energy output of a fusion reaction. NASA emerged as a sort of program manager for the worldwide effort, providing coordination and resources, and acting as the central processing unit for discoveries coming out of the group. By the time the alien's ship touched down on Mars on Valentine's Day 2218, more than 24,000 scientists and engineers were standing by to begin the work of understanding the alien technology.

The alien ship was recognizable enough, a metallic hull which opened up to reveal shelves of DecRecs. Video of the ship landing had revealed that it didn't propel anything out of it the way that our ships did to decelerate. Obviously there was more to be learned than just the DecRecs. The DecRecs matched the pictures and 3D model found on the website. Each was one meter high, resting on a tripod of adjustable feet (the manual indicated the machines worked best when level). The base was a four sided pyramid, which contained the buttons for operating the machine. On top of the base was a translucent octahedron topped by an opaque black sphere which seemed to actually trap light within it. Inside the octahedron was a ridged ellipsoid disk

which seemed to rotate freely, almost hovering inside. The back of the base pyramid took a USB-J input, which, the manual had explained, meant that the machine was digital under the hood.

The 58 days it took to get the DecRecs back to Earth would have been interminable if not for the team of scientists and engineers on Mars getting to work on deconstructing their DecRecs. The first astonishing discovery was that the ridged disk structure and the arm were not composed of atoms. It seemed to be a whole new type of matter not seen on Earth called nuclear pasta. Theorized to exist within the crusts of neutron stars, nuclear pasta shouldn't even have been stable without the extreme conditions that neutron stars created. Nonetheless, the matter existed as an elliptical disk within the translucent octahedrons of the DecRec. Notably, all four disks of nuclear pasta were different, indicating that it was the material that mattered, and not the layout specifically.

The translucent octahedron was polished reinforced glass, similar to what was used in personal electronics, but stronger by a factor of twelve. The base was made of a flexible, compressible, silicon-based material that absorbed movement from the octahedron while the machine was in operation, and housed the motor that spun the disk in the octahedron. The sphere at the top was made of the same material as the base, and contained what appeared to be the bulk of the electronics needed to make the machine work. The alien electronics looked recognizable as well, different to be sure, but definitely based on the same principles. Below the disk in the octahedron was a blue glowing gemstone that the manual called Nineum, with a brief explanation saying it was a stone infused with a form of a mixture of light and dark matter, and a note that it was not necessary for teleportation.

When a DecRec was turned on, an arm made of the same nuclear pasta material as the disk extended from within the base. With a simple voice command, the decoherence process could be started. A translucent light blue glow marked the edge of a sphere that surrounded the DecRec. The sphere marked the ACE and within it nothing could be seen, so what one saw looking into the sphere was the same as looking straight through it if it were empty. The sphere only lasted 2.3 seconds before dissipating, and the machine shutting itself off. Experiments that set things touching the arm verified that the machine decohered and then recohered the objects in the sphere. It seemed the only thing missing for teleportation was moving a decohered object to another DecRec, something the aliens had left some instructions for in the manual, but had restricted in the programming of these DecRecs.

When the rest of the world started getting their hands on the DecRecs in April of 2218, papers came pouring forth at a rate not seen before. The materials of the DecRec were broken down through spectroscopy, and people got to work recreating the base silicon and the octahedron's glass. Scientists at CERN took particular interest in the nuclear pasta disk, and its accompanying arm. The exotic material had only ever been theorized, and now they held a stable example of it in their hands. The Nineum got everybody talking. For something to contain stable light and dark matter *within* it, was not something that anyone had considered. The small gem-like crystals glowed blue, and speculation ran wild as to why the aliens would have included them with the DecRecs.

The first insight came from Werner Bernt, and a team of seven other researchers working on the upgraded DaMaGe project at CERN. Bernt had become obsessed with answering why the objects placed in the DaMaGe accelerator decohered, but the materials of the accelerator itself

did not. This fact had of course been known for decades, and was an accepted phenomenon of acceleration-based decoherence, but Bernt was not happy to let the phenomenon continue unexplained. He conjectured that since the materials of the accelerator were connected to parts that did not accelerate as much, or at all (if you think of an arm spinning around its center, the parts of the arm nearest the center are barely moving at all), that decoherence wouldn't take place. This led to a series of experiments to test this hypothesis of connectedness. By 2218, scientists were convinced that the connectedness of rigid bodies played a part in decoherence.

Now with DecRecs available to decohere without accelerating objects in a chamber, Bernt was asked to devise experiments to test his theory. Through those experiments, it was found that objects connected to objects outside of the ACE at the time of decoherence would not decohere, nor would objects not touching the nuclear pasta arm in the ACE. This was of great benefit to those testing decoherence as they could setup cameras and sensors within the ACE to monitor what was going on while the machine was operational. It seemed that this property of connectedness was inherent to the decoherence process, and that the nuclear pasta arm was somehow able to confer on anything touching it a sense of connectedness to the decohered object except when that object extended outside of the ACE.

A second theory was confirmed by MESC and ISRO in a series of experiments with recoherence. Back in 2184, a group of scientists led by Hemani Asadayal had held experiments with objects made from elements above iron on the periodic table of elements. Elements with a higher atomic number than iron don't fuse. Asadayal and her cohort were trying to determine if information from before decoherence persisted after recoherence. It took building a reactor capable of small sustained fusion reaction in an attempt to create an environment that would

allow the objects to persist through the recoherence energy production. The results were inconclusive, but observations of the reactor after recoherence suggested that at the very least the chemical composition of the objects persisted through the decoherence-recoherence loop. Now with DecRecs, it was clear that objects' makeup was persisting through recoherence.

Once it was understood that information persisted between decoherence and recoherence, a group of scientists split off from their DecRec groups to explore if information could be changed between decoherence and recoherence. These scientists called themselves The Alchemists after the ancient debunked art that had tried to turn lead into gold. Transmutation was, of course, an extreme example of what the alchemists were trying to accomplish, but if properties of matter could be changed while decohered, then why not transmute things? The alchemists' goals overlapped with a big goal of the DecRec project, namely how to apply some velocity to decohered objects. Doing so would be a change of the inherent properties of an object while in a decohered state. Since this change of velocity was essential for actual teleportation to take place, support for the alchemist effort grew over the years following the DecRecs' arrival on Earth.

As the alchemists worked on how to change the information of a decohered object, electrical engineers and computer scientists around the world were trying to figure out the electronics contained in the sphere of each DecRec. One of the most puzzling pieces is why the machine needed a digital component in the first place. As far as anyone could tell, the machine was mechanical, a spinning disk encased in an exotic glass housing. The electronics looked like something you might need for a high end computer, but without any input or output, scientists and engineers were left guessing as to what its purpose was. The manual stated that the DecRec

computer was necessary for the transmission of decohered material to a paired DecRec, but left the how of that as a big open question.

There were two clues. Next to the on/off switch on the base of the DecRec was a button that the manual said could be used to give the DecRec a name. It was by using these names that DecRecs were able to transmit decohered material for the purpose of teleportation. The presence of this switch, and the voice command used to start the decoherence process, indicated that part of the purpose of the DecRec's computer was for a vocal interface. If you could set the name by speaking it, it stood to reason that other commands were possible. The second clue was that the computer contained a series of extremely powerful antennas for the size of its housing. These antennas were more than was necessary for simply listening into something like the world's wifi networks, they must be transmitting antennas as well. It stood to reason that DecRecs would be able to network to each other in order to facilitate the teleportation signals necessary for coordination. Next to each antenna was a tube of the nuclear pasta material of similar size. The manual stated their purpose was "transmission in the Dark Field," a purpose that was as yet unknown.

Zayla Toren was born in 2185 to two parents working in the medical field. Growing up in Topeka, Kansas, Toren took an early interest in science and engineering. By the age of twelve she was building robotics at a level that saw her joining the local high school's robotics team, which went on to take state in competition. She gained a reputation for pushing the envelope, often to the chagrin of the adults charged with watching her. She graduated a year early and headed to college at Carnegie Mellon in Pittsburgh, Pennsylvania. After getting her Doctorate from CalTech in 2210, she joined NASA working on developing the next generation of

autonomous probes and rovers. By 2218 she had distinguished herself, and earned a position on the team of scientists and engineers that would get their hands on NASA's DecRecs.

It wouldn't be until 2219 that Toren would get her turn with the DecRecs, but when she did she tried something that no one had tried before, she issued voice commands beyond the start command. The first one she tried was to move the object being decohered ten centimeters up. When the object recohered it fell ten centimeters to the ground. Toren's group members eagerly joined her in trying out new voice commands. They found that so long as they were careful to make sure the object stayed within the ACE, the machine could do pretty much anything kinematically. This led Toren to wonder as to the extent of what was possible to do with the DecRec. She and her group branched out into other realms of Physics. They found they could adjust the temperature of the object, confer an electric charge to it, blow it up, crush it under pressure, and even change its chemical composition. Essentially if you could dream it up within the confines of the laws of Physics, the DecRec could do it.

Toren and her group published their findings in March of 2019. It had an immediate impact on the Alchemists' research, as it demonstrated that information could be added to a decohered object to change it when it recohered. Remember the nuclear pasta antenna used for "transmission to the Dark Field," the alchemists surmised that there must be some way of transmitting information to the Dark Field which could influence decohered objects. It would take seven years before scientists were able to figure out exactly how that mechanism worked.

Yaeko Noguchi was born in Nagano, Japan in 2164. Her parents had met while working on a game at Nintendo, and instilled in Noguchi an appreciation for the medium. She took particular interest in game physics, specifically in the digital recreation of the laws of Physics. In

2226, at the age of 62, Noguchi looked and felt like she was in her late twenties, a beneficiary of science's latest advances in longevity treatments. By then she had been teaching for eight years at the University of Tokyo while researching simulated physical systems. Not originally selected to join the Japanese contingent of DecRec researchers, she monitored the world's progress in that regard closely. In 2220 she joined the Alchemists with the belief that the DecRec's vocal interface conferring properties to decohered matter likely meant there was some digital to dark field conveyance mechanism that she could use in her research on simulated physical systems. Her novel insight was to start from the premise that some interface between digital and dark field mechanics existed, and working backwards on how that might work. Something her work on simulated physical systems uniquely prepared her for. She recruited a team of Alchemists to join her, and together they got to work.

First her and her team conjectured that there must be some way to interface with decohered objects as a whole—not only would it be too complicated to have to interface with every individual atom of a decohered object, but the fidelity with which objects recohered would indicate that the whole of the object occupied some preferred spot in dark spacetime. This conjecture led the group to pursuing wave mechanics as the principle source of interaction. Work done over a hundred years ago on the quantum mechanical underpinnings of dark spacetime had led most to believe that dark matter existed as a sort of wave that had lost the ability to render as a particle. Noguchi and team began to question just what kind of wave the decohered dark matter could be.

To try and answer that question, Noguchi had to reach all the way back to undergraduate Physics. She remembered that two properties affected by Heisenburg's Uncertainty Principle,

like position and momentum, can be shown to occupy frequency spaces of each other. For example, the more accurately we know a particle's position, the less accurately we can know its momentum, and vice versa. That is because, mathematically, position and momentum are related in such a way that knowing one affects our ability to know the other. Noguchi wondered if this mathematical quirk had a real physical meaning in the wave dynamics of the dark field. She led her team through the construction of a simulated physical system where the properties of matter effected by the uncertainty principle were represented by standing waves within a decohered object. Through manipulation of those waves, they were able to affect the underlying properties, much like the DecRecs were able to do with actual decohered objects. Noguchi and team published their findings in the fall of 2226 to a mixture of enthusiasm and skepticism that simulated physical systems could actually mimic the nebulous world of dark matter. It had been eight years since the DecRecs had arrived, however, and this was the first real attempt at explaining how dark matter manipulation worked. Most of the scientific community chose to run with it.

While Noguchi and team were working out the what of dark matter manipulation, scientists around the world were trying to figure out the how. It was clear that the nuclear pasta material allowed some interface between the light and dark fields, but attempts to figure out that interface had thus far failed. That didn't stop people from trying to make the stuff though. CERN was leading that effort, and though slow, progress was being made. Central to the effort was a question that lay outside of Physics—what were the aliens of Planet Nine's technological capabilities? Answering that would give insight into what manufacturing capabilities they had,

and thus what capabilities we would need to develop. Luckily there were groups trying to answer that very question.

Space 5

The receipt of an alien message on January 3rd, 2218 set into motion an extraordinary series of events. NASA, as the receivers of the message, became the epicenter of the effort to understand who and/or what were the owners of Planet Nine, and what that ultra-dense spaceship really was. Within minutes of the message unfolding on screens at NASA, rooms were being filled up with all manner of space-focused scientist, awash in an enthusiastic buzz at the possibilities unlocked by finding that manner of technology in our own solar system. Questions flew out like bats appearing at dusk. Was Planet Nine occupied by aliens? If not aliens, then robots and AI? How did it survive in tact for billions of years if it was manufactured? Why was it so dense? So the questions poured forth, and leaders at NASA captured them on white boards. At the bottom of each was written the most important question, “do they come in peace?”

One room coalesced around an effort to enumerate what we could know just from the message, and our observation of the spaceship formally known as a planet. The Advanced—this was the name that had risen up to describe whatever had sent the message, a nod both to “The Advancement” which the message had mentioned, and the fact that whoever had sent the message was clearly further along the tech tree than humans—were not only observing us, but capable of interacting with Earth technology as they had set up a website. The density of Planet Nine was such that its internals had to be some sort of star, most likely a neutron star, which put their technological capabilities thousands, or more likely, millions of years ahead of ours. The message had appeared on a screen similar to phosphorescent screens of the 1980s, and written in a known Earth font indicating that its design was likely recently updated as opposed to having

been built when Planet Nine was created indicating some sort of active upkeep on the vessel. The most recent research on trans-neptunian objects affected by Planet Nine's gravity showed that the planet had been in our solar system for at least 2.5 billion years, it was believed this meant Planet Nine was a rogue planet captured by our sun after the solar system had formed, but was also consistent with a spaceship showing up.

These last two facts, that Planet Nine was billions of years old, and that there had been some sort of recent upkeep, seemed to stick in the collective consciousness of those trying to put together the puzzle of Planet Nine. Could it be that aliens were actually living and working on Planet Nine? If so, had they been there in some capacity for billions of years? If not, then how were they getting there? And if there were no aliens, what was doing the upkeep? Could it be robots and AI? The prospect of some autonomous system operating for billions of years without incident in a foreign solar system was a stretch even to the most enthusiastic proponents of robotics and AI. According to the message, The Advanced had access to teleporters, could it be that the caretakers of Planet Nine weren't permanent residents, but rather teleported in and out of the spaceship when needed? This question lead to a natural followup question, where might they be teleporting from? To answer that we needed to get a handle on just how fast teleportation teleported, and to answer that we needed to know the speed of Dark.

Bell's Theorem is one of the most beautiful results of Quantum Mechanics. It is possible for two or more particles to become entangled, connected in a way where one measured quantity on one of the particles guarantees that another quantity will be measured on the other particle. Separating these entangled particles has no effect, measuring one entangled particle to be spin up will guarantee the other particle will be spin down regardless of the distance between them.

Twentieth century physicists struggled to show how this sort of information exchange (Einstein called it, “spooky action at a distance”) could occur. One suggestion was that the particles contained hidden information, which allowed them to be imbued with their respective spins, but was hidden until observed. Bell came along and proved that no such hidden information was possible. The result, now proven regularly in undergraduate Physics courses, raised more questions than it answered: if there was no hidden information, then how did one particle “know” what was going on with the other?

One solution, derided by physicists for centuries, had recently come back into favor, that the two particles do in fact exchange information, but that they do so faster than the speed of light. Faced with the reality of half-working alien teleporters, even the most ardent maintainers of the absoluteness of the speed of light had to admit the possibility that the speed of dark might be faster—faster, but still finite. Any finite speed should be measurable provided two entangled particles could be separated by sufficient distance. By the twenty-third century, terrestrial distances had been exhausted and demonstrated no limit on the speed of the information exchange so a space-based distance was needed. To do that required somehow storing an entangled particle in a superposition state through a rocket launch and some number of years of spaceflight. NASA decided to dedicate some space on their upcoming return mission to Planet Nine. At 482 AU from Earth, if the particle’s exchanged information at the speed of dark, and the speed of dark was less than around one million times faster than the speed of light, the experiment should notice the lag in information. That of course was assuming that Earth-bound scientists could coordinate with space-based clocks, a problem not at all trivial even to twenty-third century calculations.

NASA had been developing a new fusion drive for long haul space voyages, and felt this new mission—dubbed Epidosis, the ancient Greek word for advancement—would be the perfect opportunity to show it off. The engine would reduce the flight time to Planet Nine by a third down to only fourteen years. Fourteen years was a long time to keep something in space from interacting from anything at all, but two centuries of experiments on entangled particles had made scientists pretty adept at keeping superpositioned particles separate from a universe full of interference.

The plan was to ship sixteen sets of sixteen entangled particles towards Planet Nine, and measure them at evenly spaced intervals starting at a pre-determined time and ending when light signals reached Earth. The experiment would be good enough to tell if the speed of dark was within a million times faster than the speed of light or not. Twelve was the number of particles considered statistically necessary to give a reasonable chance of detecting variance from the expected results of entanglement, and sixteen was the number considered sufficient for redundancy should a particle or two be disrupted during flight. In a terrestrial experiment, those numbers would have been increased a thousand fold, but there's only so much room on a spacecraft flying seventy-two billion kilometers.

While NASA was preparing for Epidosis, astrophysicists around the world were trying to answer questions about who, or what, the Advanced were. It stood to reason that any group that could place a planet-massed spaceship in orbit around a star, and lie in wait for billions of years, likely had a much larger presence in the galaxy than we had previously thought possible. The first order of business was to check the catalog of nearly one million exoplanets that had been found for any planets fitting Planet Nine's description. Such planets, dubbed neptunian by

exoplanet hunters, were fairly common finds in the stars we had studied, but there was one big problem. Exoplanets are much easier to find the closer they are to their star, thus most neptunian planets were found much closer to their star than Planet Nine. In fact only a handful, less than one hundred, of exoplanets had been found at an orbital distance over 300 AU. Was it possible that Advancement planet ships could be found closer in to their star? Astrophysicists weren't taking any chances.

The Erin Dalton space telescope (EDST), named for the astrophysicist that had pioneered the telecascading method for exoplanet atmospheric study, was set to launch in 2225. The message from Planet Nine raised the priority on that launch, and ESA was able to get the space telescope into position a year ahead of schedule in 2224. Making use of the best spectroscopy setup mankind had ever devised, the EDST would be able to measure the atmospheric contents of exoplanets up to 1,500 light years away under optimal conditions. After the message from Planet Nine, its first year of operation was dedicated to checking as many neptunian atmospheres as possible.

The neptunians were somewhat of a forgotten group when it came to astrophysical time and effort. Smaller than Jupiter-like planets, they were harder to study as a rule, but lacking the life-giving features of Earth-like planets, they also lacked a compelling reason to study them. This meant that the new effort to catalog the atmospheres of neptunian planets was replete with specimens to observe.

After 18 months of cataloging, astrophysicists found that 100% of the neptunian planets had atmospheres. None of them were just glassy black spheres like the Advanced's Planet Nine was. Some felt this was due to the fact that all of the neptunians were far closer to their stars than

Planet Nine was to our sun. The experimentally inclined continued their cataloging of neptunian exoplanets, but the more theoretically focused astrophysicists got down to the business of trying to figure out how, using twenty-third century technology, they could go about detecting planets out past 300 AU from their star.

Igor Popov was born in 2173 in St. Petersburg, Russia. A beneficiary of the education reforms implemented by the UN in the mid twenty-second century, Popov was able to attend high school in Switzerland, undergraduate university in Belgium, and earn his PhD in astrophysics at Cambridge. In 2206 he joined Roscosmos as an expert in extrasolar planetary formation. By 2225 he was leading a group of around thirty scientists on the hunt for neptunian exoplanets at large distances from their stars. In 2226 he had an epiphany.

It's convenient to think about the planets orbiting the sun, but when things orbit each other, they actually orbit around a common center of mass. This means for each planet, the sun orbits a center of mass with the same period as the planet spends orbiting that same center of mass. This is true for every star and its planets out in the universe. Astrophysicists can measure this rate of rotation through the doppler effect on the light given off by a star. Doing so leads to the radial velocity method for exoplanet discovery, the most widely used method over the last two hundred years of planet hunting. The short radial distance of the star's orbit combined with the long orbital period of a planet at the distance of Planet Nine made using radial velocity unviable for finding the far out neptunian planets like Planet Nine. A second method for finding exoplanets is called astrometry. Astrometry looks at a star's position in the sky, and measures how it changes over time. Astrometry's drawback was that it lacked the precision of a method

like radial velocity, but it had a big benefit that made it useful in the hunt for far out neptunians: it could measure variances over long lengths of time.

Popov's insight was to use two hundred years worth of astrometrical data to inform radial velocity measurements to find differences consistent with far out neptunes. He led a year long survey of data collected all the way back to the Gaia spacecraft launched by ESA in 2013. The survey ranged over seven generations of space-based astrometrical observers looking for variations consistent with planets with orbital periods on the order of 10,000 years. It was an enormous amount of data to work through, looking for the astrophysical equivalent of a needle in a haystack. But by 2227, Popov and his team had found six needles, all within 150 light years of Earth. If the speed of dark was sufficient, they might be close enough to allow teleportation between star systems that could support something like human life, a prospect that was as mind boggling to think about as it was to write down.

Scientists the world over worked to piece together what these six star systems might tell us about the Advanced. Four of the star systems were younger than ours with two of them being less than two billion years old, which meant that the Advanced were a society capable of surviving for billions of years. For most of human history, we've been largely focused on simply surviving tomorrow. Climate change posed the first serious existential threat which required planning with a time horizon beyond just the next decade. Thinking of how a civilization could survive through billions of years was well outside of human experience, but presented with the evidence, scientists now had to determine what survival on that scale might mean. Billions of years meant surviving planet destroying asteroids, nearby supernova, giant stellar flares, and

other cosmic-level disasters. All of which humans had so far managed to avoid in the fleeting time they had existed.

Scientists looked for similarities in the star systems with far out neptunes and found that each one of them contained at least one planet suitable for life. This in and of itself wasn't all that remarkable, as most stars observed had at least one planet that might sustain life, but the six star systems' life-supporting planets were particularly close to Earth-like conditions. Three of them had already been found to have atmospheres made up of nitrogen, oxygen, water vapor, and carbon dioxide. They were similar to Earth in size and mass, and orbited within the goldilocks zone of stars not all that dissimilar to Earth's sun.

These findings began to coalesce in the form of a solution to the Fermi Paradox, the paradox which wondered where all the extraterrestrial life was. It was the so-called Galactic Federation, where advanced civilizations join together and treat more primitive civilizations the way second millenia humans treat unsullied civilizations in the Amazon. The Advanced's message about "the four things Fours must do" was some sort of technological threshold for civilizations to join the galactic federation they called The Advancement. It was a federation that existed for billions of years, which likely had seen countless civilizations reach a technological level comparable to twenty-third century human beings.

The answer to what is the technology level of a civilization capable of supplying and supporting multiple planet-massed spaceships, and which has survived for billions of years, was everything. Everything that humans could imagine, or would imagine, was within the capabilities of such a civilization. The only limitation would be the laws of Physics, but not the laws that humans understood, but the actual laws of Physics in their entirety.

But with the speed of dark experiment looming, and the DecRec physicists trying to figure out how dark matter manipulation worked, a contingent of scientists decided to focus on one very basic question. It was one that Fermi had first asked when devising his paradox, namely if there are extraterrestrials everywhere, why can't we observe them? At the time, Fermi was limited by only knowing about communication in the electromagnetic spectrum, but with our increased knowledge of the dark field, we now knew that the Advanced likely communicated via dark waves. Understanding the nuclear pasta antennas the DecRecs were outfitted with became the focus of study.

As scientists everywhere dove in on the nuclear pasta antennas, the Langston Conjecture was growing in popularity. Angela Langston was an astrophilosopher, a branch of human thought that began to grow in popularity with the first images sent back from Proxima Centauri b by the WCIP probe. She operated a popular VisiTV channel that speculated on the reach of human-level intelligence throughout the cosmos, and what that meant for the future of mankind. The Langston Conjecture stated that the Advanced must have seen many civilizations climb the technology ladder, and as such everything they would give us to work on would be of a technology level within years, decades at most, of current human technology. Put another way, the fact that the Advanced asked humans to make teleporters implied that humans were sufficiently close to making teleporters.

The Langston Conjecture, when applied to the nuclear pasta antennas that ostensibly interacted with the dark field meant two things. First that humans would soon be capable of manufacturing nuclear pasta, a premise that seemed a bit far-fetched to engineers of the time. Second, that there was some interaction capable of going between light and dark fields, which

we could hope to understand. Remember that quantum theory seemed to imply that matter could cross between the two fields, but not reside in both.

As scientists and engineers were trying to figure out what was going on with the nuclear pasta antennas, a series of experiments at the Azania Space Agency Center in southern Namibia had found something interesting. The DecRecs were able to recohere things out of the ambient dark matter that permeates space around the Earth. Using a series of undocumented commands, scientists were able to recohere a lump of iron, making the substance appear seemingly out of nowhere. The ramifications of being able to recohere dark matter into useful light matter objects was huge. This finding captured the imagination of those already working on the antennas, and lead to the next big breakthrough in the history of teleportation.

Dark Field 6

By the 2220s, Physicists had gotten pretty confident in their ability to deal with fields and forces. Between gravity, electromagnetism, the strong nuclear, and the weak nuclear forces, we had a pretty good handle on what was going on in the part of the universe we could see. For most of human history, this was the only part of the universe we cared to think about so it was natural to be a little overconfident in our abilities. Most physicists believed that dark matter followed many of the same rules, that there were fields and particles in those fields, which interacted with each other through some set of knowable rules. So when it came time to really try and figure out what was happening in the dark field, we started with what we knew—trying to get dark matter to attract or repel.

First, let's revisit what we knew about dark matter at that point. We knew that light matter could decohere into dark matter in the presence of a strong gravitational field or a strong accelerating reference frame. We knew that connectedness mattered—objects that were connected in the light field persisted through the decoherence and recoherence process. We knew that the light matter properties of objects could be manipulated while the object was dark matter. We knew that dark matter could also recohere from the ambient dark matter around a DecRec. And we knew that it was possible to transform digital information into a form that could interact with dark matter via nuclear pasta antennas.

So scientists tried to get two dark matter objects to attract each other by imbuing them with a series of physical properties. The setup put one DecRec at the center of the estimated Diallo Radius for the DecRecs, a second DecRec a touch more than two radii away. Both

DecRecs would decohere the same object, and then the DecRec within the Diallo Radius would attempt to recohere both objects. If the team, led by the tenacious Aurelia Veracruz, an experimenter who had gained prominence working on fusion recoherence experiments throughout the South American Economic Zone, could recohere both objects in a single DecRec, they would know that their objects were attracting each other.

Unsurprisingly, charge did not work. Nor did temperature. When the experimenters imbued both objects with opposite angular momentums, however, they got what they were waiting for—two objects recohering at the first DecRec. It was the first successful teleportation of any kind on Earth, an astonishing result. Theorists all over raced to try and explain this incredible property of dark matter that somehow angular momentum translated into an attractive force similar to electromagnetism. Meanwhile, Veracruz and team got onto the business of experimenting with angular momentum and the DecRecs, trying to determine how far apart the attractive force worked, and how strong it was. Teleportation was now something experimentally verified, it was hard to imagine anything more exciting to work on.

This first teleportation opened up a whole new line of inquiry into how the recoherence mechanism worked in a DecRec. When objects were simply recohering where they decohered, it was believed that whatever property of the spinning nuclear pasta that kept the object connected to the DecRec was the same thing that allowed the object to recohere. With the recoherence happening simply by the object entering the Diallo Radius, it became necessary to rethink how recoherence could happen.

Amaya Clopin was born in Trenton, New Jersey in 2201. They grew up going to the nearby Visitor Great Adventure (Visitor had taken over the six flags brand of amusement parks in

2168). The park used a proprietary system of bluetooth enabled devices, which allowed park goers to interact with installations, queue up for rides and attractions, perform payments in park, and engage with in-park games and scavenger hunts. Central to the proprietary system was the protocol used by devices in the network dubbed Multi-device Asynchronous Generic Input/output Consensus (MAGIC). As Clopin matriculated through high school, they became fascinated by how the MAGIC protocol worked, going so far as to get an internship with Visitor Parks to work directly with the protocol. This interest led them to pursue a dual Physics and Computer Science degree at Brown University before coming back home to Princeton for their PhD.

When the DecRec groups were apportioned after receipt of the Advanced's message, Clopin was invited to join Fermi Lab's group. The American particle physics lab was smaller than CERN, but still mighty. Since the 1960s when the first dark matter particles were decohered at CERN, Fermi Lab had been focused on recoherence experiments. The work helped drive advancements in energy production in the US and the world, but also provided a fully funded lab to go after the problem of recoherence with teleportation.

When the results of the 2226 teleportation and subsequent papers on the problem of recoherence were published, Clopin joined their group in pouring over the material trying to understand how the DecRecs were able to recohore disconnected objects.

The first clue came from Fermi Lab's partner in recoherence study, Agencia Especial Mexicana in Mexico. There a team of theorists led by Ernesto Gomez had worked out that the conference of physical properties by the DecRecs implied that arbitrary information could be encoded into dark matter. Gomez and team hypothesized that some of this encoding could be

used to confer onto dark matter a type of identifier. Waves interacting with this identifier could be used to pinpoint dark matter objects and change their physical properties.

Clopin wasted no time messaging Gomez after the publishing of his team's paper on this encoding mechanism. The two started a correspondence and collaboration through their shared work on recoherence. Gomez was also a Physicist, but had also majored in Math at Instituto Tecnológico Autonomo de México before getting his PhD from Universidad Nacional Autónoma de México. Born in 2179, he had a number of papers and results under his belt.

The two scientists knew that the key to understanding the DecRecs' recoherence mechanism, was figuring out how the digital information from the DecRecs' computers got transferred into information in the dark field. Dark matter being identifiable implied that the DecRecs were able to target individual dark matter objects, and of course that's what we saw in practice with being able to change physical properties of decohered material. To find that out, Clopin and Gomez started to work on how electromagnetic waves, the very things that define light matter, could manage to influence the dark field, the thing that by definition didn't interact with electromagnetism at all. The key, they felt, had to lie in the strange material of nuclear pasta that the DecRecs had introduced us humans to.

We have two examples of how waves propagate in the light field. A moving electric charge creates a magnetic field, and a changing magnetic field induces an electric current. Put these together and an electromagnetic wave propagates itself through space at the speed of light. Gravity does much the same thing, but gravity produces gravity so the gravitational wave propagates itself. For the past two hundred or so years, it was assumed that whatever a dark wave was would propagate in much the same way, propelling itself forward at the speed of dark. We

had just found that angular momentum worked as a sort of analog to electric charge, and so long as the speed of dark was finite that would necessitate, via dark relativity, that something like magnetism must also exist. So the question was, how to transform the moving electric charge of the light field to the angular momentum of the dark field.

It would be Clopin that would have the critical breakthrough, but Gomez who would provide the mathematics to support it. We like to talk often about how the speed of light is constant, and it is: *in vacuo* (in a vacuum). The speed of light actually changes in mediums that it travels through. We see this in every day life when we see something in water looking disjointed from the part of it sticking above the water. In everyday materials, this change in the speed of light is not that dramatic, but, Clopin asked, what is the speed of light in nuclear pasta? If it was slow enough, then the acceleration that light would experience leaving the nuclear pasta antenna might be sufficient to decohere it into a dark wave.

With a plausible explanation for how electromagnetic waves could decohere into dark waves, Clopin and Gomez next had to figure out how the dark waves interacted with dark matter. With their suspicion that dark matter was addressable, they tried to think of ways that dark waves could hook into dark matter affect a dark matter object's properties. Waves can combine. Two waves combining will form a wave that's the sum of the two that combine. If the waves are in phase, they'll combine to form a larger wave. If they're out of phase, the combined wave will be smaller, and if they're perfectly out of phase the two waves will cancel each other and disappear. This latter scenario caught Clopin's attention, captured here in a messaging exchange between them and Gomez.

Clopin: What if it doesn't?

Gomez: Doesn't what? Doesn't destroy the wave?

Clopin: Yes, exactly. Think about it. Dark matter, as far as we know, is made up of superpositioned particles existing as a sort of standing wave in the dark field. A simple phase inversion of part of a dark matter object's standing wave can't destroy its part of the wave because then other parts of the superposition would have to be similarly destroyed, and that violates locality.

Gomez: Ok, so if it doesn't destroy it, what does it do?

Clopin: Well if it's making our jobs easier, it'll absorb the information contained in the wave and incorporate it into its superposition. You encode the object's address and the properties you want to change and send that message out. It's almost like how ethernet works.

Gomez: Seems worth a try. Can you rig up an antenna to encode a message from us?

Clopin: Yes I think so.

Gomez: Ok, I'll work up a proposal to get us the space and equipment we need for an experiment.

International politics, and other experiments being run, led the two to setup in the US at Fermi Lab for their experiments. On February 7th, 2228, after a lengthy period of trial and error, the first successful recoherence of manipulated dark matter using Earth computers was performed, much to the delight of the team of 35 scientists led by Clopin and Gomez.

Clopin put their computer science background to work leading a small team on encoding all the various properties that DecRecs had been found to affect. Within just a few months, they, and, as a result of their papers, the rest of the world, were operating nuclear pasta antennas which covered a significant fraction of the DecRecs' capabilities. But there was still one thing gnawing

at Clopin. The Alchemists had shown that DecRecs were able to recombine arbitrary chunks of dark matter. Without a known address, how were the DecRecs able to make such finely tuned manipulation of dark matter? The question burrowed its way into the back of their mind as they worked out the encodings.

The discovery that nuclear pasta could decohere light into dark matter caused many to ask if the reverse could be true. Hooking up nuclear pasta antennas to a receiver was, of course, one of the first things humans had thought to do upon deconstruction of a DecRec, but the resulting static had been considered regular ambient noise at the time, and scientists quickly moved on to other things. With the publishing of Clopin and Gomez's paper on the speed of light in the nuclear pasta medium, however, interest turned back to whether or not nuclear pasta antennas could pick up dark waves sufficiently to enable communication.

Again scientists tried to pick up a signal through the noise of a DecRec's antenna, and again they found just incoherent static, this time, however, to rule out Earth-based interference with possible dark wave signals, they decided to perform the experiment on Mars. When the martian team fired up their antenna, their signal was very different. Fed through software designed to look for patterns within the staticky patterns of electromagnetic waves, the martian data was absolutely chock full of patterns that looked like communication. When analysis had removed all of the known electromagnetic signals, unaccounted for communications made up five times as many signals as known. Here, finally! was a solution to Fermi's Paradox—we couldn't "hear" other civilizations because they communicated using dark waves, once we figured that out we were awash in their chatter. It's still an open question as to why Earth seems

to filter out this communication, perhaps some combination of atmosphere and magnetic field, some future scientist will no doubt figure it out.

With dark wave listening a novel capability of 2227, space agencies around the world coordinated the launch of dark wave antennas into orbit to start picking up some of the signals that Mars had started to receive. By 2230 there were four space-based nuclear pasta antennas harvested from DecRecs around the world. With the parallax these satellites created, astrophysicists could start determining where signals were coming from. The short answer: everywhere. Some directions were fainter than others, indicating the signals were coming from farther away, but there was little variation in the volume of signals, except when looking at the galactic center, which was understandable because of the greater density of stars in that direction. Around the clock monitoring of these dark wave antennas was running through around the clock processing by some of the most powerful computers on the planet. If there was rhyme or reason to the signals, we were determined to find it.

Once there was 24/7 monitoring of different dark wave antennas, people started noticing the differences between antennas. Most signals came from far away, and were thus picked up in some capacity by the different antennas, but there were low energy signals that were reaching each individual antenna. These signals were different at each antenna, indicating the cause was local, as opposed to far off. A highly anticipated experiment on Mars was carried out to get a baseline on how far these local signals were traveling. They found that the signals were localized to a sphere of about a kilometer in radius. Wave signals that were this localized were quite the head scratcher. First off, no one knew what could be giving off these waves, as far as anyone knew, dark matter sat in a decohered state until something interacted with it. Secondly no one

had an idea of what these signals could be. In the light field, everyday objects don't sit around giving off light. Light comes from stars and is reflected off those objects. Could it be that there was something in the dark field that was giving off dark radiation which was being reflected, or were dark matter objects giving off radiation of their own?

Clopin had an idea as to what the signals could be. Having studied the MAGIC protocol the Visitor Parks used, they knew that to enable the multi-device communication, devices using the protocol had to advertise their existence on shared networks. What if dark matter objects somehow advertised their existence to the DecRecs, and that's how the DecRecs were able to recombine ambient dark matter? The dark matter wouldn't be arbitrary then, it would be known to the DecRec. They and their team dove into the data of the low energy signals, looking for some sort of pattern that they could hook into. At the same time, now that they knew what to look for, terrestrial scientists were trying out nuclear pasta antennas and realizing that they were picking up the same local dark wave signals. Whatever prevented long-range dark wave signals from reaching Earth wasn't interfering with the local waves.

While their team was sifting through the copious amounts of data provided by nuclear pasta antennas, Clopin turned their attention to the last piece of the recombination puzzle—namely how did anything recombine absent the extreme conditions of a nuclear fusion event. This was the specific question that Fermi Lab and AEM had been working on solving since we discovered Planet Nine was a spaceship. The two groups had been working without many clues. We knew that recombination must be possible without extremes of energy, temperature, pressure, etc. since the DecRecs were capable of recombining dark matter at ambient conditions. Years of experiments

had yielded little progress, but being colleagues with the scientists working the problem helped Clopin to figure it out.

They started with an assumption: since it was something in the light field—an accelerating reference frame—that decohered light matter, it should be something in the dark field—?—that would recohere dark matter. To get after that question mark, Clopin got back to basics. Remember what dark matter is. Light matter can enter a state called a superposition. Normally matter in a superposition will return to its normal state when it interacts with something. This interaction “collapses the wave function,” a phrase which speaks to the quantum mechanical nature of the superpositioned particle. When an object decoheres from the light field, it somehow loses the ability to interact. It is a wave function that can no longer collapse. Clopin figured that if they could reverse whatever process it was that caused dark matter to lose its ability to interact, then interaction could happen and dark matter would recohere.

There was a clue. We had been able to decohere by building a giant machine capable of spinning matter fast enough to mimic the gravitational field near a black hole, but the DecRecs didn’t need to spin what they were decohering. They only needed to spin the nuclear pasta disk in their glass octahedrons. It was theorized that the DecRec was somehow able to convey information about its accelerating reference frame to its nuclear pasta arm, which in turn conferred the condition necessary for decoherence to any object touching it. Clopin, after working out the details of light and dark wave reception with Gomez, felt that the material’s ability to interact with both waves must be the key to this conveyance.

In August of 2233, Clopin invited Gomez and team members from Fermi Lab and AEM for a demonstration. The group met in the gymnasium of a high school in Batavia, Illinois, near

Fermi Lab. Clopin explained that they wanted to be away from established scientific equipment to better demonstrate what they had found. The gymnasium was empty save for the rows of bleachers that the invited guests took seats on. Clopin rolled in a cart with a laptop wired up to a nuclear pasta antenna. In their hand they carried a stand with a ball of cloth on it. They parked the cart at one free throw line and walked the stand over to the other. When they had walked back to the cart they announced that they were about to recombine dark matter sufficient to light the ball on top of the stand on fire. They counted down from five, and then hit a key on the laptop. A fireball appeared over the cloth ball, igniting it. Near center court, an orange glowing crystal fell to the ground with a clink.

The demonstration was filmed from several angles. What follows below is a transcript of that video:

Gomez: You figured out addressability between the light and dark fields?

Clopin: Yes indeed. The light field seems to imprint on dark matter, and the dark matter broadcasts the imprinted address back to the light field through nuclear pasta antennas. By sending out decohered instructions with the address we can manipulate arbitrary dark matter. It seems that even though matter cannot occupy both fields at once due to the splitting of Time dimensions, that some manner of back and forth is still allowed. I've been calling the back and forth Asynchronous Coordination.

Team Member 1: What light field properties have you tried?

Clopin: As many as I could think of. As far as I can tell, any input you give to the dark matter has some corresponding effect when the material recombines.

Team Member 1: Can we see another one?

Clopin nodded, and turned their attention to the laptop. After keying in some instructions they called for everyone to return their attention to the stand. They counted down from five again and a bolt of electricity arced down towards the stand from above. Another small crystal, yellow this time, fell to the ground near center court.

Gomez: What is that that keeps dropping near center court?

Clopin walked over, picked up the two gems, and walked them over to the stands for the audience to pass around.

Clopin: As far as I can tell, it's Nineum.

Team Member 2: I thought Nineum was blue.

Clopin: I thought so too, but I've now seen pink, yellow, and orange. It looks the same as the Nineum from the DecRecs save for the color. Seems to be some sort of byproduct of the MAGIC process.

Gomez: MAGIC?

Clopin: Multi-field Asynchronous Generic Input/Output Coordination—MAGIC for short. That's what I call this whole process. Do you have a better name for conjuring fireballs and lightning bolts out of thin air?

Space 6

The discovery of MAGIC was obviously a momentous occasion in human history, but it's a story that we're still writing, and is better told in a future chronicle. For now we need to continue on our path to understanding the history of teleportation. With the mechanism behind recoherence of arbitrary dark matter being understood, there were two questions left to be answered on the path to human engineered teleportation. The first was solved by MAGIC. Nuclear pasta, material thought impossible to manufacture with Earth-based technologies, was trivially created by the manipulation of dark matter through the MAGIC process. By 2234, nuclear pasta was being created out of dark matter, and being used for dark wave antennas, and the beginnings of human-made DecRecs. This left one more open question: how to move dark matter from one DecRec to another.

How do you move something that doesn't interact with the electromagnetic field? For all of human ingenuity, we had really only figured out that to move something you need to push it with something else. Even nuclear powered rocket ships were just highly sophisticated machines for pushing a spaceship with something else. Pushing works due to the electromagnetic force between atoms in the materials doing the pushing. Without electromagnetism, the atoms would slip right through each other leaving both materials motionless. So complete is dark matter's disconnection from the electromagnetic force, that it doesn't even get trapped by black holes. To make matters worse, not only did we need to compel it to move, but we had to accelerate it to a speed that made teleportation faster than regular travel, something not even nuclear powered rocket ships could do.

In May 2235, a mere nine months after Clopin and team's MAGIC paper had been published, a pivotal paper hit the wires. The lead author was a dark field theorist named Hakim Bennis, a post-doc at the University of Zurich. Born in 2209 in Tangier, Morocco, Bennis developed an early love for math and physics, excelling in schools throughout the North African Economic Zone, before getting his PhD from Copenhagen University and then landing an associate professorship in Zurich. There, perhaps in honor of Einstein, the university's most famous alumnus, Bennis became enamored with endeavoring to work out the meaning of a speed of dark greater than the speed of light. Laboring to try and provide a prediction of the speed of dark in advance of the result of the Epidosis mission had led him to publish this paper entitled *On Dark Dynamics and Massome Matter*.

The universe of the light field is made up of particles, and, as scientists, we endeavor to categorize these particles in such a way as to grant us the ability to make predictions about their behavior. One such division is between massive and massless particles. Einstein's famous equation, $E = mc^2$ and its corresponding relativistic transformations sets up the speed of light as a cosmic speed limit for light matter. As massive objects approach the speed of light, it takes an unbounded amount of energy to increase their velocity. A useful quirk of this reality though, is that massless particles are able to travel at the speed of light. That's a good thing too, as the only known massless particles, photons, *are* light. This division, between massive and massless particles, encapsulates all of the light matter that we know. In Bennis's paper, he introduced a third type of particle, one that he called *massome*, and used it as a hypothetical description for dark matter, and a possible explanation for the speed of dark.

Perhaps the main reason for physicists historical reluctance to accept anything traveling faster than the speed of light, was that when you started plugging superluminal speeds into the equations of the universe, things started going haywire. The mathematics of an additional dimension of time provided some relief from these problems, but scientists still had to resolve what happened in the realm that the light and dark field shared—Gravity. Gravity waves propagate at the speed of light, so what exactly, if the speed of dark was faster than the speed of light, happened when dark matter traveled faster than the speed of light? Sonic booms had become a rather regular occurrence on a planet sending as much into space as we were in the beginning of the twenty-third century. Sonic booms happen when something travels faster than the speed of sound, was there some proportionally larger boom that happened when dark matter exceeded the speed of light?

Bennis argued that we could get around these problems if dark matter only had mass while moving at subluminal speeds. Here we must introduce a twentieth century idea, the principle of least action. Put simply, objects in the universe follow the simplest path when traveling between two points. If the universe were flat, this would mean objects follow a straight line between two points. In the curved spacetime of our universe, this meant that objects followed the shortest curve between two points (my undergraduate Physics professor would want me to point out here that straight lines in a curved space *are* curves, and that objects just always follow straight lines). Bennis showed that any deviation from these curves of least action led to infinities for any dark matter that had a non-zero mass. Infinities are usually a quick way of knowing that you're wrong about something. The interpretation was that any deviation from the

curves of least action meant that dark matter must have zero mass. The implication was that dark matter could transition between dark matter and dark waves with just a little push.

Bennis continued his line of reasoning in his Dark Dynamics paper, and argued that just as massless photons carry the electromagnetic force in the light field, massless dark matter particles were the force intermediaries of a Dark Force. Dark Forces of various kinds had been proposed throughout the two centuries since Diduch's paper on Dark Time. This time, however, Bennis provided a reasonable, and testable, framework for the Dark Force. Here we had a force propagated by massless particles that were also waves, moving at a finite speed, and interacting with matter in knowable ways. Since the speed of dark was finite, some relativistic analog to magnetism was needed for the dark wave to propagate. It was electromagnetism, just with different values for the permeability and permittivity of free space.

The permeability and permittivity of free space are functions of the same four units: kilograms (mass), meters (length), seconds (light time), and Amperes (electric current). The Dark Field shared two of these units: kilograms and mass, and had a linear transformation to change seconds to dark seconds. The open question was what was the dark force analog to the amp. One amp is the amount of charge in about 6×10^{18} electrons moving past a point in a second. Bennis had no way of determining what this amp analog could be, but it was bounded, he argued, with a lower limit established experimentally by studies of recoherence already performed on earth of one hundred times the speed of light, and an upper limit provided by the fact that space is not littered with dark matter black holes of about ten million times the speed of light.

In July of 2236, Bennis and the rest of the physics community got their long awaited answer. The speed of dark was on the order of magnitude of 100,000 times the speed of light

right smack in the center of Bennis's limits. The establishment of a baseline for the speed of dark had a rather large effect on the popular zeitgeist. Similar to the first picture of the earth taken from space, it captured the imagination in a way that hypothesis and conjecture could not. Sending objects through the dark field as dark waves at the speed of dark, humans would be able to travel to nearby stars in a matter of hours. A trip from one side of the Milky Way to the other would only take a year. Travel within the solar system would be so fast as to seem to be instantaneous pretty much anywhere one would want to go. All that was standing in our way to the stars was figuring out how to give dark matter a push.

Dark Field 7

Aurelia Veracruz was born in 2114 in Montevideo, Uruguay. By 2236, at the age of 122, she had lived two lifetimes worth of careers in the fast paced world of high energy physics, and space age engineering. In 2152, she had joined the South American Economic Zone's leadership as the scientific lead for the development of the SAEZ Space Agency. She stayed at the SAEZSA through 2174 before returning for a stint at the University of Chile. Wanting another change with the change of the century, she returned to the SAEZ Space Agency to lead their research and development. Still there in 2218, she was a natural choice to lead South America's DecRec program.

Veracruz led with the wisdom of her years, and the enthusiasm of someone much younger, a combination that rejuvenation programs around the world were making more common. I've already recounted how her group was the first to make a successful teleportation of any kind, and by 2236, with the speed of dark estimated, MAGIC discovered, and massome dark force particles theorized, Veracruz was determined to win the race to perform the first directed teleportation.

The last barrier to directed teleportation, if Bennis was correct, was being able to confer to dark matter a direction of motion off of a path of least action. Veracruz brought together her team at SAEZSA for a brainstorming session in November of 2236 of which we have video. What follows is a transcript of that meeting where Veracruz, standing formidably in front of white boards, gathered the thoughts of her team.

Veracruz: If we apply the Langston Conjecture, we can say that how a DecRec provides direction and impetus to a decohered object must be knowable with current technology and scientific knowledge. What do we know, or think we know about how we can interact with dark matter?

SAEZSA Team Member 1: We know we can confer onto it arbitrary light field properties, which will persist through recoherence.

Veracruz: Good, how do we confer those properties?

SAEZSA Team Member 1: Dark matter contains addressable wave components, which we can use to hook dark wave messages into.

Veracruz: Alright let's target that as one of the avenues to pursue. What else do we know about dark matter?

SAEZSA Team Member 2: We know that angular momentum creates a force that is sometimes attractive and sometimes repulsive, analogous somewhat to electromagnetism in the light field.

Veracruz: Good good. What else?

SAEZSA Team Member 3: We know that connectedness matters. That objects that decohere at the same time will stay together through recoherence. And I guess along those lines we know that the connected objects have to be touching the nuclear pasta arm when they decohere.

Veracruz: Good again. Anyone have anything else?...No? Ok so invoking the Langston Conjecture once more, our solution should be something that combines some, or perhaps all of

what we know about MAGIC, the Dark Force, and Connectedness. This is our focus now people. All ideas are welcome.

Teams formed around the three pillars Veracruz had identified, doing deep dives into the literature on MAGIC, the Dark Force, and Connectedness. Despite being very new to the world of Physics, all three had generated such substantial interest that simply aggregating all the papers written on them was a hefty endeavor. A spirit of collegial competition sprang up between the teams. Amongst the non-DecRec groups at SAEZSA, bets were placed as to which group would find the answer to directed teleportation.

What is a photon? The carrier of the electromagnetic force, which travels at the speed of light, has all sorts of curious properties that separate it from other particles in the Standard Model. For one, photons aren't conserved. This means that an energetic body can give off an arbitrary amount of photons, limited only by energy and quantum mechanics. Photons are also their own antiparticles, but because they don't have mass they can't annihilate into real particles. These two properties of photons allow them to make up all the light that we see from our sun and from the stars around us.

Dark force particles traveling at the speed of dark are also massless, but if Bennis was right, those massless particles *were* conserved since they came from dark matter to begin with. Similarly, though the verdict was out on any sort of anti-particle for dark particles, dark force particles were believed to create massive dark matter should they annihilate. These two hypotheses were enough for teams to go on in exploring what Dark Matter, and Dark Force particles should behave like.

The Dark Force group, which was working on these hypotheses, had heard that the connectedness group was working on something of use. The connectedness group had been investigating the ramifications of force carrying particles which were transmuted from matter, like we believed dark force particles were, rather than emitted, like light particles are, when someone asked what that meant for connectedness. Working up the math for what should happen within a complex system with regards to connectedness, it seemed to indicate that should one part of a dark matter object be transmuted into a dark force particle, then all of it should. In other words, if we remember wave-particle duality, if part of a dark matter object was turned into a dark wave, then all of it should be. To test the hypothesis, the group needed to find a way to transmute a dark matter object, and for that they turned to the MAGIC group.

The world had been furiously cataloging what messages it was possible to encode in dark matter using the MAGIC process. The SAEZSA MAGIC group, when presented with the other two groups' findings, turned their attention to a message that could provide connectedness of the message to the dark matter object. The effect was easy enough to test for, normally an attached message would manifest itself when an object recohered, but if hypotheses were correct and the connectedness transmuted a dark matter object into a dark wave, the object should simply disappear as it would find itself traveling away from the group at the speed of dark.

The MAGIC and Dark Force groups turned back to the connectedness group for insight into what that message needed to be—what exactly was the property of connectedness. It was something more than just chemical bonds since complex objects with multiple parts could decohere and recohere. The connectedness group had to try hard to keep it from becoming a metaphysical question.

Veracruz had a critical insight. The Action Containment Enclave or ACE that each DecRec created ostensibly removed whatever was in it from interacting with anything outside of it. Perhaps it was the ACE that allowed for connectedness with objects within it. If so, then perhaps any action involving the object would be considered connected at the time of decoherence. If that were the case then a message as to the desired kinematics of the object directed at the object at time of coherence might just do the trick.

I got my invitation to join Veracruz and her team for a demonstration of teleportation on March 19, 2238. Witnessing the first successful light field decoherence, and spatially directed recoherence of a living organism has been one of the greatest moments of my life. It has ushered in a new age of human development, one that promises to take the human endeavor to new heights.

I started this chronicle shortly after witnessing the first teleportation of a living organism, but as my writing velocity is only so high, I can share one more piece of information. Three days after Veracruz's demonstration, the room in NASA dedicated to monitoring Planet Nine once again sprang to life. The Advancement, it would seem, was aware of our progress. The shiny black screen of the planet-sized space ship once again blinked with its green cursor before spelling out a simple message,

Good work humans. We'll see you soon.